

Policy Implementation and Electoral Accountability

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Abstract

Policies that provide public services are expected to improve incumbents' electoral prospects. In practice, such policies do not always benefit incumbents and can even reduce their support. We argue that public-service policies can enable voters to observe incumbents' efforts to provide public goods and whether those efforts succeed or fail. Unsuccessful efforts may generate electoral losses that outweigh the rewards for success. We study a Philippine policy that randomly assigned municipalities access to public-goods funding. In municipalities with access, villages competed for grants through a public application process. We use this assignment and funding cutoffs to estimate the effects of policy access and of receiving a grant. Policy access reduces village incumbents' reelection and their likelihood of seeking another term. Among incumbents who run, narrowly receiving funding increases vote share. These findings suggest that policies can deliver public goods yet generate electoral losses on average, without implying weak electoral accountability.

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1 Introduction

By the early 2020s, about 4.5 billion people were not fully covered by essential health services, 2.2 billion lacked safe drinking water, and 3.5 billion lacked safely managed sanitation ([World Bank, 2020](#); [World Health Organization, 2023](#)). At the same time, citizens rank public services among their highest priorities across developing democracies ([Grossman and Slough, 2022](#)).

In standard theories of electoral accountability, voters use observed public-goods outcomes to evaluate incumbents and decide whether to retain them, which can give politicians incentives to respond to these priorities ([Ferejohn, 1986](#); [Fearon, 1999](#); [Besley, 2006](#); [Ashworth, 2012](#)). Yet, a growing empirical literature finds that policies that expand public-service provision do not consistently benefit incumbents and can even reduce their electoral support ([De Kadt and Lieberman, 2017](#); [Sandholtz, 2025](#); [Habyarimana et al., 2021](#); [Goyal, 2024](#)). Understanding why these policies do not consistently translate into electoral benefits, and whether such effects indicate weak electoral accountability, therefore remains an open question.

In this article, we examine a new explanation for why public-service policies do not always generate electoral gains for incumbents: visible efforts to secure public goods can expose incumbents to electoral losses when those efforts fail. Public goods are rarely delivered automatically. Even when policies are initiated and funded by national governments, local politicians may need to secure resources, coordinate officials, organize delivery, or maintain services. These implementation efforts may succeed or fail to secure the intended benefit.¹ When citizens observe these efforts, non-delivery tells them that the incumbent has tried but failed to secure the intended benefit. If they believe the incumbent could influence the outcome, this failure may reduce electoral support. Under this account, the policy’s average electoral effect depends on how frequently efforts succeed or fail and on citizens’ responses to each outcome. Losses following unsuccessful efforts may outweigh the rewards associated with successful projects. A policy can therefore deliver public goods while reducing incumbent support, without implying weak electoral accountability.

To examine our argument, we need a setting in which citizens can observe politicians’ efforts

¹We define policy implementation as actions intended to put prior policy decisions into practice ([Van Meter and Van Horn, 1975](#), pp. 447–448). For public-service policies, this may include efforts to secure resources, organize delivery, and maintain services. Visible implementation refers to politicians’ publicly observable actions within this process.

to provide or sustain public goods, learn whether those efforts succeed, and attribute at least some responsibility for the outcome to the incumbent. We study KALAHÍ–CIDSS, a community-driven development program in the Philippines, through which villages in participating municipalities could apply for government funding to provide local public goods.² Elected village mayors, known as barangay captains, were required to convene the first public assembly, often helped recruit residents, and sometimes served on the teams presenting funding proposals. Residents selected the projects, while volunteer teams prepared the proposals. The funding was substantial: the annual grant budget per village equaled about 44 percent of median annual village revenue from central-government transfers and local sources (Beatty et al., 2015). Yet completing the application process did not guarantee funding. At municipal forums, projects were ranked and funded until the available budget was exhausted. Funding decisions were publicly announced before the October 2013 village election. The findings therefore speak most directly to participatory project-funding programs.

Our empirical design exploits randomized municipal access to KALAHÍ and funding cutoffs among ranked proposals. In 2011, 198 municipalities were grouped into 99 matched pairs, and public lotteries assigned one municipality in each pair access to the program. We use the lottery to estimate the average electoral effect of program access and to instrument recorded implementation participation. Under the IV assumptions discussed below, this identifies the complier average treatment effect (CATE) of participation, including any subsequent funding and delivery, among villages induced to participate by assignment. We then use a regression discontinuity design to estimate the local effect of funding by comparing proposals that barely received funding with those that narrowly missed it. For each comparison, we examine incumbent reelection, candidacy, and vote share among incumbents who run.

Three findings emerge. Assignment to KALAHÍ reduces incumbent reelection by 2.5 percentage points, approximately 5 percent of the control-group reelection rate. Candidacy falls by 2.1 percentage points, while the estimated effect on vote share among incumbents who run is close to

²Similar policies include application-based social funds, matching grants requiring local contributions, and competitive intergovernmental transfers. These arrangements condition resources for local public goods on proposals, contributions, or approvals (Barron et al., 2024; World Bank, 2002; OECD, 2022). Similarly, EU grants also require substantial effort from local politicians to apply to apply for them (Veiga et al., 2026).

zero. For villages induced by program assignment to complete recorded implementation participation, the estimated effect is a 3.4-percentage-point reduction in reelection and a 2.9-point reduction in candidacy, again with little change in conditional vote share. Among ranked proposals near the funding boundary, barely receiving funding raises incumbent vote share among those who run by approximately 5 percentage points relative to narrowly missing funding. The corresponding funding estimates for candidacy and reelection are positive but imprecise.

Together, these findings show that positive local electoral returns to funding can coexist with negative average electoral effects within the same policy. This pattern is consistent with our argument that visible efforts to secure public goods can expose incumbents to losses when those efforts fail, even when success brings electoral rewards. The candidacy results also point to incumbents' decisions before the election as an important part of the policy's electoral consequences.

Our preferred explanation is that some village mayors anticipated an unfavorable voter response when their efforts did not secure a project. Consistent with this explanation, the estimated effect on candidacy is more negative among incumbents who previously won narrowly. We complement this analysis with interviews we conducted and survey and qualitative evidence from earlier KALAHİ evaluations. These sources include accounts of residents blaming village leadership for unsuccessful proposals and expressing frustration after sustained participation without funding. We also consider whether implementation burdens made another term less attractive independently of voters' responses. Earlier surveys document favorable assessments of KALAHİ among many local officials, despite reported burdens. Together, the quantitative and qualitative evidence provides suggestive support for an accountability interpretation.

This article contributes to research on electoral accountability in two ways. First, the argument connects models of electoral accountability to the policy-feedback literature. Models of accountability explain how citizens use observed government performance to evaluate incumbents ([Ferejohn, 1986](#); [Fearon, 1999](#)), while policy-feedback research shows that policies themselves can reshape citizens' information, expectations, and experiences ([Schattschneider, 1935](#); [Pierson, 1993](#)). Previous work has begun to connect these traditions by showing that citizens' experiences during policy rollout can shape electoral support, pointing to perceived corruption and shortcomings in service quality as possible explanations ([De Kadt and Lieberman, 2017](#); [Habyarimana et al.,](#)

2021). Whereas this work emphasizes what implementation reveals about a policy, we focus on how politicians' visible actions during implementation connect them to whether the intended benefit is secured. This argument also builds on research showing that voters respond more strongly to government performance when they can identify who is responsible (Powell Jr and Whitten, 1993; Tavits, 2007; Harding, 2015; Martin and Raffler, 2021). We connect these literatures by treating visible implementation as one process through which responsibility becomes clear. When politicians visibly work to secure a benefit, citizens can evaluate them not only on what is delivered but also on whether their efforts succeed. Public-service policies can therefore expose incumbents to electoral losses as well as gains, depending on success or failure, helping explain why policies that deliver valued goods can produce null or negative average electoral effects.

Second, we contribute to empirical research on the electoral returns to public goods and service delivery in low- and middle-income democracies (Harding, 2015; De Kadt and Lieberman, 2017; Habyarimana et al., 2021; Croke, 2021; Goyal, 2024).³ By studying program access and funding within a common institutional setting, we distinguish a policy's average electoral consequences from the local electoral returns to securing a project.

Within this literature, our study is most closely related to Cruz and Schneider (2017), who examine an earlier iteration of KALAH-CIDSS. They show that uncertainty about project selection allows municipal mayors to claim electoral credit despite having little formal control over funding decisions. Our findings complement this result by examining village mayors, whom KALAH publicly associates with local mobilization and proposal preparation. Funding decisions are subsequently announced to participating communities before the election. Taken together, the two studies show how program design can shape which politicians become associated with public benefits and the electoral consequences of that association.

2 Visible Implementation and Electoral Accountability

In standard theories of electoral accountability, citizens observe government performance and use it to decide whether to retain the incumbent or select a challenger (Ferejohn, 1986; Fearon, 1999; Besley, 2006; Ashworth, 2012). Citizens in low- and middle-income countries frequently identify public goods and services among their most important concerns (Grossman and Slough,

³Appendix Table A1 reviews the existing empirical scholarship on the electoral effects of public-service policies and describes the policy treatment examined in each study.

2022). Consistently, studies of public goods have accordingly examined whether voters reward incumbents for delivering projects, transfers, and improvements in public services (Harding, 2015; Huet-Vaughn, 2019; Croke, 2021). Yet the electoral returns to public-service provision are not consistently positive. Some valuable and visible programs generate no incumbent reward (Goyal, 2024), while others reduce incumbent support despite expanding access or improving service quality on average (De Kadt and Lieberman, 2017; Sandholtz, 2025; Habyarimana et al., 2021). If voters value public services, why can policies that provide them leave incumbents no better off, or even worse off?

Our argument begins from a simple observation: public-service provision often depends on politicians' efforts to secure resources, organize delivery, or maintain services. In decentralized systems, for example, national governments often retain authority over policy design and overall funding while relying on local governments to carry out these tasks (Bardhan, 2002; Faguet, 2014).⁴ When local politicians must perform these tasks publicly, visible implementation can influence electoral outcomes. Citizens can observe not only whether a benefit arrives, but whether the incumbent's visible efforts to secure it succeed or fail. When citizens believe these efforts could affect the outcome, failure to secure the benefit may reduce their support for the incumbent. Public-service policies can consequently expose incumbents to electoral losses as well as gains, helping explain why their average electoral return need not be positive.

2.1 Implementation as Public Performance

The idea that public policy can shape subsequent politics has a long tradition. The policy-feedback literature treats policies not only as consequences of politics but also as potential causes of political change. Policies can alter the resources and incentives facing political actors, and they can provide information and interpretive cues through which citizens understand government (e.g. Schattschneider (1935); Pierson (1993)). We connect the policy-feedback literature to theories of electoral accountability by arguing that visible implementation can clarify an incumbent's responsibility for whether the intended benefit is secured and thereby affect electoral support.⁵

⁴Community-driven development is one setting in which external financing supports community-level planning and implementation. As of June 2023, the World Bank supported community and local development projects in 98 countries (Barron et al., 2024).

⁵Related work has examined citizens' electoral responses to policy implementation. De Kadt and Lieberman (2017) document a negative association between expanded service provision and governing-party support and provide suggestive evidence that delivery exposes citizens to perceived corruption and changes what they expect government to

We define policy implementation as actions intended to put prior policy decisions into practice (Van Meter and Van Horn, 1975, pp. 447–448). For public-service policies, we include local efforts to secure resources, organize delivery, and maintain services. These activities may begin before individual projects receive funding. Visible implementation refers to politicians’ publicly observable actions within this process. These actions may include applying for funding, coordinating administrative actors, mobilizing beneficiaries, satisfying program requirements, securing local resources, or supervising delivery. The particular activities may differ across policies. What unites them is that they publicly associate the politician with the implementation process, giving citizens a basis for connecting the eventual outcome to that politician. This argument builds on research showing that voters respond more strongly to performance when responsibility is clear (Powell Jr and Whitten, 1993; Tavits, 2007; Harding, 2015; Martin and Raffler, 2021).

The argument applies when citizens value the objective being pursued, observe politicians’ efforts and their outcomes before an election, and believe those efforts could influence the result. Citizens must also attribute at least some responsibility for that result to the incumbent.⁶ These conditions can hold both when politicians seek resources and when they organize delivery or maintain services after funding has been secured.

While we remain agnostic about the precise mechanism through which visible implementation affects electoral behavior, we consider three possibilities. First, implementation may change beliefs about incumbent competence. In models of electoral selection, citizens use past performance to form beliefs about traits that predict future performance, such as effort, competence, integrity, or organizational capacity (Fearon, 1999; Ashworth, 2012). If more capable incumbents are more likely to convert implementation effort into success, the outcome provides information about incumbent quality. Because outcomes also depend on scarce budgets, bureaucratic decisions, and luck, this signal is imperfect. It can nevertheless inform voters whenever incumbent competence affects the likelihood of success. Voters need not value the past outcome for its own sake. They may support the incumbent because it changes their expectations of future performance. Second,

provide. Habyarimana et al. (2021) develop a broader account of temporally dynamic policy feedback in which electoral responses can change as implementation reveals information about benefits, costs, service quality, state capacity, and politicians’ effort, or changes expectations and political participation.

⁶We focus on how citizens respond to implementation efforts and their outcomes, abstracting from politicians’ decisions about whether to participate. We return to this limitation in the discussion below.

implementation can change the standards citizens use to evaluate the incumbent. Voters cannot sanction poor performance without some benchmark for what government could reasonably have achieved. If resources appear unavailable or the incumbent appears uninvolved, non-delivery provides little reason for sanction. Visible implementation can therefore raise the standard against which voters judge performance and make poor outcomes more electorally consequential, even without changing beliefs about incumbent quality (Gottlieb, 2016). Consistent with this possibility, De Kadt and Lieberman (2017) provide suggestive evidence that service expansion can raise citizens' expectations and alter their subsequent electoral responses. Third, citizens may respond directly to costs incurred during implementation. This possibility fits most closely with the reward-and-punishment account of retrospective voting. Rather than treating the implementation outcome only as information about the incumbent's type, citizens may condition their support on policy consequences they attribute to the incumbent (Healy and Malhotra, 2013). Participatory programs can require meetings, paperwork, labor, and contributions of time or money, while delaying access to the intended benefit (Habyarimana et al., 2021). When implementation does not secure that benefit, these costs are not offset by the anticipated return. Citizens may therefore withdraw support without changing their beliefs about incumbent competence or the standard by which they evaluate performance. This response still depends on attribution because citizens must connect the burdens to the incumbent before holding that incumbent responsible.

The three pathways are not mutually exclusive. Under the first two, visible political involvement changes what citizens infer about the incumbent or the standard by which they judge the outcome. Under the third, citizens respond directly to costs incurred during implementation. Visibility may still matter by helping citizens assign responsibility for those costs, but the response need not involve learning or raised expectations.

2.2 Observable Implications

The argument yields three observable implications. First, the average electoral effect of a public-service policy need not be positive. If voters respond only to delivered goods, receiving a valued public good may benefit the incumbent, while receiving no good should leave support unchanged. The average effect should therefore be positive or null. If voters also respond to implementation, however, unsuccessful efforts can reduce support. These losses may offset the rewards associated

with successful projects, producing a null or negative average effect.

Second, if implementation contributes to a negative average effect, the losses should be concentrated among communities that enter the implementation process. Where no visible implementation occurs, citizens observe neither a delivered good nor an implementation outcome associated with the incumbent. Once implementation begins, non-delivery can change assessments of the incumbent, fall short of the standard citizens apply, or leave participation costs without the anticipated benefit. Negative effects should therefore be greater among communities that undergo implementation.

Third, among otherwise comparable communities that enter implementation, success should produce more favorable electoral outcomes than failure. All three mechanisms predict this ordering, but so does an explanation based only on the value of receiving a public good. This comparison can therefore show whether the outcome of implementation matters, but it cannot establish whether success produces a reward, failure produces a penalty, or both. Nor can it distinguish among the mechanisms.

Electoral responses may appear before as well as at the election. Incumbents who anticipate a less favorable voter response may decide not to seek another term, while voters may change their support for incumbents who remain on the ballot ([Grossman et al., 2023](#); [Szakonyi, 2022](#)). Because candidacy precedes vote choice, these two margins need not move together. Conditional vote-share estimates describe only the selected group of incumbents who decide to run and therefore do not recover the electoral response that exiting incumbents would have faced.

3 Institutional Setting

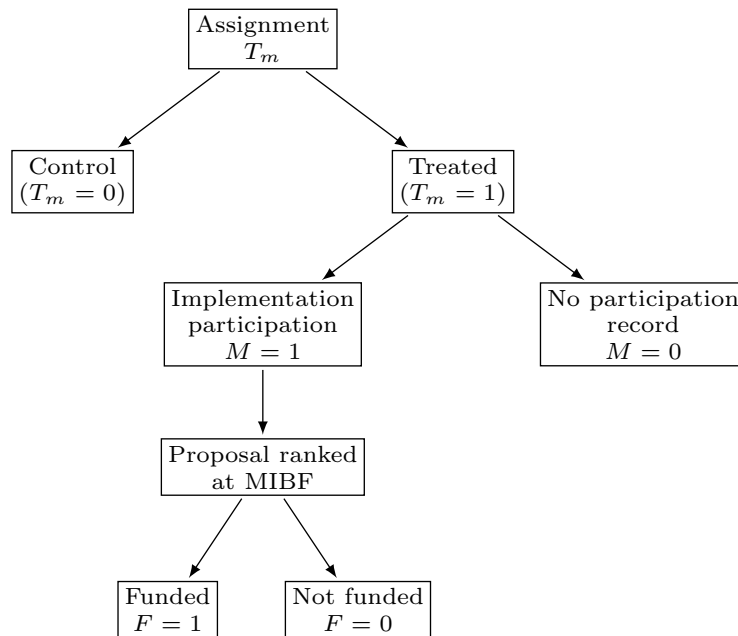
Public-service responsibilities in the Philippines are divided across tiers of local government. Provinces provide hospitals and infrastructure serving multiple municipalities. Municipalities administer primary health care, social-welfare programs, environmental services, and municipality-wide infrastructure. Barangays maintain village-level facilities and services, including barangay health and day-care centers, sanitation and waste collection, local roads and bridges, and water-supply systems ([Atienza, 2004](#); [Llanto et al., 2012](#); [Sidel, 1999](#)). Barangays, the lowest tier of government, are headed by *Punong Barangay* (village mayors), who are elected every three years and limited to three consecutive terms. As the barangay's chief executive, the village mayor convenes and presides over the barangay assembly and is responsible for ensuring the delivery of

services assigned to the barangay.⁷

3.1 KALAHI-CIDSS

KALAHI-CIDSS (KC)⁸ is a community-driven development program that finances village-selected public goods through a multistage process of community mobilization, proposal preparation, and competitive funding. It was initiated by the president, co-funded by the central government and international donors, and implemented by government ministries. The first phase (KC1) took place from 2003-2009. KC2 (2011–2014) was financed by a US\$120m MCC grant and a US\$59m World Bank loan with national counterpart funds. Within targeted provinces, municipalities with poverty $\geq 70\%$ automatically received KC, those with $\leq 33\%$ were excluded. Among municipalities with 33–69% poverty, 198 were randomly assigned in public lotteries (May 23–June 30, 2011) to 99 treatment and 99 control municipalities (Beatty et al., 2018). These 198 municipalities constitute the population of this study,

Figure 1: KALAHI Implementation



The implementation followed six steps. (I) Municipal mayors applied to participate. (II) Public lotteries (May 23-June 30, 2011) randomly assigned treatment and control municipalities (Beatty et al., 2018).⁹ (III) In treated municipalities, DSWD community facilitators led social

⁷Republic Act No. 7160, secs. 17 and 389.

⁸Kapit Bisig Laban sa Kahirapan ("Linking Arms Against Poverty") – Comprehensive and Integrated Delivery of Social Services.

⁹Funding flowed only in provinces where "guaranteed" municipalities had not exhausted KC funds and if a municipal

preparation and explained the program at village meetings. The village mayor formally convened the first assembly, while village mayors and other village officials often helped recruit residents to attend. Interviews describe some village mayors and councilors going house-to-house. This local recruitment supplemented the formal mobilization and facilitation role of DSWD staff.¹⁰ At the first assembly, residents elected volunteers to conduct a participatory assessment of community needs. A second assembly validated that assessment and elected two three-member volunteer teams: the Project Preparation Team (PPT) and the Barangay Representation Team (BRT). (IV) With technical assistance, the PPT developed the subproject concept, which a subsequent assembly reviewed and validated. The BRT served as the village's official delegation to the municipal inter-barangay forum (MIBF) (Beatty et al., 2018). Village mayors were not automatically members of either team. The volunteer records show, however, that they were frequently elected to the BRT and, when selected, took a direct role in presenting and defending the village proposal (Appendix D.3). Assemblies sought broad household participation, and interview respondents described an attendance target of around 80%. (V) Before the municipal forum, the elected BRT members set the ranking criteria. At the forum, each BRT presented its village's proposal, representatives from the other villages could ask questions, and the voting representatives scored proposals against the agreed criteria. The consolidated scores produced a ranked list. Some barangay representatives actively sought to persuade other representatives—as one village official stated, *“It is up to us to make others understand and convince them to vote for us.”*¹¹ (VI) Once a subproject was prioritized, funds were transferred directly by DSWD to a community account to limit political capture. Municipalities and barangays provided a Local Counterpart Contribution, often from the Local Development Fund and in cash or in-kind (DSWD, 2015, pp. 33–34).

Up until the village election in 2013, KALAHI financed local public goods investments, stretching from social services (i.e., health, education, water), infrastructure (roads, bridges), facilities (community production, economic support, and common service facilities), to environmental protection and conservation. Each participating municipality received an annual funding envelope equal to PHP 450,000 (approximately US\$11,250) times its number of barangays. This amount

mayor or representative attended the lottery.

¹⁰See Appendix C.3.

¹¹See more details on interviews in Appendix K.2.2.

was large relative to barangay finances. The per-barangay component used to calculate the envelope equaled about 44 percent of median annual revenue from the IRA and locally generated funds.¹² In an earlier evaluation of the economic impacts of KALAHİ, Beatty et al. (2018) found sizable welfare gains, but those improvements did not translate into broad household-level growth.¹³

Mobilization and funding were highly visible to voters. First, barangay assemblies sought broad participation, and village mayors and other village officials often helped recruit residents. According to survey evidence, roughly 80% of respondents named the village mayor as the individual with the most influence on subproject selection (ADB, 2012). Second, MIBF results were publicly announced by LGUs/DSWD through press releases, Facebook posts, or local news, making funding outcomes salient.¹⁴

These features map KALAHİ to the scope conditions of the framework. Village mayors were institutionally associated with mobilization because they convened barangay assemblies, and the interviews describe many of them as actively recruiting participants. DSWD facilitators and elected volunteers nevertheless performed much of the facilitation, proposal preparation, and day-to-day implementation. Village mayors' involvement became more direct when residents elected them to the BRT. Citizens learned the funding outcome before the election. Residents also identified local priorities and validated the resulting proposal in public assemblies. This participatory process makes it more likely that the proposed project reflected a locally valued objective, although it does not imply unanimous support. Finally, the regularly scheduled 2013 village election gave citizens an opportunity to respond to the implementation process and its outcome. KALAHİ therefore provides a useful setting in which local political involvement, a community-selected objective, and

¹²Our calculations from the interim barangay survey give median combined IRA and internally generated revenue of PHP 1,033,000 in 2013 among 73 barangays with valid information on both sources (Beatty et al., 2015). The envelope was pooled at the municipal level rather than transferred equally to every barangay.

¹³In particular, barangays that received KC funding experienced (i) shorter and cheaper trips to services and markets and lower crop-haulage costs after road investments, (ii) a rise in primary and secondary enrollment where school buildings were financed, and (iii) large reductions in both the time and money households spent obtaining water following water-system projects. Yet household consumption, assets and labor earnings remained flat, and rice yields fell in road-project barangays.

¹⁴For example, the local news site Boracay Island News reported six winning proposals in New Washington, Aklan, noting that they were selected through Municipal Development Council–Participatory Resource Allocation scoring, with barangay and LGU cash counterparts already pledged: <https://www.boracayislandnews.com/kalahi-cidss-to-fund-six-community-projects-in-new-washington-aklan/>. The detailed ranking sheet is usually kept at the municipality where the MIBF was held and is accessible only to officials and community members who took part in the forum.

a publicly observed result occur before an election.

4 Data

We combine village-level electoral returns with administrative records from the KALAH-CIDSS (KC2) program to estimate how program access, recorded implementation participation, and project funding affect incumbent reelection, candidacy, conditional vote share, and the number of candidates. The empirical section brings together several data sets: the random assignment status of municipalities obtained from the public registry, project-cycle monitoring files, village-level electoral returns, and the 2010 population census.

Random Assignment and Treatment Status

The starting point is the public registry compiled by [Beatty et al. \(2018\)](#), who were also responsible for the randomization. It lists all 198 municipalities that cleared the national poverty screen (poverty incidence between 33% and 69%) and indicates which member of each matched pair won the lottery ($T_m = 1$) and which one served as control ($T_m = 0$). The file also reproduces the four variables—population, number of barangays, land area, and poverty incidence—that were used to form pairs. These covariates are included in the main specification to increase precision.

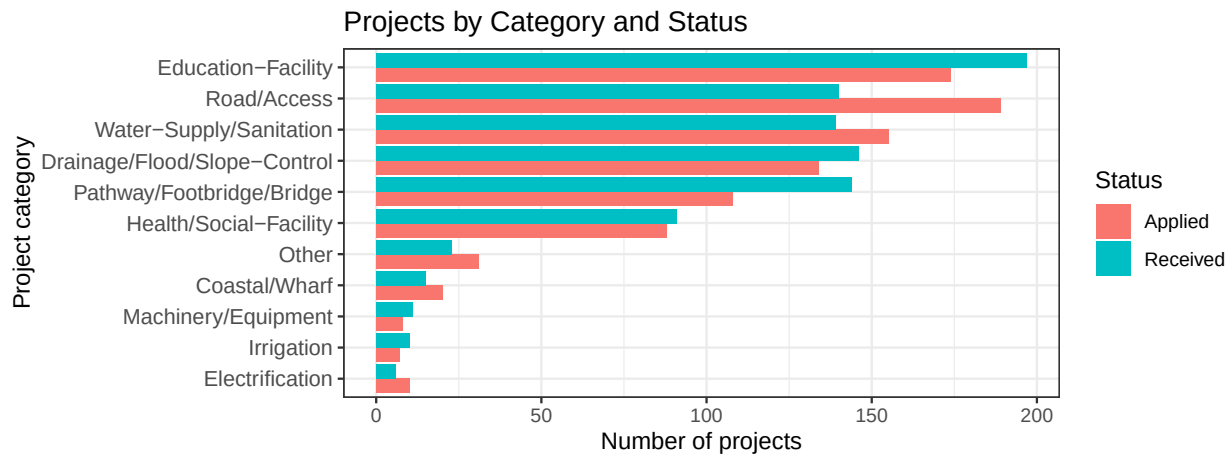
Recorded Implementation Participation

To measure implementation participation, we combine two administrative sources from the KALAH-CIDSS Monitoring and Evaluation Unit. First, the proposal-ranking data identify villages whose proposals reached a municipal forum and received ranking. Second, the volunteer-team data identify villages with a recorded project team even when no corresponding ranking record is available. We code a barangay as participating ($M_{im} = 1$) when at least one project appears in either source before the election and as having no recorded participation ($M_{im} = 0$) when it appears in neither.¹⁵ Overall, 2,170 of 4,819 villages are coded as participants. Of these, 1,543 appear in both sources, 366 only in the volunteer records, and 261 only in the ranking records. Using the union avoids classifying a village as a nonparticipant solely because one administrative source is incomplete. The indicator should therefore be interpreted as *recorded implementation participation*. It does not capture every early contact with KALAH. In particular, a village appearing in neither source may still have experienced initial information, social preparation, or assembly activity. Thus, $M_{im} = 0$ means that we observe neither a volunteer-team nor a ranking record, not that the village necessarily

¹⁵Appendix Table A16 shows that the results are similar when participation is defined using either source separately.

had no program exposure whatsoever.

Figure 2: Subprojects started in the 99 treatment municipalities between May 2011 and November 2013



Notes: Projects financed by Millennium Challenge Corporation (MCC) and Additional Financing (AF) from the World Bank in the 99 treatment municipalities until November 2013. Source: Department of Social Welfare and Development, Republic of the Philippines.

Subproject Ranking and Funding

The KALAH-CIDSS Monitoring and Evaluation Unit records each proposal's rank, funding status, KC grant amount, local counterpart contribution, total project cost, funding source, and funding-decision date.¹⁶ Lower numerical ranks indicate higher priority within a municipality and funding cycle. Let r_{imc} denote the raw rank of proposal i in municipality m and cycle c , and let F_{imc} equal one when the proposal received funding. For each clean municipality-cycle sequence, let r_{mc}^F be the last funded rank and r_{mc}^U the first unfunded rank. We locate the funding boundary at their midpoint, $c_{mc} = (r_{mc}^F + r_{mc}^U)/2$, and define the centered running variable as $X_{imc} = c_{mc} - r_{imc}$. Consequently, $F_{imc} = 1$ if and only if $X_{imc} > 0$. The last funded proposal lies at $X_{imc} = 0.5$, the first unfunded proposal lies at $X_{imc} = -0.5$, and no proposal lies at zero. The clean-sequence rules and resulting RD sample are described in Section 5.3. The dataset also includes the short titles of all ranked projects. Using natural-language classification, we assign approximately 1,700 project names to predefined infrastructure categories, such as water and sanitation, irrigation, and roads. Figure 2 presents the resulting distribution. Roughly half of the applications received funding.

¹⁶If two or more villages applied jointly for a project, they received the same rank.

Electoral Outcomes

The lottery registry covers 4,886 villages in 198 municipalities. We match village-level returns from the 2010 and 2013 Commission on Elections (COMELEC) files for 4,839 villages in 197 municipalities.¹⁷ Because assignment occurred within matched pairs, we exclude Mauban and its treated partner, San Antonio. The experimental analysis therefore contains 4,819 unique villages in 196 municipalities and 98 matched pairs.¹⁸ The ranking data can contain more than one proposal for a village. The ITT and implementation-participation analyses retain one observation per village, while the funding RD remains at the proposal level.

Village mayors are elected in first-past-the-post elections every three years. The October 2013 election was the first contest after KC2 implementation began.¹⁹ The returns report all candidates and their votes. We construct four outcomes. Reelection equals one when the middle and last names of the 2013 winner match those of the 2010 winner. Candidacy equals one when the incumbent's middle and last names match any candidate in 2013. Conditional vote share is the incumbent's share of valid votes among incumbents who run. The final outcome is the total number of candidates running for village mayor in 2013.²⁰ Because our election records begin in 2010, we cannot identify incumbents barred by the three-term limit. Term-limit status was fixed before random assignment and is therefore balanced across assignment arms in expectation. The treatment-control difference in candidacy consequently captures changes in reelection-seeking among incumbents who were eligible to run.

Covariates: Village Characteristics

From the 2010 Census, we construct pre-treatment measures of population, education, age, urbanization, Catholic and married shares, Philippine citizenship, and ethnic concentration. The 2010 election returns provide the prior incumbent vote share, winning margin, and number of candidates. Proposal records report KC grant amounts, local counterpart contributions, and total project costs, which we use only for balance checks at the funding boundary.

¹⁷Municipality Mauban is absent from the 2013 COMELEC file.

¹⁸Please refer to Appendix G for details.

¹⁹The Philippines was severely affected by Typhoon Yolanda in November 2013, after the village elections.

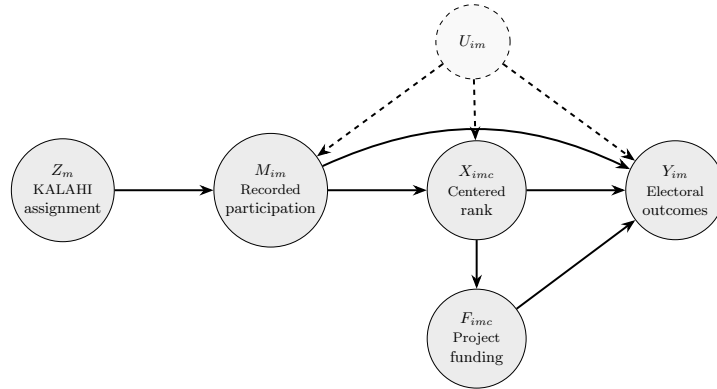
²⁰Reelection, candidacy, and candidate count are observed for all 4,819 villages. Conditional vote share is observed for 3,197 villages. Of the remaining villages, 1,613 incumbents did not run and nine runners have missing vote totals.

5 Empirical Strategy

Our framework predicts that electoral responses depend on whether incumbents' visible efforts to secure a public good succeed. Under a benchmark in which the policy affects incumbent support only through the valued goods it delivers, average electoral effects should be nonnegative. Our argument allows visible but unsuccessful implementation to generate electoral losses even where no good is delivered. If implementation accounts for these losses, negative effects should also appear among communities that go through the process. Among participating communities, securing funding should improve the incumbent's electoral prospects relative to failing to secure it.

KALAHİ supplies only part of this variation. Municipal access was randomly assigned, participation was voluntary, and project funding was determined by rank and the available municipal budget. We use the lottery to estimate the effect of program access and as an instrument for recorded implementation participation. We then use the funding cutoff to estimate the effect of receiving funding among ranked proposals. Figure 3 summarises the assignment process and the sources of selection that motivate these designs.

Figure 3: Schematic Causal Structure of Program Access, Implementation Participation, and Funding



Notes: Z_m denotes random assignment of municipality m to KALAHİ. M_{im} indicates recorded implementation participation by village i , X_{imc} is proposal rank centered at the municipality–cycle funding boundary, F_{imc} indicates project funding, and Y_{im} denotes an electoral outcome. Solid arrows show the program sequence and possible causal paths. Dashed arrows show selection arising from latent village and incumbent characteristics, U_{im} . The arrow from X_{imc} to Y_{im} allows electoral outcomes to vary smoothly with rank independently of funding. In the clean sharp-RD sample, funding is determined by which side of $X_{imc} = 0$ a proposal falls on. The absence of a direct arrow from Z_m to Y_{im} represents the exclusion restriction required for the IV.

5.1 Effect of Program Access

The first analysis asks whether access to a policy that finances valued public goods can nevertheless reduce incumbents' chances of reelection on average. The original design paired 198 municipalities using poverty incidence, population, land area, and number of villages, then randomly assigned one municipality in each pair to KALAH. After excluding the complete pair containing Mauban because its 2013 electoral returns are unavailable, the analysis contains 4,819 villages in 196 municipalities and 98 matched pairs. Let Z_m indicate assignment to KALAH, and let $Y_{im}^Z(z)$ denote an electoral outcome for village i in municipality m under assignment z . The intention-to-treat effect is

$$\text{ITT} = \mathbb{E}[Y_{im}^Z(1) - Y_{im}^Z(0)] . \quad (5.1)$$

Random assignment within matched pairs identifies this effect in the experimental population. The ITT includes every pathway through which program access may affect electoral outcomes and does not require an exclusion restriction between assignment and the outcome. A negative ITT for reelection would establish an average electoral loss from program access, but would not identify which implementation experiences account for it.

5.2 Effect of Implementation Participation

The second analysis estimates the effect of recorded implementation participation among villages induced to participate by municipal assignment. Participation was not randomly assigned, so a direct comparison of participating and nonparticipating villages could also reflect differences in their residents and incumbents. Let $M_{im} = 1$ when village i appears in either the volunteer-team records or the proposal-ranking records, and let $M_{im}(z)$ denote participation under assignment z . With $Y_{im}^M(d)$ denoting the electoral outcome under participation status d , including its downstream consequences, the Complier Average Treatment Effect is

$$\text{CATE}_M = \mathbb{E}[Y_{im}^M(1) - Y_{im}^M(0) \mid M_{im}(1) > M_{im}(0)] . \quad (5.2)$$

This quantity describes the effect of recorded team formation or proposal participation, including subsequent funding and delivery, among villages whose recorded participation was changed by assignment.

5.3 Effect of Project Funding

The third analysis asks whether securing funding improves incumbent electoral outcomes relative to failing to secure it among villages that entered implementation. Simply comparing funded and unfunded proposals would be misleading. Proposals may rank higher because their teams are stronger, their representatives perform better at the municipal forum, or their village mayors are better connected. These characteristics may also predict subsequent electoral outcomes. We address this selection problem using the municipal funding boundary as a regression-discontinuity design.

At each municipal inter-barangay forum, proposals were ranked and funded in order until the available budget was exhausted. The main sample contains 53 municipality–cycle sequences that begin with a funded proposal, have contiguous rank support, and switch only once from funded to unfunded. Together, these sequences contain 1,162 ranked proposals. Appendix D.1 displays the sequences and sample restrictions.

Let r_{mc}^F denote the rank of the last funded proposal and r_{mc}^U the rank of the first unfunded proposal in municipality m and cycle c . The cutoff is the midpoint $\bar{r}_{mc} = (r_{mc}^F + r_{mc}^U)/2$. We define the centered score as $X_{imc} = \bar{r}_{mc} - r_{imc}$. Funded proposals have $X_{imc} > 0$, unfunded proposals have $X_{imc} < 0$, and the closest support points are 0.5 and -0.5 . No proposal lies at zero. In the clean sharp-RD sample, $F_{imc} = \mathbf{1}\{X_{imc} > 0\}$. Our third causal quantity is the funding effect at this boundary:

$$\tau_F = \mu_1(0) - \mu_0(0), \quad (5.3)$$

where $\mu_f(x)$ denotes the smooth continuation to x of the conditional mean potential outcome $\mathbb{E}[Y_{imc}^F(f) \mid X_{imc} = x]$ under funding status f . The quantity τ_F is the effect of barely receiving funding rather than narrowly missing it among ranked proposals at the normalized funding boundary.

5.4 Relating the Estimands to the Theory

The theories distinctive prediction is that unsuccessful implementation can produce worse electoral outcomes than no implementation. We assess whether the estimates follow the pattern implied by the theory. An ideal design would randomly assign communities to no implementation, implementation without funding, and implementation followed by funding. It would thereby identify the effect of unsuccessful implementation relative to no implementation and the effect of funding among

otherwise comparable participants.

KALAHl provides two sources of causal variation. The municipal lottery identifies the ITT and, under the IV assumptions, the CATE of recorded implementation participation and its downstream consequences. The funding cutoff identifies the local effect of barely receiving funding rather than narrowly missing. Together, they allow us to assess whether a negative average program effect, interpreted under the IV assumptions as a negative participation effect among compliers, coexists with a positive local effect of funding. This is the pattern implied by the framework, although the funding comparison alone is also consistent with the material value of funded projects.

The strategy does not identify unsuccessful implementation relative to no implementation. The ITT averages across different participation and funding experiences, the CATE includes the downstream consequences of funding, and the RD has no nonparticipation benchmark. Moreover, because the CATE rescales the same lottery-based reduced form by the first stage, its greater magnitude than the ITT is not independent evidence that losses are concentrated among participants. We therefore treat the estimates as complementary evidence about the predicted pattern rather than as a causal decomposition of success and failure.

6 Results

6.1 Effect of Program Access

We estimate the effect of program access by regressing each electoral outcome on municipal assignment, matched-pair fixed effects, and predetermined covariates:

$$Y_{im} = \alpha + \tau Z_m + \mathbf{X}'_{im} \boldsymbol{\gamma} + \mu_p + \varepsilon_{im}, \quad (6.1)$$

where τ is the ITT effect, $\mathbf{X}_{im}$ includes municipal matching covariates and census and political characteristics, and μ_p denotes matched-pair fixed effects. Standard errors are clustered by municipality, the unit of random assignment.²¹ We examine reelection, whether the incumbent runs, and the number of candidates, and separately report incumbent vote share conditional on running.

²¹ Appendix H.1 reports the complete specification sequence, including estimates without the full covariate set and inference clustered by matched pair.

Table 1: Electoral Effects of Program Access

	Program access			
	Pr(reelection)	Pr(run)	Vote Share	No. candidates
Control mean	0.490	0.677	0.590	2.234
ITT	-0.025** (0.010)	-0.021* (0.011)	0.004 (0.008)	-0.013 (0.021)
Matched-pair fixed effects	✓	✓	✓	✓
Covariates	✓	✓	✓	✓
Municipality-clustered SE	✓	✓	✓	✓

^a Notes. This table reports the raw control-group mean and the ITT effect of KALAHİ assignment. Term-limit eligibility predates assignment, so legally required exits are balanced across assignment arms in expectation. The ITT estimates include matched-pair fixed effects, municipal matching covariates, and census and political covariates, with standard errors clustered by municipality. The binary and candidate-count models contain 4,781 observations; vote-share models contain 3,170. Robust standard errors are in parentheses. Significance: *** 1 percent, ** 5 percent, * 10 percent.

Table 1 reports the control-group mean and ITT estimate in separate rows for each outcome. Assignment to KALAHİ reduces incumbent reelection by 2.5 percentage points (s.e. 1.0 percentage point), approximately 5 percent of the control-group reelection rate of 49 percent. This corresponds to about 25 fewer incumbents reelected per 1,000 villages offered program access. Assignment also reduces the probability that the incumbent runs by 2.1 percentage points (s.e. 1.1 percentage points). By contrast, the estimated effects on vote share conditional on running and the number of candidates are close to zero.

We return in Section 6.4 to whether the reduction in running is larger among incumbents who previously won by narrower margins.

Identification Random assignment balances observed and unobserved characteristics across treatment and control municipalities in expectation. Appendix Table A9 examines pre-treatment balance. None of the four municipal matching variables differs significantly between treatment and control municipalities. At the village level, the married share is the only one of eleven characteristics that differs at the 5 percent level.²²

²²It is 0.525 percentage points higher in treated municipalities after pair adjustment ($p = 0.030$). All other village-level p -values exceed 0.30.

6.2 Effect of Implementation Participation

We estimate the effect of recorded implementation participation by two-stage least squares, instrumenting participation M_{im} with municipal assignment Z_m and retaining the ITT specification's controls, fixed effects, and clustering:

$$Y_{im} = \alpha_M + \theta_M M_{im} + \mathbf{X}'_{im} \gamma_M + \mu_p + \varepsilon_{im}, \quad (6.2)$$

$$M_{im} = \pi_0 + \pi_1 Z_m + \mathbf{X}'_{im} \eta_M + \mu_p + v_{im}, \quad (6.3)$$

where θ_M is the CATE. Participation is recorded when a village appears in either administrative source, although only the ranking records establish that a proposal reached the municipal forum. The coefficient π_1 measures the first-stage effect of assignment on recorded participation. To interpret the magnitude of the estimates, we also report the complier control mean: the estimated mean outcome that these villages would have experienced without recorded participation. Appendix I.1 explains how we estimate this mean and reports the complete IV specification sequence.

Table 2: Electoral Effects of Implementation Participation

	Implementation participation			
	Pr(reelection)	Pr(run)	Vote Share	No. candidates
Complier control mean	0.504	0.682	0.590	2.215
CATE	-0.034** (0.014)	-0.029* (0.015)	0.006 (0.011)	-0.017 (0.028)
Matched-pair fixed effects	✓	✓	✓	✓
Covariates	✓	✓	✓	✓
Municipality-clustered SE	✓	✓	✓	✓

^a Notes. This table reports the estimated mean under no recorded participation for compliers and the Complier Average Treatment Effect (CATE) of recorded implementation participation, using assignment as the instrument. Complier means follow [Abadie \(2003\)](#), with the implementation adapted from the public replication code for [Goodman et al. \(2020\)](#). Term-limit eligibility predates assignment, so legally required exits are balanced across assignment arms in expectation. The CATE estimates include matched-pair fixed effects, municipal matching covariates, and census and political covariates, with standard errors clustered by municipality. The binary and candidate-count models contain 4,781 observations; vote-share models contain 3,170. Robust standard errors are in parentheses. Significance: *** 1 percent, ** 5 percent, * 10 percent.

Table 2 reports the complier control mean and CATE estimate in separate rows for each outcome. Recorded implementation participation reduces incumbent reelection by 3.4 percentage points (s.e.

1.4 percentage points), approximately 7 percent of the estimated complier control mean of 50.4 percent. The probability of running falls by 2.9 percentage points (s.e. 1.5 percentage points), while the estimated effects on vote share conditional on running and the number of candidates are close to zero. As in the ITT analysis, the negative reelection effect is accompanied primarily by fewer incumbents seeking another term.

Identification Interpreting these estimates as the causal effect of recorded implementation participation requires random assignment, a nonzero first stage effect of the lottery on participation, monotonicity, and the exclusion restriction.

The first stage is strong: all effective F -statistics reported in Appendix I.3 exceed 250. Random assignment within matched pairs makes the instrument independent of potential outcomes. Monotonicity is plausible because assignment provides opportunities and support for village participation, but a positive first stage does not establish that assignment increased participation for every village. Monotonicity is plausible because assignment offered funding opportunities and facilitation that should encourage participation. We assume these opportunities did not deter any village that would otherwise have participated.

The more demanding question is whether assignment could affect electoral outcomes independently of recorded village participation. First, the lottery result itself might change citizens' expectations or their assessments of the village mayor. The 2012 baseline survey for the impact evaluation in the same experimental sample provides indirect evidence about this possibility. Roughly one year after assignment, 30% of households in sampled treatment barangays had never heard of KALAH! while another 38% knew of the program but reported no household participation (Beatty et al., 2013). This uneven awareness gives some reason to doubt a widespread electoral response to the lottery alone. The survey does not, however, measure awareness of the lottery specifically or rule out responses among those who knew the result (Appendix I.4.1).

Second, assignment may induce early village activities that our participation measure does not capture. At the baseline interviews, 54 of the 99 sampled treatment barangays had held an information assembly, but only one had progressed further (Beatty et al., 2013). Our measure includes villages appearing in either the volunteer records or the proposal rankings, but villages

absent from both may still have attended initial meetings.²³ If these activities affected electoral outcomes independently of recorded participation, exclusion would fail. Their likely magnitude nevertheless matters. Most attendees participated only once, and financial contributions were uncommon. Villages stopping at these stages may therefore have incurred fewer costs and developed less concrete expectations than those that prepared proposals but failed to secure funding. Early activity alone is consequently a less convincing explanation for the full negative estimate, although smaller effects could still weaken statistical significance. Appendices I.4.2 and I.4.5 discuss this evidence and assess sensitivity to such effects.

Third, assignment could affect village mayors through municipal politics or spending. Municipal mayors benefited electorally from an earlier phase of KALAH (Cruz and Schneider, 2017), and village mayors often serve as their local political brokers (Hicken et al., 2018; Cruz et al., 2017). Mayoral credit claiming or support for allied village mayors could therefore affect village elections independently of recorded participation. We examine whether the electoral effect of assignment varies with baseline support for the municipal mayor. The interaction is small and statistically indistinguishable from zero for incumbent vote share (Appendix Table A18). This provides no evidence of a stronger effect in mayoral strongholds, but does not rule out a municipal political effect shared across villages. Municipal spending was lower in several categories in 2011 and in health in 2012, with no statistically significant differences in 2013 (Appendix I.4.4). The 2011 figures cover the assignment year but precede implementation in 2012. They could reflect adjustments to anticipated funding rather than displacement by delivered projects, but the data do not establish whether spending changes affected village mayors' electoral outcomes.

We therefore assess sensitivity to an effect of assignment outside recorded participation, allowing for any of these pathways (Appendix I.4.5). An adverse effect of 0.50 percentage points on reelection would make the adjusted IV estimate's 95 percent confidence interval include zero, while approximately 0.82 percentage points would do so at the 90 percent level. Explaining the point estimate entirely would require an adverse effect of 2.52 percentage points. Thus, smaller violations could weaken statistical significance without accounting for the full estimate. The interpretation as

²³Estimates remain similar when either source is used separately (Appendix Table A16), but this does not address activities absent from both.

a participation effect among compliers remains conditional on exclusion.

Complier Characteristics To clarify which villages the CATE represents, we follow [Marbach and Hangartner \(2020\)](#) and compare the predetermined characteristics of three compliance groups. Compliers would have recorded implementation participation only when their municipality was assigned to KALAH. Always-takers would have recorded participation under either assignment, while never-takers would have it under neither. Random assignment and monotonicity allow us to estimate the shares and average characteristics of these groups even though a village’s compliance type is not directly observed. Our preferred estimates use the Abadie κ -weighting approach implemented by [Bowles and Larreguy \(2025\)](#). We compare them with the direct decomposition implemented in `ivdesc`. Appendix [I.5](#) describes both procedures.

Appendix Figure [A16](#) reports the estimated group shares and standardised means of eleven predetermined village and political characteristics.²⁴ The two procedures produce similar profiles. They estimate that 69–71% of villages are compliers, approximately 10% are always-takers, and 19–21% are never-takers. Complier villages are somewhat larger and more educated than never-taker villages, and they have a larger Catholic share than either noncomplier group. Prior incumbent vote share and winning margin, by contrast, are not statistically distinguishable across the three groups. These comparisons describe the villages represented by the CATE but do not establish that the effect generalises to noncompliers.

The ITT and CATE estimates include both funded and unfunded villages and therefore do not establish how receiving funding affects electoral outcomes. We next compare applicants whose proposals barely received funding with those whose proposals narrowly missed it.

6.3 Effect of Project Funding

Using the centered rank X_{imc} and clean-sequence sample defined in Section [5.3](#), the preferred specification fits local-linear models of the form

$$Y_{imc} = \alpha + \tau_F F_{imc} + \beta_- X_{imc} + \beta_+ F_{imc} X_{imc} + \mathbf{X}'_{imc} \gamma + \mu_m + \lambda_c + \epsilon_{imc},$$

²⁴Numerical estimates and bootstrap standard errors are reported in Appendix [I.5](#).

where τ_F is the funding discontinuity, $\mathbf{X}_{imc}$ contains predetermined census and political controls, and μ_m and λ_c denote municipality and funding-cycle fixed effects. The unit of observation is a ranked proposal.

Estimation with a Discrete Running Variable Proposal rank is ordinal and has a limited number of support points. Estimation must therefore extrapolate between the nearest ranks because no proposal lies at the normalized cutoff. Narrowing the estimation window cannot eliminate this problem because there are no proposals between adjacent ranks. We use two approaches to account for this.

First, we use local-linear estimates from RDHonest with a triangular kernel. The method limits how sharply the expected electoral outcome may bend on either side of the boundary. For a chosen curvature bound, denoted M , it calculates the largest bias that such bending could generate in the local-linear estimate and incorporates this worst-case bias into the confidence interval (Kolesár and Rothe, 2018). A larger value of M permits greater curvature and therefore produces a wider, more conservative interval. Following the calibration proposed by Imbens and Wager (2019) and its application in Goldsmith-Pinkham et al. (2023), we fit a quadratic among unfunded proposals within 25 ranks of the boundary and set M to four times the absolute implied second derivative.²⁵ We select the bandwidth to minimise the length of the fixed-length confidence interval (FLCI) and cluster standard errors by municipality–cycle in the preferred specification. The interval is honest in the sense that it has its stated coverage for any outcome function satisfying the maintained bound.²⁶

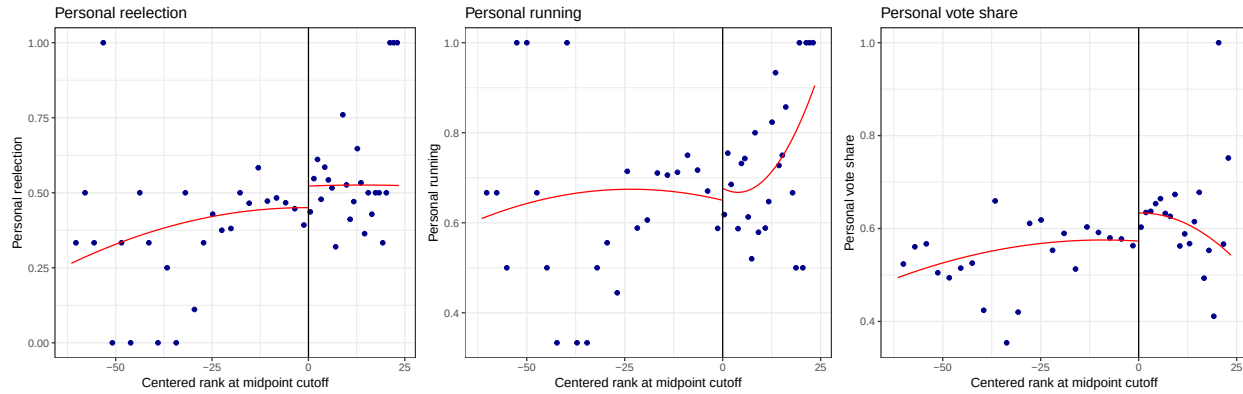
Second, we use the support-point approach recommended by Cattaneo et al. (2024). For adjusted specifications, we first account for the listed controls and fixed effects in the proposal-level data. We then collapse the outcome to one unweighted mean at each distinct score and estimate a local-linear discontinuity using these score-level means, with an MSE-optimal bandwidth and robust bias-corrected inference. This gives each observed rank equal weight rather than allowing ranks with more proposals to exert greater influence. The collapsed analysis does not use clustered standard errors and does not depend on the choice of M .

²⁵There are more proposals on the unfunded side. The robustness checks vary both the calibration range and the curvature bound.

²⁶Please refer to Appendix J.4 for sensitivity analysis to alternative bounds.

Visual Evidence Figure 4 provides a first look at the relationship between funding and electoral outcomes. It plots binned means for the complete sharp-RD sample and fits separate second-order polynomials on either side of the cutoff. Funded proposals appear to the right of zero and unfunded proposals to the left.

Figure 4: Project Funding and Electoral Outcomes at the Budget Cutoff



Notes: Points are binned means produced by `rdplot`. The panels report reelection, whether the incumbent runs, and incumbent vote share conditional on running. The score is centered at the midpoint between the last funded and first unfunded proposals. Funded proposals have positive values and unfunded proposals have negative values. The fitted curves use second-order polynomials on each side of the cutoff. The figure is descriptive. Formal estimates use the discrete-running-variable procedure reported in Table 3. The vertical line marks the funding threshold at $X = 0$.

The discontinuity is clearest for conditional vote share. The fitted curve rises at the boundary, which indicates greater support for incumbents whose proposals were just funded than for those whose proposals just missed funding. The reelection panel also suggests a positive shift, although it is less pronounced. The running panel shows no clear break at the cutoff. These broad polynomial fits are descriptive.

Main Results Table 3 reports the RDHonest estimates in Panel A and the collapsed support-point estimates in Panel B. In both panels, the columns move from an unadjusted model to the preferred specification with fixed effects and controls. Models (3)–(5) in Panel A cluster standard errors by municipality–cycle.

Table 3: RDD Estimates of KALAHI Funding on Electoral Outcomes

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: RD Honest</i>					
Pr(reelection)	0.086 (0.072)	0.075 (0.066)	0.075 (0.065)	0.084 (0.069)	0.084 (0.070)
Pr(run)	0.083 (0.079)	0.083 (0.069)	0.083 (0.076)	0.102 (0.079)	0.106 (0.080)
Vote Share	0.064*** (0.024)	0.059** (0.025)	0.059* (0.031)	0.060* (0.029)	0.049* (0.026)
No. candidates	-0.172 (0.100)	-0.145* (0.072)	-0.144 (0.086)	-0.125* (0.073)	-0.120 (0.088)
<i>Panel B: Collapsed support-point RD</i>					
Pr(reelection)	0.080 (0.110)	0.082 (0.106)	0.082 (0.106)	0.100 (0.104)	0.102 (0.101)
Pr(run)	0.119 (0.081)	0.115 (0.083)	0.115 (0.083)	0.130 (0.085)	0.128 (0.088)
Vote Share	0.046 (0.031)	0.053** (0.026)	0.053** (0.026)	0.054** (0.023)	0.050*** (0.018)
No. candidates	-0.121 (0.093)	-0.142* (0.083)	-0.142* (0.083)	-0.151* (0.078)	-0.166** (0.072)
Fixed Effects		✓	✓	✓	✓
Clustered SEs (Panel A)			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. Panel A reports RDHonest local-linear inference with a triangular kernel, Holder smoothness class, and FLCI-optimal bandwidth. The running variable is the MIBF rank centered at the midpoint between the last funded proposal and the first unfunded proposal, so positive values denote funded proposals. The number-of-candidates outcome counts all candidates for village mayor in 2013. The preferred/default smoothness bound follows the published left-side quadratic rule: fit a quadratic on the unfunded side within 25 ranks and multiply the implied absolute second derivative by four. Panel B follows the discrete-score guidance in Cattaneo, Idrobo, and Titiunik by collapsing the data to one unweighted mean outcome for each unique score value, then applying local-linear rdrobust inference to the collapsed support-point data. For covariate-adjusted Panel B columns, the outcome is first adjusted on the raw data using the listed predetermined controls and fixed effects, then collapsed to score-level means. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors in Panel A, Model (4) adds census controls, and Model (5) adds political controls. Panel B does not use clustered standard errors, so its Models (2) and (3) are identical. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Funding increases incumbent vote share conditional on running. In the preferred specification, Model (5), the estimated increase is 4.9 percentage points with RDHonest and 5.0 percentage points with the collapsed support-point approach. The RDHonest estimates are positive and statistically distinguishable from zero at the 10 percent level or better across all five specifications. The preferred collapsed estimate is more precise, with a standard error of 1.8 percentage points and a p -value of 0.005. All adjusted collapsed estimates are distinguishable from zero at the 5 percent level or better, while the unadjusted estimate includes zero.

The approximately 5-percentage-point gain in vote share is substantial relative to prior electoral margins. In the full experimental sample, 22.3 percent of contested 2010 elections were decided by 5 percentage points or less and 40.9 percent by 10 points or less. The corresponding shares are 23.6 and 42.9 percent within the preferred RDHonest window, and 23.1 and 42.1 percent within the collapsed-RD window.²⁷

The estimated effects on reelection and running are also positive, but their confidence intervals include zero in every specification. We therefore find no precise evidence that funding changes whether incumbents run or return to office. The candidate-count estimates are negative, but only one clustered RDHonest specification is distinguishable from zero at the 10 percent level.²⁸ We therefore view the evidence on candidate entry as weak and sensitive to the specification.

Identification For the RD estimates to have a causal interpretation, potential outcomes and predetermined characteristics must evolve smoothly through the funding boundary. The main concern is that barangay representation teams may have coordinated during the municipal forum to move allied proposals across the funding line. Precise manipulation would have been difficult because teams needed to anticipate both the final ranking and the point at which the municipal budget would be exhausted. It cannot be dismissed, however. In qualitative interviews, some officials described project selection as formal and rules-based, while others reported that barangays spoke before the vote, formed factions, or supported one another strategically. This evidence does not establish manipulation, but it makes inter-barangay coordination a plausible threat.²⁹ We assess

²⁷Please refer to Appendix Table A7 for more details.

²⁸The p -values in Models (3), (4), and (5) are 0.114, 0.095, and 0.198, respectively. The collapsed Model (5) estimate excludes zero, but that analysis does not provide clustered inference.

²⁹See Appendix K.2.2 for details.

three observable implications of this concern.

First, precise sorting could produce an unusual concentration of proposals at the funding boundary. Appendix [J.1.1](#) tests for a change in the distribution of the centered score and does not detect a discontinuity. We treat this result as descriptive because the running variable is an ordinal rank that is centered separately within each municipality–cycle. The absence of a density break therefore does not by itself rule out strategic coordination.

Second, sorting could place proposals from systematically different villages on opposite sides of the boundary. We test balance in Appendix [J.1.2](#) using common windows matched to the two preferred vote-share specifications. In the RDHonest window, the number of candidates differs at the 5 percent level, while Catholic share and prior incumbent vote share differ at the 10 percent level. None of these differences appears in the collapsed-RD window. The married share differs at the 5 percent level only in the collapsed-RD window and not in the RDHonest window. All four characteristics enter the covariate-adjusted specifications. Controlling for them leaves the estimated vote-share effect close to 5 percentage points under both approaches. Across 32 tests, the discontinuities do not recur across the two windows. This pattern provides no systematic evidence of sorting, although it does not imply exact balance.

Recorded project amounts follow a more systematic pattern in the collapsed-RD window. The KC grant amount and total project cost are lower on the funded side of the boundary. Both differences retain the same sign but are less precise in the RDHonest window. The source does not establish whether these amounts were fixed before funding, so these comparisons do not by themselves establish differences in pre-funding proposal costs.

Third, manipulation intended to secure funding should be most consequential for the proposals immediately bracketing the boundary. At these support points, the smallest change in rank can switch a proposal’s funding status. We therefore estimate a donut RD that drops the last funded and first unfunded mass points, $X = 0.5$ and $X = -0.5$ (Appendix Table [A31](#)). We hold the full-sample curvature bounds and collapsed-RD bandwidths fixed so that only the sample changes. The conditional vote-share estimate remains positive under both approaches. It is 5.6 percentage points under honest RD and 6.2 percentage points in the collapsed analysis, although both estimates are less precise than their full-sample counterparts. The Model (5) reelection and running estimates

rise to 22.8 and 23.2 percentage points under honest RD, and both exclude zero at the 5 percent level. The corresponding collapsed estimates rise to 32.4 and 23.1 percentage points and exclude zero at the 10 percent level. The candidate-count estimates remain negative at -0.114 and -0.200 , but both confidence intervals include zero. The donut exercise therefore shows that the positive vote-share result is not generated solely by the proposals closest to the cutoff. It also reveals that the binary-outcome magnitudes are sensitive to removing those support points. Because the donut design extrapolates across a wider gap and only addresses manipulation concentrated at the immediate boundary, we treat it as a diagnostic rather than as a preferred estimate.

Taken together, these checks do not establish that the ranking process was free from strategic coordination. They show that the vote-share discontinuity is not accompanied by a detectable density break, does not depend on covariate differences that recur across estimators, remains close to 5 percentage points after adjustment for the flagged characteristics, and remains positive when the two closest support points are removed.

Robustness We next examine whether the vote-share result depends on the clean-sequence restriction or the tuning choices used by the two estimators. First, we admit sequences with funding reversals and estimate a fuzzy RD, using the reconstructed funding boundary as an instrument for actual funding. The main fuzzy sample contains 64 municipality–cycle sequences, compared with 53 in the sharp design. Estimates with fixed effects and municipality–cycle clustered standard errors range from 4.3 to 5.4 percentage points. Two of these three estimates are distinguishable from zero at the 10 percent level, while the fully adjusted estimate remains positive but is less precise. Appendix J.9 also varies the treatment of rank gaps and the location of the reconstructed cutoff. Across these broader checks, the clustered estimates remain positive and range from 4.0 to 6.0 percentage points, although precision varies. The positive sign and approximate magnitude therefore do not depend on a single reconstruction of the funding boundary.

Second, we vary the curvature bound and the estimation window. We also recalibrate the bound over the full unfunded-side support and implement local-randomization inference (Appendix J.3 and Tables A37 and A38). The vote-share discontinuity remains positive across these exercises. We therefore interpret the two panels of Table 3 jointly. Honest RD shows that the vote-share increase

survives explicit allowance for curvature bias and municipality–cycle dependence. The collapsed analysis shows that the same approximately 5 percentage point discontinuity emerges when the data are analyzed at the score-support level, with greater precision in the adjusted specifications.

External Validity The regression discontinuity design estimates a LATE of project funding on incumbent vote share for proposals near funding cutoffs. The preferred RDHonest and collapsed support-point windows contain 94 percent and 72 percent of proposals in the clean sequences, respectively.³⁰ Appendix Table A39 describes how predetermined characteristics change as the experimental sample is restricted to ranked applicants, clean sequences, and the preferred RDHonest window, and reports an estimated covariate profile at the cutoff. We present two checks on whether the vote-share result extends to the broader clean-sequence sample. First, Appendix J.3.2 shows that the positive estimates are not confined to the preferred window when the respective RD window is widened or narrowed. Second, we compare predetermined characteristics inside and outside both the RDHonest and collapsed support-point windows (Appendix J.5). No difference is statistically significant at the 5 percent level in either comparison. Overall, we find limited empirical evidence that the positive vote-share effect is systematically smaller over the broader support available within the clean sequences. However, our findings remain representative only of proposals near clean funding cutoffs among incumbents who seek reelection. Any extrapolation to proposals outside the clean sequences, proposals farther from the cutoff, or incumbents who do not run should be interpreted cautiously.

6.4 Mechanisms

So far, the evidence shows that program access and recorded implementation participation reduce incumbent reelection primarily because fewer incumbents seek another term, while barely receiving funding increases vote share among incumbents who run. Section 2 discussed three reasons why implementation may affect electoral behavior. Outcomes may change voters' assessments of the incumbent, raise the standard by which they evaluate performance, or leave citizens bearing participation costs without the anticipated benefit. Incumbents may also anticipate these responses

³⁰The 53 clean sequences contain 1,162 ranked proposals. Applying the 24.42-rank FLCI-optimal bandwidth from Model (5) of the RDHonest vote-share analysis retains 1,095 proposals, or 94.2 percent of the clean-sequence sample. Incumbent vote share is observed for 780 proposals, of which 739 lie within this window. The 11.45-rank bandwidth from the collapsed support-point specification retains 833 clean-sequence proposals, including 550 with observed vote shares.

when deciding whether to run. An alternative explanation is that implementation makes holding office less attractive to village mayors independently of voters' reactions. We first assess these two explanations for why fewer incumbents seek another term. We then consider what the available evidence suggests about the three accountability mechanisms.

6.4.1 Electoral Accountability versus Incumbent Costs

If village mayors anticipate that unsuccessful implementation will reduce their electoral support, incumbents who previously won narrowly should be more likely to reconsider running because even a modest loss could affect their reelection prospects. This prediction is the same whether voters revise their assessment of the village mayor, apply a higher performance standard, or react to participation costs incurred without receiving a project. Direct costs to village mayors could produce the same pattern if implementation lowers the value of another term or leaves fewer resources for campaigning, especially for incumbents with a lower probability of winning.

We examine this pattern by interacting program assignment, and then implementation participation, with the village mayor's centered 2010 winning margin. Appendix Sections [K.1.1.1](#) and [K.1.1.2](#) report the estimation details, complete specifications, and figures. In both analyses, the estimated effect is more negative among previously less secure incumbents. In the ITT specification, a 10-percentage-point larger prior margin makes the effect of program assignment on running 0.8 percentage points less negative ($p = 0.088$). The implied effect is -4.8 percentage points for an incumbent whose prior winning margin was zero. In the corresponding CATE specification, a 10-percentage-point larger prior margin makes the participation effect 1.2 percentage points less negative ($p = 0.086$), with an implied effect of -6.8 percentage points at a prior margin of zero. Because both explanations can generate this pattern, the interactions do not by themselves reveal why village mayors reconsider running.

Barangay-official surveys from an earlier KALAHI evaluation show that some respondents regarded implementation as tedious, but 49 percent in 2006 and 75 percent in 2010 were satisfied or very satisfied with the program ([Labonne, 2013](#)). Appendix [K.1.2](#) reports the full results. Our interviews likewise describe substantial work by village mayors, but none links these demands to a decision not to run, and one village mayor was willing to participate in another cycle (Ap-

pendix K.1.3). These sources provide little indication that implementation made another term less attractive. We therefore turn to how citizens understood and responded to implementation outcomes.

6.4.2 Learning, Expectations, or Voter Costs

Village mayors could serve on the BRT that presented village proposals or remain outside it. We use this variation to assess whether funding effects depend on village mayor involvement. Formal responsibility may make funding outcomes more informative about the village mayor’s competence or raise the standard against which citizens judge those outcomes. If greater responsibility makes unsuccessful outcomes more consequential, the electoral difference between receiving funding and narrowly missing should be larger for more involved village mayors. Citizens’ costs alone provide no reason for this difference to vary by role.

Table 4: Funding Effects by Village Mayor’s BRT Role

	Village mayor’s BRT role		
	Not on BRT	Member	Chair
<i>RD Honest</i>	0.085 (0.066)	0.079* (0.043)	-0.015 (0.058)
<i>Collapsed support-point RD</i>	0.079 (0.056)	0.123** (0.050)	-0.030 (0.051)
Municipality and cycle fixed effects	✓	✓	✓
Census and political covariates	✓	✓	✓
Clustered SEs (RD Honest)	✓	✓	✓

^a Notes. Entries are preferred Model (5) effects on incumbent vote share among runners, re-estimated by role. RD Honest uses role-specific curvature bounds and bandwidths. Role samples contain 215, 218, and 263 observations. Because roles were not randomly assigned, comparisons are suggestive. Appendix Table A48 reports all specifications. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4 shows positive effects for members and village mayors outside the team, although the latter estimate is imprecise, and a near-zero estimate for chairs. This pattern is not what we would expect if greater formal involvement simply strengthened the electoral response to funding. It is nevertheless consistent with voter learning if citizens already knew more about chairs, making the funding outcome less informative about their competence (Arias et al., 2022; Dias and Ferraz, 2019). To assess this possibility, we compare prior electoral strength across roles. Chairs’ mean 2010 vote share exceeded members’ by 3.4 percentage points ($p = 0.011$; Appendix Table A49).³¹ The near-zero chair estimate offers less support for raised standards, which would imply a larger

³¹Prior vote share imperfectly measures voter information. BRT roles are nonrandom and role estimates imprecise.

chair effect if formal responsibility increased citizens' expectations. The role-specific pattern is also inconsistent with a simple participation-cost account, which provides no reason for the funding effect to vary with the village mayor's role.

To investigate these mechanisms more directly, we turn to interviews and survey evidence on how citizens understood implementation. Some respondents in our interviews associated unsuccessful proposals with weak leadership or insufficient effort, as voter learning would predict. Another linked a village mayor's defeat to residents tiring of waiting for projects, consistent with expectations going unmet.³² Earlier evaluations document both frustration and costs after sustained participation without funding. Some participants initially considered withdrawing from later cycles (ADB, 2012, pp. 25–26). An evaluation synthesis similarly warns that time spent on unsuccessful proposals may discourage continued participation and recommends managing expectations (Labonne, 2013). Survey evidence from a subset of villages in the same experimental sample also documents burdens. In 2015, 45 percent of respondents considered the process too slow and 11 percent too burdensome (Beatty et al., 2018, p. 26). Attribution was nevertheless uneven. Some participants described failure as the “luck of the draw” or blamed other villages rather than the village mayor (ADB, 2012, p. 26). Taken together, the funding results by BRT role and the accounts linking failure to leadership lend the strongest support to voter learning. The remaining evidence makes unmet expectations and participation costs plausible, but uneven attribution limits their ability to explain electoral responses to village mayors (Appendix K).

7 Conclusion

We argue that the electoral consequences of public-service policies depend not only on the benefits delivered but also on politicians' visible efforts to secure them. The evidence from KALAHÍ is consistent with this argument. Program access reduces incumbent reelection, primarily because fewer incumbents seek another term. A similar response appears among villages whose participation in implementation was induced by program assignment. Among incumbents who run, however, barely receiving funding raises vote share by about five percentage points relative to narrowly missing it.

The joint pattern is difficult to reconcile with an account in which the only electoral consequence of a public-service policy is to reward incumbents for goods received. One interpretation is that some incumbents anticipate an unfavorable voter response to unsuccessful implementation and choose not to run. This interpretation remains tentative. We do not observe incumbents' expectations, and

³²See Appendix K.2.2, “Perceived Electoral Responses,” for the interview accounts and exact transcript references.

the designs do not identify the effect of unsuccessful implementation relative to nonparticipation. The IV includes any funding that follows participation, while the RD compares funded and narrowly unfunded applicants in a different population. The results therefore support the broader argument without establishing that unsuccessful implementation causes incumbents to leave the ballot.

The exploratory comparisons across implementation roles offer some support for voter learning as an explanation for the funding effect. Funding estimates are smallest for chairs, despite their greater formal responsibility. Chairs also entered with stronger electoral support. This pattern is consistent with outcomes changing voters' assessments less when an incumbent is already well known, rather than greater involvement simply raising the standard citizens apply. Yet roles were not randomly assigned, prior support is not a direct measure of voters' information, and the subgroup estimates do not establish statistically significant differences across roles. This evidence concerns the funding effect only. It does not establish why program access or implementation participation reduces reelection.

The argument is relevant beyond community-driven development where local politicians must help secure resources or meet requirements for their communities to receive valued services. For example, Kenya's Urban Support Program ties infrastructure grants to local governments' compliance with institutional requirements and performance standards ([World Bank, 2017](#)). Such arrangements create opportunities for implementation to affect elections when citizens observe politicians' involvement and the resulting outcome before voting, and can plausibly connect the two. Whether they do so is an empirical question. KALAH's public competition may make these connections especially clear, so neither the sign nor the magnitude of its effects should be assumed to hold in other programs.

Future research could test the effect of unsuccessful implementation more directly through a design that independently varies participation in implementation and the allocation of funding. Comparing unsuccessful participants with nonparticipants would help establish whether a visible attempt has electoral consequences even when it produces no good. Measuring voters' beliefs, performance standards, and participation costs, alongside incumbents' reasons for running or withdrawing, would help distinguish the explanations considered here. Null or negative electoral returns to public-service policies need not imply weak accountability. Understanding those returns

requires attention to what citizens experience during implementation, as well as to the benefits they ultimately receive.

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Appendices

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A Background: Previous Research on the Electoral Effects of Public Service Policies

Table A1 summarizes the main electoral treatment or exposure in studies of public goods and publicly provided services. The review focuses on actual policy interventions and citizens' exposure to infrastructure, service access, or service quality. It excludes studies in which the electoral treatment consists only of information about public services.

Table A1: Treatments in Previous Research on Public Goods and Public Services

Study	Main electoral treatment or exposure
Adiguzel et al. (2023)	Reform-induced changes in walking time to public health clinics and clinic congestion
Ajzenman and Durante (2023)	Quasi-random assignment to vote in schools with different infrastructure quality
Akbulut-Yuksel et al. (2024)	Expressway construction across provinces and years
Araújo et al. (2024)	Rollout of rural electrification and resulting electricity access
Bardhan et al. (2024)	Reported receipt of benefits from local infrastructure programs, compared with private welfare benefits
Boudot-Reddy and Butler (2024)	Construction of rural roads under the PMGSY program
Boyne et al. (2009)	Local-government service performance measured by administrative indicators and citizen satisfaction
Bozcaga (2023)	Local public investments in health and education
Burnett and Kogan (2017)	Street quality measured through neighborhood pothole complaints
Chukwuma et al. (2019)	Proximity to primary health facilities receiving a service-delivery intervention
Croke (2021)	Distribution of insecticide-treated bed nets
Cremaschi et al. (2025)	Reform-induced loss of access to local public services
de Benedictis-Kessner (2018)	Experienced train delays, combined with information about government responsibility
Di Pietro (2019)	Local public-transport use, air quality, crime, and economic conditions
Fernandes and Tavares (2026)	Closure of a municipality's only school
Fried and Venkataramani (2017)	Clean-water provision and the health improvements it produced
Garfias et al. (2021)	Randomized and later nonrandom neighborhood infrastructure improvements
Goyal (2024)	Construction of all-weather rural roads
Guiteras and Mobarak (2019)	Random assignment to a sanitation-subsidy program that promoted household latrine investment
Habyarimana et al. (2021)	Announcement and subsequent implementation of a secondary-school expansion
Harding (2015)	Changes in local road conditions Changes in education inputs, including schools, classrooms, toilets, drinking water, pupil–teacher ratios, books, and seats

Continued on next page

Table A1 continued

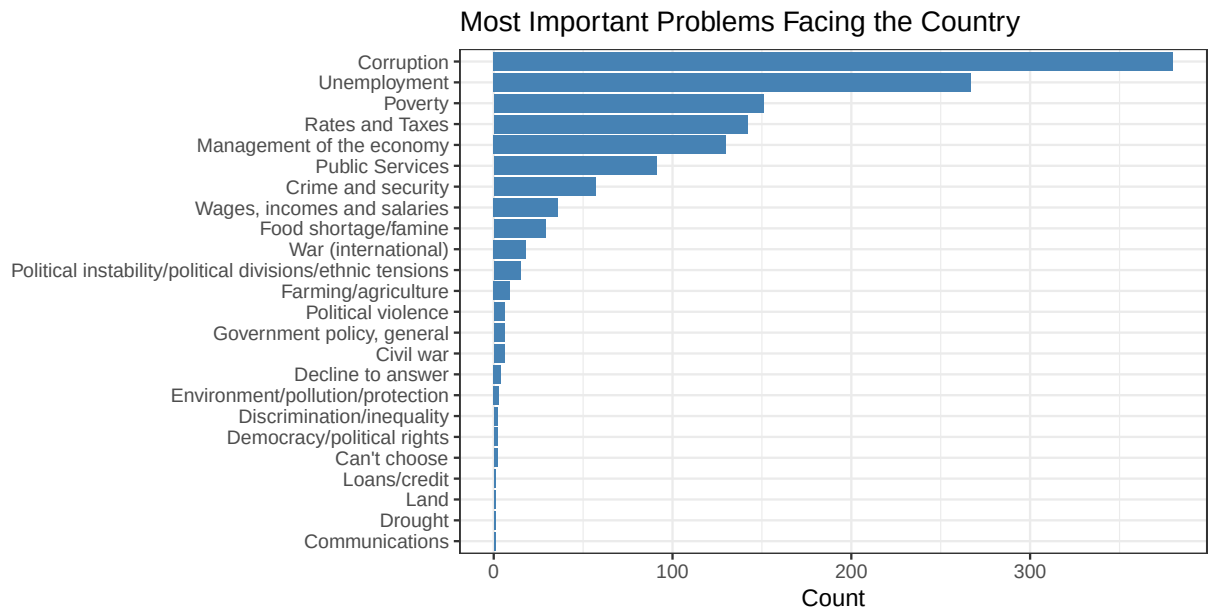
Study	Main electoral treatment or exposure
Harding and Stasavage (2014)	School-fee abolition, school inputs, attendance, and perceived school quality
Huet-Vaughn (2019)	Proximity to completed ARRA road and bridge projects
Maffioli (2021)	Ebola response resources and treatment facilities
Sandholtz (2025)	Random assignment to a school-management reform and treatment-induced changes in learning
Springman (2023)	Presence of NGO aid and service-delivery projects
Springman (2022)	Randomized access to NGO-provided health services
Takahashi et al. (2025)	Earlier receipt of a COVID-19 vaccination
Tarasenko (2020)	Roadworks, new metro stations, and housing renovation
Voigtländer and Voth (2026)	Construction of the German Autobahn

Notes: The table reports the main treatment or exposure used to estimate electoral responses to public goods and publicly provided services. It includes studies of actual policy interventions, infrastructure, service access, and experienced service quality. Studies whose electoral treatment consists only of information about public services are excluded. Studies centered on targeted private benefits and studies of general spending or fiscal transfers that cannot be tied to a specific public service are also excluded. Separate working-paper and published versions and separate supplementary appendices are treated as one study. Citations report the most recent published version located as of September 2026, and otherwise the most recent version reviewed.

B Background: Philippines

B.1 Public Service Provision: Public Opinion

Figure A1: Philippines Public Opinion: Most Important Problems



Notes: Data from Asiabarometer Wave 4.

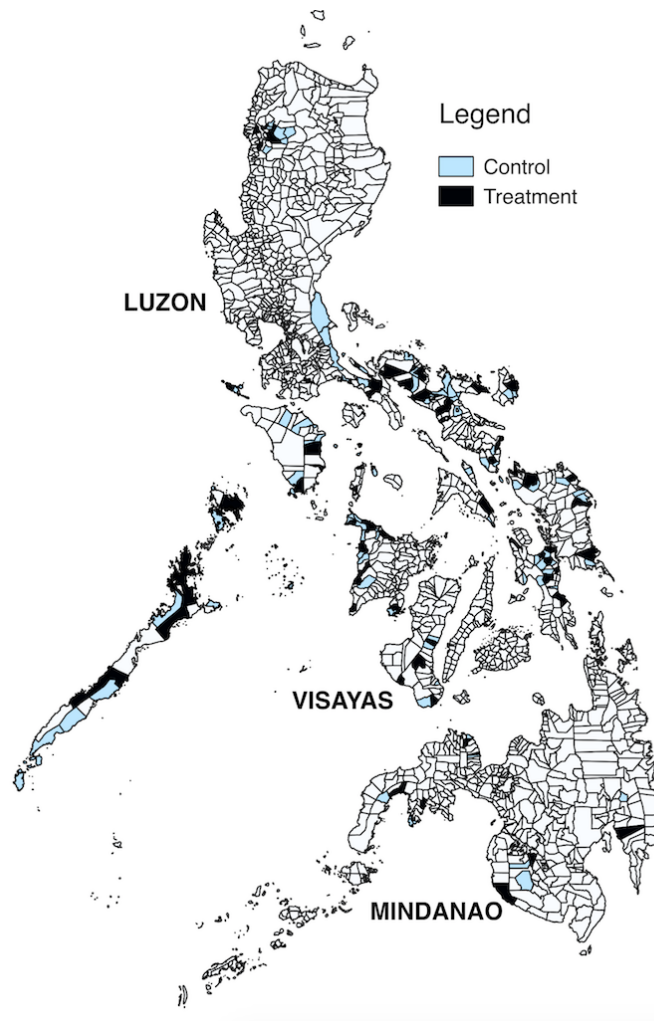
B.2 Public Service Provision: Finance

Public services are financed largely by a fiscal transfer from central government and, to a smaller extent, by local taxes. The transfer is called Internal Revenue Allotment (IRA). The IRA is allocated 23% to provinces, 23% to cities, 34% to municipalities and 20% to barangays. It is funded by a fixed percentage of national tax revenues from three years prior, transferred directly from the national government, and constitutes about 85% of municipal government revenue. Comparatively, this transfer is rather large, and the municipal governments receive, on average, 49,905 million PHP (roughly one million US\$) from IRA transfers ([Troland, 2016](#)). About 20 % of fiscal transfer are reserved for the Barangay Development Fund, giving village mayors substantial discretion over both the scale and visibility of village services.

C Background: Kalahi

C.1 Sample

Figure A2: KALAHI Sample



C.2 Timeline

By the time of the village election in 2013, all villages went through the social perpetration stage and a substantial number of village had projects started and completed (see below.)

Figure A3: KC Timeline and Local Elections

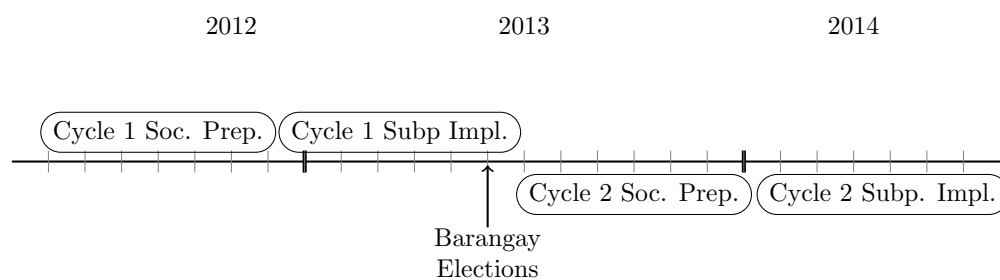


Figure A4: Example:Counterpart Contribution

RANK	BARANGAY	NAME OF SUB-PROJECT	TOTAL PROJECT COST	KC GRANT	LOCAL COUNTERPART CONTRIBUTION		
					BLCA	PLCA	COMMUNITY COUNCIL
1	CAMBAND	IMPROVEMENT OF 1.5 KM SUBURBAN MAJALAYTE FROM TO MARKET ROAD	2,457,891.00	2,456,500.00	226,000.00	306,000.00	76,190.00
2	MALAZARTE	CONSTRUCTION OF 1.5 KM SUBURBAN FED WATER SYSTEM	481,061.00	480,000.00	90,000.00	59,306.00	28,700.00
3	STA ROSA	CONSTRUCTION OF 1.5 KM SUBURBAN SECONDARY SCHOOL BUILDING	1,342,719.00	1,340,500.00	90,000.00	136,331.00	42,110.00
4	CANDILERIA	CONSTRUCTION OF HEALTH CENTER	748,990.00	748,000.00	60,964.00	78,999.00	48,700.00
5	BULAK	IMPROVEMENT OF 1.5 KM SUBURBAN CEMETERY FROM TO MARKET ROAD	2,070,606.00	1,709,000.00	40,000.00	243,308.00	40,300.00
6	SAN SEBASTIAN	CONSTRUCTION OF 0.5 KM SUBURBAN CEMETERY FROM TO MARKET ROAD	1,396,760.00	1,100,000.00	180,000.00	198,872.00	46,900.00
7	MACABA	IMPROVEMENT OF 1.5 KM SUBURBAN CEMETERY FROM TO MARKET ROAD	3,023,363.00	1,613,000.00	500,000.00	501,737.00	60,600.00
	TOTAL		11,847,171.00	9,403,000.00	900,964.00	1,126,878.00	349,800.00

C.3 Visibility and Attribution

The argument developed in the main text requires implementation to involve publicly observable action and citizens to associate its outcome at least partly with the village mayor. The program rules establish an institutional basis for both conditions. Village mayors formally convened the first barangay assembly and often helped recruit residents, while village teams presented proposals in public municipal forums. This subsection complements that institutional description with evidence from an earlier KALAH! assessment and our interviews. The evidence documents whether implementation was visible and how citizens assigned responsibility. It does not establish that visibility or attribution caused an electoral response.

C.3.1 Evidence from an Earlier KALAH! Assessment

An external assessment asked 180 respondents which groups or individuals had the greatest influence on subproject selection. Multiple answers were permitted. Respondents most often named the village mayor or neighborhood leader (79.4%), other village or neighborhood officials (63.9%), and ordinary community residents (60.6%). Community volunteers were named by 30.6% and the municipal mayor by 23.9% (ADB, 2012, p. 25). The survey therefore supports the institutional claim that citizens associated village leaders with project selection. It also shows that they regarded responsibility as shared among several local actors.

C.3.2 Our Interviews

To complement the quantitative analysis, we conducted semi-structured interviews in October 2019 with officials and residents from eight *out-of-sample* barangays in Nueva Ecija. The material

used here consists of 11 interview files covering 13 respondents: 6 citizens, 4 village mayors, 2 barangay secretaries, and 1 barangay councilor. Although these conversations post-date the study period (2010–2013), the institutional structure of KALAHÍ–CIDSS—including mobilisation assemblies, inter-barangay ranking, and municipal sign-off—remained broadly similar across cycles and municipalities. The interviews are therefore informative about how the program operated locally. They should nevertheless be interpreted cautiously when discussing citizens’ knowledge of KALAHÍ. By 2019, the programme had already undergone multiple cycles, so respondents were likely more familiar with it than citizens would have been during the study period. In that sense, the interviews may provide an upper bound on the program’s salience to ordinary voters in 2010–2013. Quoted passages reproduce the English translations in the transcript. The corresponding Filipino passages are retained in the fieldwork archive.

The clearest theme in the interviews is that mobilisation was local and publicly observable. Village mayors repeatedly described persuading residents to attend assemblies, coordinating volunteers, and keeping the process moving. One village mayor explained:³³

“I really went around the barangay, convincing people to accept this program. And we weren’t disappointed. The barangay assemblies require 80% attendance, and we easily exceed that. Sometimes we even reach 96% attendance.”

When asked whether this involved visiting residents at home, the same village mayor answered:

“Yes, we really visit houses. I make it a point not to get tired, after the barangay assembly we still visit the houses and tell them that there are more meetings and that they have to attend because this is really important.”

Other officials gave similar accounts. A second village mayor reported attending barangay assemblies while leaving project work to volunteers. Another described the village mayor and volunteers as the actors who carried out local implementation and emphasized that volunteers often worked without financing.³⁴ These accounts show that village mayors were publicly associated with mobilisation even when volunteers performed much of the subsequent work.

The interviews also show that citizens often recognized KALAHÍ projects without identifying a single political sponsor. One resident said, “Yes, we are familiar. They constructed something like that in our place.” When asked whose program it was, she answered, “I don’t know whose program it is.” A second resident identified DSWD as responsible and said that the village mayor also helped. A third associated the program with foreigners and was unsure who oversaw implementation.³⁵ Other responses likewise described village mayors and volunteers as helping to carry out the

³³Interview with a village mayor, transcript file ddb3be, pp. 23–24.

³⁴Interviews with village mayors, transcript files d8861f, p. 5, and 2d611b, p. 4.

³⁵Citizen interviews, transcript files 2ef321, pp. 27–28, 2f2f08f, pp. 28–29, and 0353c3cc, pp. 31–32.

program rather than as the direct source of projects. Attribution to village mayors was therefore plausible but neither automatic nor universal.

Respondents generally described the municipal mayor as supportive, supervisory, or connected to counterpart funding, while describing village mayors and volunteers as the actors closest to implementation. One village mayor explained, “If it is not acceptable to the Mayor, it won’t even reach our barangay. ... [A]ll of these projects have counterpart funding from the municipal government.” The same village mayor later described the division of labor in simpler terms: “The municipality is just for the approval. But we are really the ones who do the work.” A barangay secretary similarly said that the village mayor was “more directly involved since he is closer to the project” and described the municipal mayor’s role as supervisory.³⁶ Taken together, the interviews indicate that citizens could associate village mayors with visible local implementation while also recognizing the roles of volunteers, municipal officials, and DSWD.

³⁶Interviews with a village mayor, transcript file ddb3be, pp. 24–26, and a barangay secretary, transcript file d82222, pp. 20–22.

C.4 Compliance

The table below shows the municipalities that experienced issues of non-compliance according to Beatty et al. (2013). The two Abra drop-outs Peñarrubia and Pidigan were not included in the data because they never reached baseline.

Table A2: Dropout Municipalities in KC Evaluation (Beatty et al.)

Province	Municipality	Assignment	Reason for Dropout or Non-Compliance
<i>Dropouts and Replacement</i>			
Abra	Peñarrubia	Treatment	Unable to provide required counterpart funding.
Abra	Pidigan	Control	Dropped with paired treatment municipality (Peñarrubia).
Palawan	Taytay	Treatment	Replacement for Peñarrubia.
Palawan	San Vicente	Control	Replacement for Pidigan.

Additionally, there were four municipalities assigned to treatment that did not comply with the randomization and thus did not participate in KALAHÍ. There were also four municipalities originally assigned to the control group were nonetheless allocated funding. Table A3 lists every municipality that violated its random-assignment status in the original KC experiment and spells out the specific mechanism. We distinguish four analytically separate groups:

- treatment-assigned municipalities that never managed to launch KC
- control municipalities that kept their control status but lost their treated partner when that partner dropped out
- control municipalities that crossed over and received KC funds
- pairs in which irregular reallocations left *both* municipalities treated.

Table A3: Non-Compliant Municipalities in the KC Impact Evaluation

Province	Municipality	RCT Assignment	Detailed Reason for Non-Compliance
<i>Treatment-assigned Municipalities that Never Implemented KC</i>			
Leyte	Calubian	Treatment	Failed to mobilise counterpart funds.
Sultan Kudarat	Palimbang	Treatment	Security conflict halted assemblies.
Sultan Kudarat	Lambayong	Treatment	Governance problems prevented Stage 1 mobilisation.
Abra	Lagangilang	Treatment	Withdrew after missing counterpart-funding deadlines.
<i>Control Municipalities that Lost Their Treatment Match (Still KC-Free)</i>			
Leyte	Santa Fe	Control	Lost treatment partner when Calubian exited.
Sultan Kudarat	Bagumbayan	Control	Lost treatment partner when Palimbang exited.
Sultan Kudarat	Esperanza	Control	Lost treatment partner when Lambayong exited.
Abra	Villaviciosa	Control	Lost treatment partner when Lagangilang exited.
<i>Control Municipalities that Crossed Over (Received KC)</i>			
Sorsogon	Santa Magdalena	Control	Successfully appealed for KC; funded in Cycle 1.
Oriental Mindoro	Pinamalayan	Control	Successfully appealed for KC;.
Abra	Luba	Control	Received KC funds after Lagangilang exited.
Abra	Malibcong	Control	Funded when DSWD replaced a non-sample drop-out.
<i>Pairs with Irregular Funding – Both Municipalities Received KC</i>			
Sorsogon	Irosin	Treatment	Funded, but its control (Santa Magdalena) also funded.
Oriental Mindoro	Roxas	Treatment	Funded, but its control (Pinamalayan) also funded.
Abra	La Paz	Treatment	Funded, but its control (Luba) also funded.
Abra	Langiden	Treatment	Funded, but its control (Malibcong) also funded.

C.4.1 Compliance, Data on Baranagay Level

The tables below present the compliance status at the village or municipality level as of November 2013. A municipality is coded 1 for each category if at least one village within the municipality fall into the category. For example, when at least on village within a treatment municipality received a project it is coded “Treatment: applied, received”. Only if villages applied and none within a municipality received a project the municipality if coded ”Treatment: applied, not received”.

Table A4: Distribution of Villages by Assignment and Implementation Status

Status	Count	Percentage (%)
Control: never applied	2157	44.76
Control: applied, not received	63	1.31
Control: applied, received	121	2.51
Treatment: never applied	858	17.80
Treatment: applied, not received	823	17.08
Treatment: applied, received	797	16.54
Total	4819	100.00

^a Notes. The table uses one observation per village in the 4,819-village experimental analysis sample. Treatment and control refer to original lottery assignment. A village is classified as received if any proposal was funded, as applied but not received if it has an unfunded ranked proposal and no funded proposal, and as never applied otherwise.

D Data

D.1 Data: Rankings

Figure A5 presents all municipality-cycle funding sequences with observed project ranks. Each horizontal strip is one municipality-cycle; projects are ordered by their original rank, and tiles indicate actual funding. The 93 ranked sequences include clean funded-to-unfunded steps, missing ranks, funding reversals, and cycles in which every ranked proposal has the same funding status. Figure A6 displays the 53 clean sequences used by the main sharp RD. These sequences begin with funded proposals, have contiguous rank support, and contain a single transition from funded to unfunded proposals.

The fuzzy analysis admits progressively broader sets of observed sequences. Table A5 reports the number of municipality-cycles under each rule. Fifty-three sequences are monotone and contiguous. The main fuzzy rule retains 64 sequences that begin funded and have contiguous support, allowing actual funding to depart from the rank assignment after the boundary. Allowing gaps while retaining monotonicity yields 58 sequences; allowing both gaps and reversals among sequences that begin funded yields 71. The least restrictive rule retains all 72 sequences with at least one funded and one unfunded proposal. These rules are not all nested because monotonicity and contiguous support restrict different features of the data.

Table A5: Municipality-Cycles under Alternative Fuzzy RD Inclusion Rules

Cycle-inclusion rule	Cycle 1	Cycle 2	Total
Strict: monotone and contiguous	51	2	53
Current: starts funded and contiguous	62	2	64
Monotone; rank gaps allowed	56	2	58
Starts funded; gaps or reversals allowed	69	2	71
At least one funded and one unfunded	70	2	72

^a Notes. Counts refer to municipality by program-cycle funding sequences with observed project ranks. The least restrictive rule requires at least one actually funded and one actually unfunded proposal. The rows are alternative admissibility rules and are not all nested because contiguity and monotonicity impose different restrictions.

Figure A5: All Funding Sequences by Municipality-Cycle

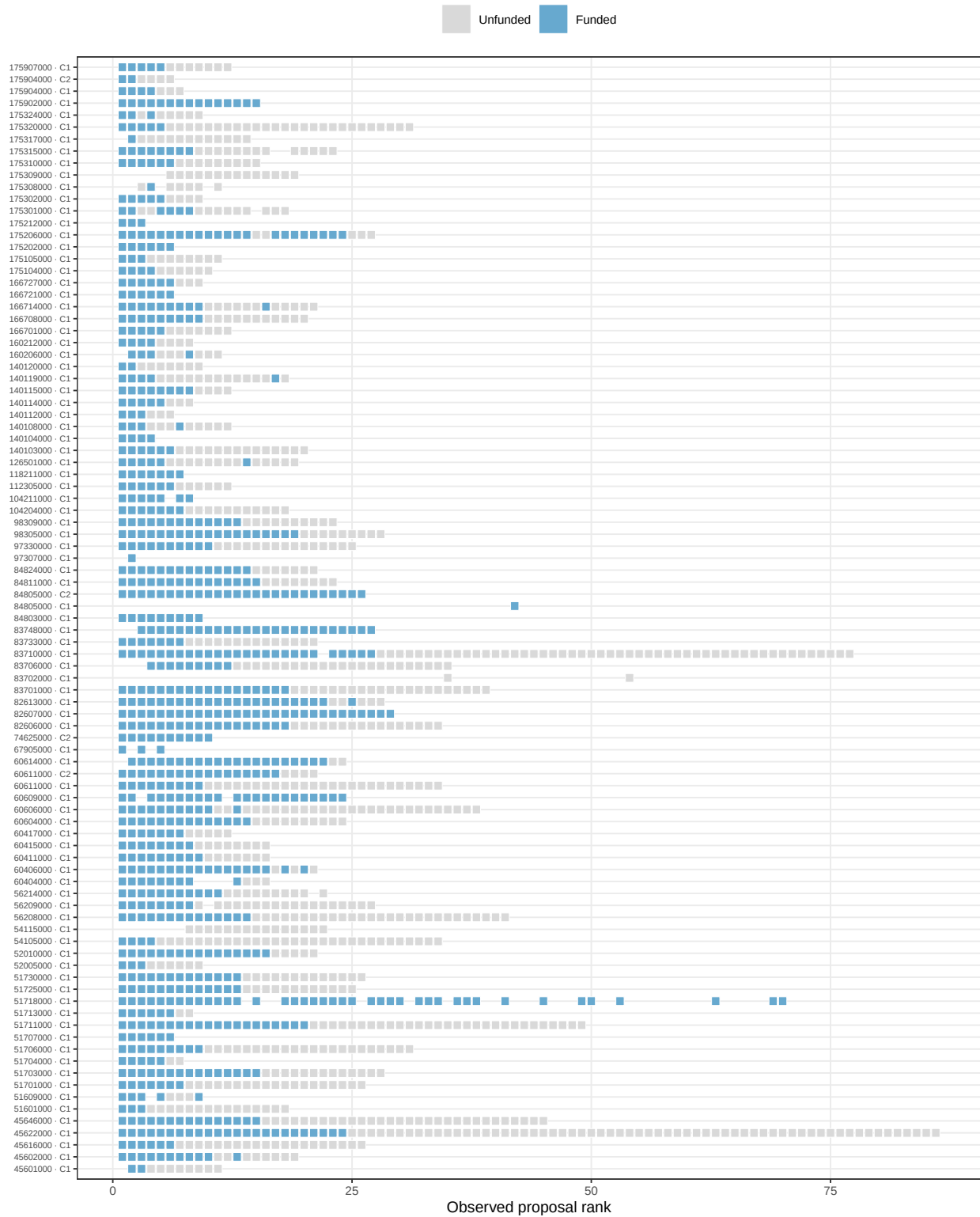
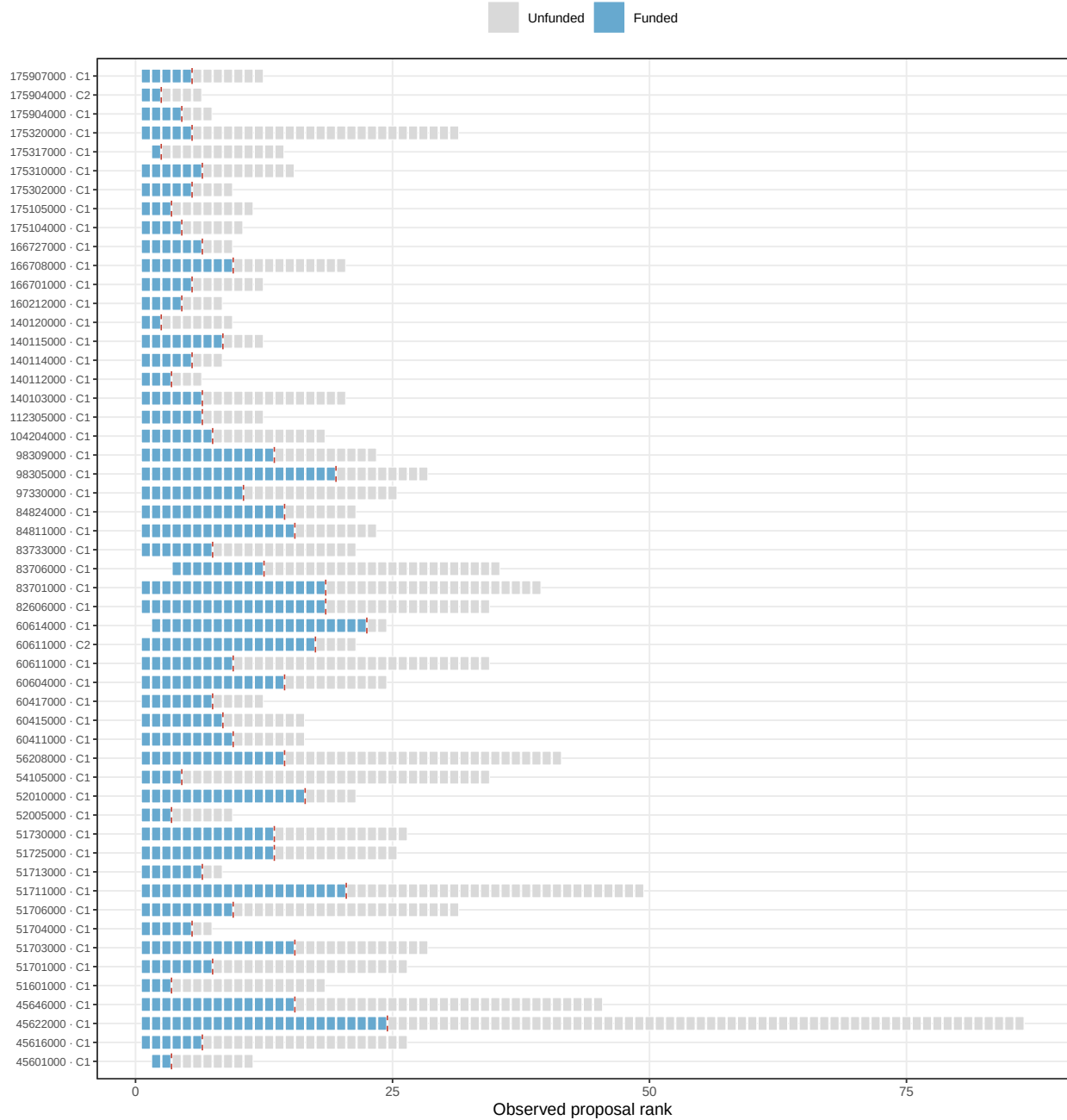


Figure A6: Funding Sequences in the Main Sharp RD Sample

Each row is one of the 53 municipality-cycles included in the main sharp funding RD. Projects are ordered by their observed rank. Blue tiles denote funded proposals and gray tiles denote unfunded proposals. The sample retains sequences that begin funded, have contiguous rank support, and contain no funding reversal after the first unfunded proposal. The dashed red segment marks the midpoint between the last funded and first unfunded ranks. Centering proposal rank at this midpoint places the closest funded and unfunded support points at 0.5 and -0.5 , respectively.



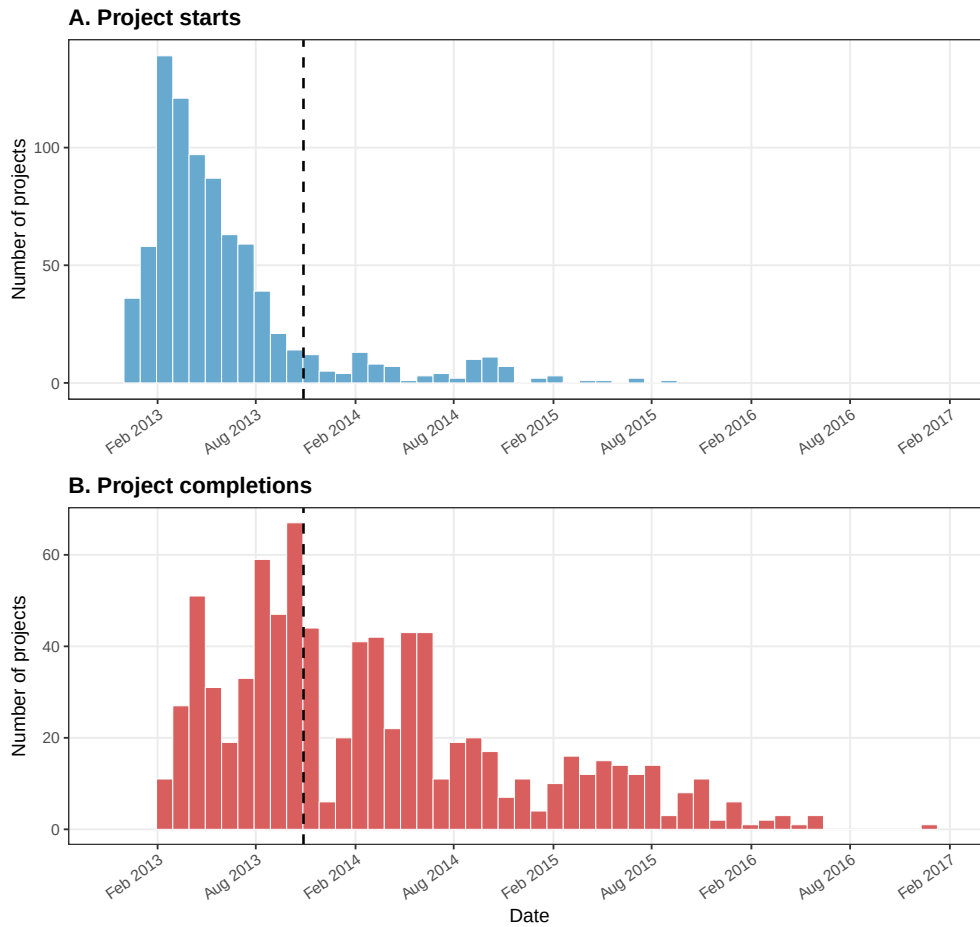
D.2 Funded Projects Started and Completed

We complement the ranking and volunteer data with final administrative project data from the Philippine Department of Social Welfare and Development (DSWD). Specifically, we use DSWD's *List of All CDD Projects*, a project-level administrative workbook dated July 2017. The workbook reports each funded subproject's location, financing phase and cycle, project type and cost, construction start date, and completion date. We retain the KC-MCC and KC-AF phases corresponding to the ranked competitions, including the crossover phase, and exclude PAMANA because it is a separate funding stream.

We first link the project records to every ranked proposal coded as funded, whether or not its municipality–cycle sequence enters the clean funding RD. We match on standardised province, municipality, barangay, and compatible funding cycle. For the few village–cycles containing more than one funded proposal or project record, we use the difference between total project costs in the ranking and project records to establish a one-to-one link. This procedure uniquely matches 831 of the 922 ranked funded proposals. The matched sample includes proposals from both lottery arms, including crossover municipalities. All matched projects began after the corresponding MIBF ranking date, providing an additional check on the linkage. The remaining 91 proposals could not be linked uniquely using the available identifiers.

Figure [A7](#) plots start and completion dates for this broader sample. By the October 2013 election, 734 of the 831 matched projects had started and 345 had been completed. Among the matched projects, 389 were underway but not recorded as complete and 97 had not yet started.

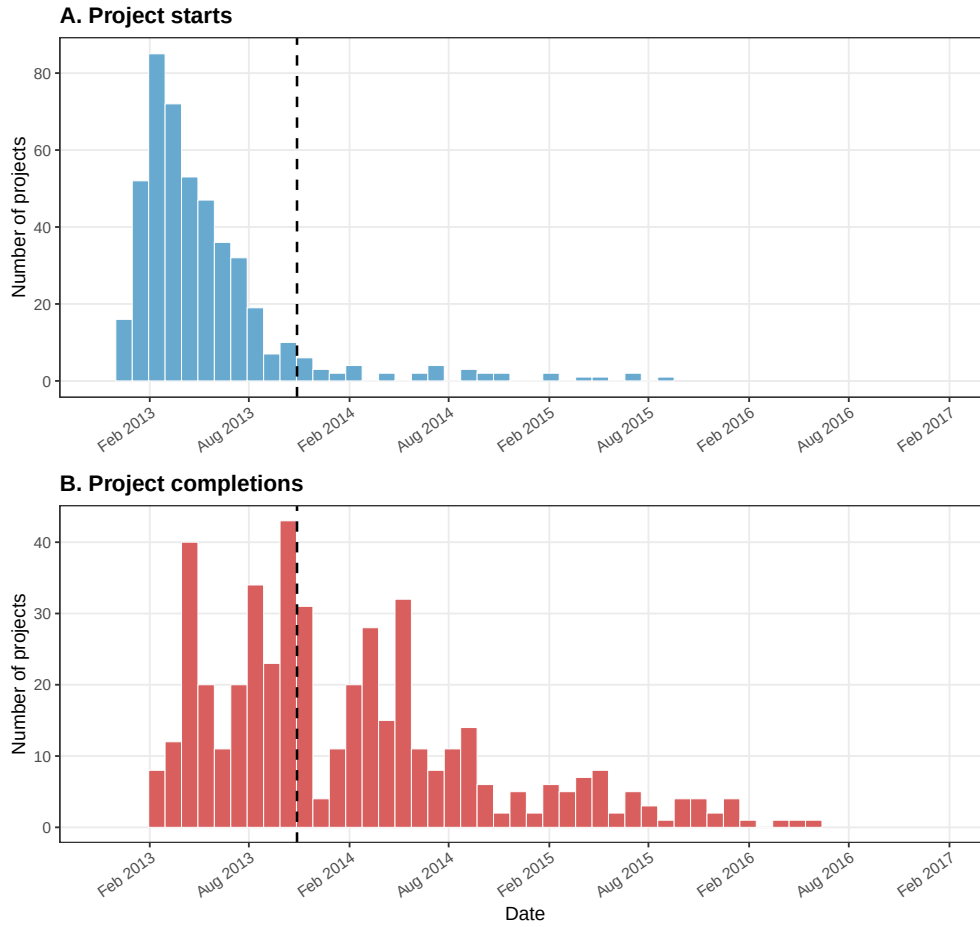
Figure A7: Funded Projects Started and Completed: All Ranked Funded Proposals



Notes: Source: DSWD, *List of All CDD Projects*, July 2017. The sample contains 831 of the 922 ranked proposals coded as funded that can be linked one-to-one to DSWD project records. We match on standardised province, municipality, barangay, and compatible funding cycle. When identifiers repeat within a village-cycle, we retain the proposal-project pair with the closest total project costs in the ranking and project records. The sample includes both lottery arms and crossover municipalities. PAMANA projects are excluded. All 831 projects have recorded start dates and 829 have recorded completion dates. The dashed vertical line marks the October 28, 2013 barangay election. By that date, 734 projects had started and 345 had been completed. Ninety-one funded proposals could not be linked uniquely and are omitted.

Figure A8 repeats the exercise for the clean funding-RD sample. We uniquely match 466 of its 494 funded proposals. By the election, 429 had started and 211 had been completed. A further 218 were underway but incomplete and 37 had not yet started. The two figures use the same monthly bins and date range. The RD sample therefore resembles the broader funded-proposal sample in implementation timing, although its projects were somewhat more likely to have started or been completed by election day.

Figure A8: Funded Projects Started and Completed: Clean Funding-RD Sample



Notes: Source: DSWD, *List of All CDD Projects*, July 2017. The sample contains 466 of the 494 funded proposals in the clean sharp-RD sample that can be linked uniquely to DSWD project records by standardised province, municipality, barangay, and funding cycle. PAMANA projects are excluded. All 466 matched projects have recorded start and completion dates. The dashed vertical line marks the October 28, 2013 barangay election. By that date, 429 projects had started and 211 had been completed. Twenty-eight funded proposals could not be linked uniquely and are omitted. The axes and histogram bins match Figure A7.

Table A6 reports the same composition within the preferred Model (5) vote-share windows. The 24.42-rank RDHonest window contains all 494 funded proposals in the clean sequences and therefore reproduces the clean-sample composition. The 11.45-rank collapsed-RD window contains 401 funded proposals, of which 381 can be linked to project records. Among these linked projects, 28 had not started by the election, 178 were underway, and 175 had been completed. Thus, the timing composition changes little when the sample is restricted to either preferred window.

Table A6: Implementation Status within the Preferred RD Windows

Sample	Bandwidth	Funded proposals	Matched projects	Not started	Underway	Completed
Clean funding-RD sample	–	494	466 (94.3%)	37 (7.9%)	218 (46.8%)	211 (45.3%)
RD Honest vote-share window	24.42	494	466 (94.3%)	37 (7.9%)	218 (46.8%)	211 (45.3%)
Collapsed RD vote-share window	11.45	401	381 (95.0%)	28 (7.3%)	178 (46.7%)	175 (45.9%)

Note:

The sample contains funded proposals in the clean sharp-RD sequences. The score is centered at the midpoint between the last funded and first unfunded proposals, and each preferred window retains observations with $|X| \leq h$. The 24.42-rank RD Honest bandwidth is FLCI-optimal, and the 11.45-rank collapsed-RD bandwidth is MSE-optimal. Both are read directly from the Model (5) vote-share estimator output. Matched-project entries report counts and percentages of funded proposals. Implementation-status entries report counts and percentages of matched projects. All proposals are included regardless of incumbent candidacy or observed vote share. The compositions are descriptive and do not condition an RD estimate on realized implementation status.

The figures and table show that the funding comparison frequently includes visible construction and, for many funded proposals, completed delivery. The funding RD should therefore be interpreted as the electoral effect of barely receiving funding together with the implementation activity observed by election day, rather than as isolating an informational signal that necessarily precedes delivery. Because project start and completion are realised after funding and have no corresponding realised status for unfunded proposals, we do not condition the RD estimates on these outcomes. The timing evidence instead describes the treatment bundle represented by barely crossing the funding threshold.

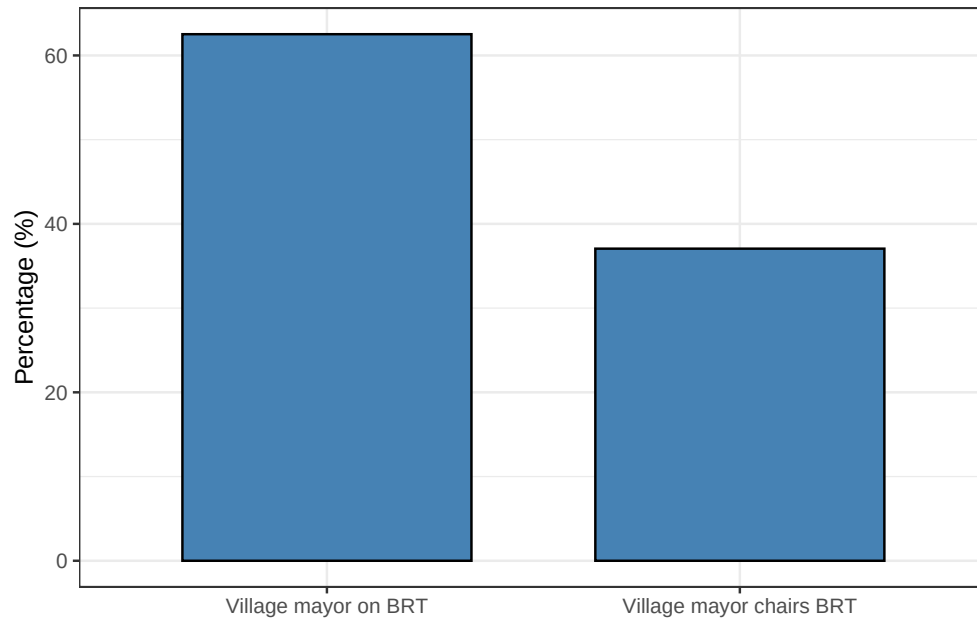
The *KC Operations Manual* requires physical implementation to finish within six months of the first tranche release and no later than twelve months after an approved extension.³⁷ Because first-tranche dates are unavailable, completion by October 2014 is only a proxy. We retain later completions and use this proxy descriptively rather than to define the timing sample.

³⁷KC Operations Manual v1.3, §4.3.2(c): “The physical implementation of any sub-project shall be completed within six (6) months from the date of first tranche release to the community account. In exceptional and fully documented cases the RPMO may grant a single extension; total duration shall in no case exceed twelve (12) months.”

D.3 Volunteer Data

The DSWD volunteer records identify whether a village mayor served on the three-member Barangay Representation Team and whether the village mayor chaired it. They do not record the village mayor's broader involvement in village assemblies or mobilisation outside the team. Figure A9 shows that village mayors served on the BRT in over 60% of the observed village teams and chaired it in almost 40%.

Figure A9: Village Mayor Participation in Barangay Representation Teams



Notes: Source: DSWD Monitoring and Evaluation Unit. The figure reports the percentage of observed village teams in which the village mayor served on the BRT and the percentage in which the village mayor served as chair. BRT participation does not capture the village mayor's broader involvement in village mobilisation.

E Research Design: Implementation Participation

Following [Felton and Stewart \(2024\)](#), we consider four main IV assumptions. (1) *Relevance* implies that the instrument has a non-zero average causal effect on treatment uptake. (2) *Unconfoundedness* assumes that the instrument shares no common causes with the treatment or outcome. (3) The *exclusion restriction* assumes that the instrument has only an effect on the outcome through the treatment. (4) *Monotonicity* assumes that there are no defiers in our sample.

Define:

$$M_{im}(z) \in \{0, 1\}, \quad Y_{im}(d) \in \mathbb{R},$$

where

- $M_{im}(z)$ is the recorded implementation-participation status village i in municipality m would have if $Z_m = z$.
- $Y_{im}(d)$ is the electoral outcome (e.g. incumbent re-election) under participation status d .

Under monotonicity ($M_{im}(1) \geq M_{im}(0)$), there are three compliance types:

Compliers: $M_{im}(0) = 0, M_{im}(1) = 1,$

Never-takers: $M_{im}(0) = 0, M_{im}(1) = 0,$

Always-takers: $M_{im}(0) = 1, M_{im}(1) = 1.$

Type	$M_{im}(0)$	$M_{im}(1)$	Interpretation
Complier	0	1	Villages that participate if and only if assigned ($Z_m = 1$)
Never-taker	0	0	Villages that do not participate under either assignment
Always-taker	1	1	Villages that participate under either assignment
Defier	1	0	Villages that participate only when assigned to control (ruled out by monotonicity)

F Summary Statistics

F.1 Narrow Election

Table A7 describes prior electoral competition in the full experimental sample and among villages within the preferred vote-share bandwidths for RDHonest and the collapsed support-point analysis. Winning margins are the differences between the two leading candidates' vote shares in contested 2010 elections. Each village appears once per sample, regardless of whether its incumbent ran in 2013. These margins provide a benchmark for interpreting the vote-share estimates, but do not establish how many elections funding changed.

Table A7: Prior Electoral Competitiveness

	Full experimental	RD Honest	Collapsed RD
Villages	4,819	1,063	816
Contested elections	4,202	942	718
Single-candidate elections	579	121	98
Missing or incomplete returns	38	0	0
Median winning margin (pp)	12.7	11.9	12.0
Elections decided by 5 pp or less (%)	22.3	23.6	23.1
Elections decided by 10 pp or less (%)	40.9	42.9	42.1

Note:

Each sample counts villages once, including incumbents who did not run in 2013. The RD columns use the preferred vote-share bandwidths of 24.42 ranks for RDHonest and 11.45 ranks for the collapsed support-point RD. Winning margins and close-election percentages refer to contested 2010 elections with recorded returns. Single-candidate elections with recorded returns and missing or incomplete returns are reported separately. Recorded zero votes are not treated as missing.

G Experimental Sample Construction and Balance

Table A8 reconciles the samples used by the lottery and implementation-participation analyses. The original assignment registry contains 198 municipalities and records 4,886 villages. Matching villages across the 2010 and 2013 COMELEC returns yields 4,839 villages in 197 municipalities. The 47 unmatched villages comprise all 40 villages in Mauban, four in Laua-an, and one each in Patnongon, Sibunag, and Arteche. The reason is that Mauban was missing from the sample we obtained from COMELEC. Therefore, we exclude its entire matched pair, which also removes the 20 observed villages in San Antonio. The resulting experimental sample contains 4,819 villages in 196 municipalities and 98 matched pairs.

The merged proposal file has 4,861 rows because 42 villages have two proposal-ranking records. These are distinct proposals rather than additional villages. The experimental analyses aggregate the implementation-participation indicator within village and retain one village record. The funding RD continues to use proposal records because the proposal is its unit of observation. The binary electoral outcomes and candidate count are observed for every village in the experimental sample. Conditional vote share is observed for 3,197 villages. The fully adjusted IV models additionally lose 38 villages with missing 2010 incumbent vote share and winning margin. Thirty-six are in Laua-an and two are in Catubig. No census or municipal matching covariate is missing.

Table A8: Construction of the Experimental Analysis Sample

Sample	Municipalities	Pairs	Villages
Lottery registry	198	99	4,886
Matched 2010 and 2013 election returns	197	99	4,839
Experimental analysis sample	196	98	4,819
Conditional incumbent vote-share sample	196	98	3,197
Fully adjusted IV sample	195	98	4,781
Fully adjusted IV vote-share sample	195	98	3,170

^a Notes. The lottery registry contains 198 municipalities and 4,886 villages, where the village count is the sum of the registry variable recording the number of barangays in each municipality. The election panel retains villages observed in both the 2010 and 2013 COMELEC files. The 47 unmatched villages are all 40 villages in Mauban, four in Laua-an, and one each in Patnongon, Sibunag, and Arteche. Because Mauban has no 2013 returns, matched pair 7, including Mauban and San Antonio, is excluded from the experimental analysis. The proposal-ranking merge creates 4,861 rows because 42 villages have two ranked proposal records. ITT and implementation-participation models retain one row per village, yielding 4,819 unique villages. Conditional vote share is observed for 3,197 villages. Of the remaining villages, 1,613 incumbents did not run and nine runners have missing vote totals. The fully adjusted IV specification additionally excludes 38 villages with missing 2010 incumbent vote share and winning margin.

Table A9 reports pre-treatment balance. Panel A uses all 198 municipalities as randomised. None of the four variables used to construct the matched pairs differs significantly between treatment and control municipalities. Panel B compares the village sample (N=4,819) across treatment and control. Only the married share is 0.525 percentage points higher in treated municipalities after pair adjustment ($p = 0.030$) and is the only one of eleven village-level differences that is statistically significant at the 5 percent level. All other village-level p -values exceed 0.30.

Table A9: Pretreatment Balance in the Experimental Sample

Characteristic	Control mean	Treatment mean	Std. diff.	Pair-adjusted diff.	(SE)	p -value	N
<i>Panel A: Municipality-level lottery sample</i>							
Poverty incidence	47.162	47.446	0.030	0.285	(1.090)	0.794	198
Number of villages	24.101	25.253	0.077	1.152	(2.089)	0.583	198
Population	30,159.990	31,250.747	0.053	1,090.758	(2,998.903)	0.717	198
Land area	215.936	220.174	0.017	4.238	(34.354)	0.902	198
<i>Panel B: Village-level experimental analysis sample</i>							
Population	1,345.252	1,308.701	-0.028	-3.829	(47.375)	0.936	4,819
Average education	2.771	2.753	-0.036	-0.010	(0.021)	0.645	4,819
Average age	26.194	26.248	0.021	0.079	(0.128)	0.535	4,819
Catholic (percent)	83.490	85.647	0.098	0.722	(1.205)	0.550	4,819
Urban	1.965	1.971	0.036	0.005	(0.006)	0.428	4,819
Ethnicity HHI	0.836	0.832	-0.020	-0.010	(0.010)	0.318	4,819
Philippine citizenship (percent)	99.963	99.966	0.029	0.004	(0.004)	0.325	4,819
Married (percent)	33.608	33.883	0.054	0.525	(0.240)	0.030	4,819
Incumbent vote share, 2010	0.618	0.616	-0.008	-0.001	(0.010)	0.907	4,781
Incumbent winning margin, 2010	0.289	0.287	-0.004	0.000	(0.018)	0.988	4,781
Number of candidates, 2010	2.241	2.240	-0.001	0.010	(0.039)	0.807	4,819

Notes: Panel A uses all 198 municipalities in the original lottery registry. Panel B uses one observation for each of the 4,819 villages in the experimental analysis sample after excluding matched pair 7. Standardized differences use the pooled treatment-control standard deviation. Pair-adjusted differences come from regressions of each predetermined characteristic on treatment assignment with matched-pair fixed effects. Panel A clusters standard errors by matched pair. Panel B clusters them by municipality, the level of assignment. The two 2010 vote-based variables are observed for 4,781 villages. P-values below 0.05 are shown in bold.

H Further Results: Average Effect of the KALAHÍ Policy

H.1 Complete ITT Specifications

Table [A10](#) reports the complete ITT specification sequence underlying the preferred estimates in Table [1](#). The first column reports the outcome mean among villages assigned to the control group in the preferred Model (3) sample. All four models include matched-pair fixed effects and clustered standard errors. Model (1) clusters at the municipality level, the unit of treatment assignment. Model (2) adds the four characteristics that entered the pair-matching algorithm—population, number of villages, poverty index, and land area—to increase precision. Model (3) adds pre-treatment census and political covariates. Following [De Chaisemartin and Ramirez-Cuellar \(2024\)](#), Model (4) retains the fully adjusted specification but clusters standard errors at the matched-pair level.

Table A10: Intention-To-Treat Effects of KALAHİ on Electoral Outcomes

	Control mean	Model (1)	Model (2)	Model (3)	Model (4)
<i>Electoral outcomes</i>					
Pr(reelection)	0.490	-0.023** (0.010)	-0.024** (0.010)	-0.025** (0.010)	-0.025* (0.014)
Pr(run)	0.677	-0.024** (0.010)	-0.024** (0.011)	-0.021* (0.011)	-0.021 (0.015)
Vote Share	0.590	0.010 (0.009)	0.008 (0.008)	0.004 (0.008)	0.004 (0.011)
No. candidates	2.234	-0.020 (0.028)	-0.015 (0.023)	-0.013 (0.021)	-0.013 (0.029)
Pair Fixed Effects		✓	✓	✓	✓
Municipality-Clustered SE		✓	✓	✓	
Mun. Cov			✓	✓	✓
Census and Political Cov.				✓	✓
Pair-Clustered SE					✓
Num.Clusters		196	196	195	98

^a Notes. This table reports the effects of the KALAHİ policy on electoral outcomes using one observation per village. Reelection and running are measured by exact last- and middle-name matching. Term-limit eligibility predates assignment, so legally required exits are balanced across assignment arms in expectation. The number-of-candidates outcome counts all candidates for village mayor in 2013. The first column reports the raw outcome mean among villages assigned to the control group in the preferred Model (3) sample. Each remaining column presents an OLS regression of the dependent variable on an indicator for whether the village was assigned to a treatment or control municipality. All models include matched-pair fixed effects and clustered standard errors. Model (2) adds the municipal characteristics used for pair matching: population, number of villages, poverty index, and land area. Models (3) and (4) add census and political covariates. Models (1)–(3) cluster by municipality, while Model (4) clusters by matched pair. Models (1) and (2) contain 4,819 observations for reelection, running, and candidate count and 3,197 for vote share. Models (3) and (4) contain 4,781 and 3,170 observations, respectively. The full sample loses 38 villages and the conditional vote-share sample loses 27 observations because the 2010 vote-based controls are missing. Robust standard errors are displayed in parentheses. Significantly different from zero at 99 (***), 95 (**), 90 (*) % confidence.

The ITT coefficients for reelection are stable across specifications and range from –0.023 to –0.025. They are statistically distinguishable from zero at the 5 percent level in all three municipality-clustered specifications and at the 10 percent level when standard errors are clustered by matched pair. The corresponding candidacy estimates range from –0.021 to –0.024. They exclude zero at the 5 percent level in Models (1) and (2) and at the 10 percent level in the fully adjusted municipality-clustered specification. Pair-level clustering produces a wider interval that

includes zero. Conditional on running, the vote-share coefficient remains close to zero throughout. Assignment also has no detectable effect on the total candidate field. The fully adjusted municipality-clustered candidate-count estimate is -0.013 and statistically indistinguishable from zero.

The following tables reproduce this specification sequence separately for each outcome and report the corresponding sample sizes, cluster counts, and dependent-variable means.

Table A11: Appendix: Reelection

Pr(reelection)	Model (1)	Model (2)	Model (3)	Model (4)
Treatment	-0.023** (0.010)	-0.024** (0.010)	-0.025** (0.010)	-0.025* (0.014)
Observations	4,819	4,819	4,781	4,781
Clusters	196	196	195	98
Mean dependent variable	0.475	0.475	0.476	0.476
Pair fixed effects	✓	✓	✓	✓
Municipality-clustered SE	✓	✓	✓	
Municipal covariates		✓	✓	✓
Census and political covariates			✓	✓
Pair-clustered SE				✓

^a Notes. This appendix table reports OLS estimates of the treatment effect for the outcome 'Reelection'. All columns include pair fixed effects and clustered standard errors. Column (2) adds the municipal covariates used for pair matching. Columns (3) and (4) add census and political covariates. Columns (1)–(3) cluster by municipality, while column (4) clusters by matched pair. Standard errors are shown in parentheses. *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$.

Table A12: Appendix: Run for reelection

Pr(run)	Model (1)	Model (2)	Model (3)	Model (4)
Treatment	-0.024** (0.010)	-0.024** (0.011)	-0.021* (0.011)	-0.021 (0.015)
Observations	4,819	4,819	4,781	4,781
Clusters	196	196	195	98
Mean dependent variable	0.665	0.665	0.665	0.665
Pair fixed effects	✓	✓	✓	✓
Municipality-clustered SE	✓	✓	✓	
Municipal covariates		✓	✓	✓
Census and political covariates			✓	✓
Pair-clustered SE				✓

^a Notes. This appendix table reports OLS estimates of the treatment effect for the outcome 'Run for reelection'. All columns include pair fixed effects and clustered standard errors. Column (2) adds the municipal covariates used for pair matching. Columns (3) and (4) add census and political covariates. Columns (1)–(3) cluster by municipality, while column (4) clusters by matched pair. Standard errors are shown in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table A13: Appendix: Vote share

Vote share	Model (1)	Model (2)	Model (3)	Model (4)
Treatment	0.010 (0.009)	0.008 (0.008)	0.004 (0.008)	0.004 (0.011)
Observations	3,197	3,197	3,170	3,170
Clusters	196	196	195	98
Mean dependent variable	0.592	0.592	0.592	0.592
Pair fixed effects	✓	✓	✓	✓
Municipality-clustered SE	✓	✓	✓	
Municipal covariates		✓	✓	✓
Census and political covariates			✓	✓
Pair-clustered SE				✓

^a Notes. This appendix table reports OLS estimates of the treatment effect for the outcome 'Vote share'. All columns include pair fixed effects and clustered standard errors. Column (2) adds the municipal covariates used for pair matching. Columns (3) and (4) add census and political covariates. Columns (1)–(3) cluster by municipality, while column (4) clusters by matched pair. Standard errors are shown in parentheses. *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$.

Table A14: Appendix: Number of candidates

No. candidates	Model (1)	Model (2)	Model (3)	Model (4)
Treatment	-0.020 (0.028)	-0.015 (0.023)	-0.013 (0.021)	-0.013 (0.029)
Observations	4,819	4,819	4,781	4,781
Clusters	196	196	195	98
Mean dependent variable	2.216	2.216	2.218	2.218
Pair fixed effects	✓	✓	✓	✓
Municipality-clustered SE	✓	✓	✓	
Municipal covariates		✓	✓	✓
Census and political covariates			✓	✓
Pair-clustered SE				✓

^a Notes. This appendix table reports OLS estimates of the treatment effect for the outcome 'Number of candidates'. All columns include pair fixed effects and clustered standard errors. Column (2) adds the municipal covariates used for pair matching. Columns (3) and (4) add census and political covariates. Columns (1)–(3) cluster by municipality, while column (4) clusters by matched pair. Standard errors are shown in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10.

I Further Results: Implementation Participation

This section reports the complete specifications for implementation participation and then examines their robustness, identifying assumptions, and complier characteristics.

I.1 Complete Specifications for Implementation Participation

Table A15: Complier Average Treatment Effect of Implementation Participation on Electoral Outcomes

	Control mean	Model (1)	Model (2)	Model (3)	Model (4)
<i>Electoral outcomes</i>					
Pr(reelection)	0.504	-0.032** (0.014)	-0.033** (0.014)	-0.034** (0.014)	-0.034* (0.019)
Pr(run)	0.682	-0.034** (0.014)	-0.033** (0.015)	-0.029* (0.015)	-0.029 (0.020)
Vote Share	0.590	0.014 (0.012)	0.011 (0.011)	0.006 (0.011)	0.006 (0.016)
No. candidates	2.215	-0.028 (0.038)	-0.020 (0.032)	-0.017 (0.028)	-0.017 (0.039)
Pair Fixed Effects		✓	✓	✓	✓
Municipality-Clustered SE		✓	✓	✓	
Mun. Cov			✓	✓	✓
Census and Political Cov.				✓	✓
Pair-Clustered SE					✓
Num.Clusters		196	196	195	98

^a Notes. This table reports the Complier Average Treatment Effect (CATE) of recorded implementation participation on electoral outcomes using one observation per village. A village is coded as participating when it appears in either the proposal-ranking records or the volunteer-team records. Reelection and running are measured by exact last- and middle-name matching, and the number-of-candidates outcome counts all candidates for village mayor in 2013. The first column, labelled Control mean, reports the estimated mean under no recorded participation for compliers in the preferred Model (3) sample. Following [Abadie \(2003\)](#), we instrument one minus participation with assignment in a 2SLS regression of the outcome multiplied by one minus participation on one minus participation. This implementation adapts the public replication code for [Goodman et al. \(2020\)](#). The auxiliary regression uses the Model (3) controls, pair fixed effects, and municipality-clustered standard errors. All outcome models include matched-pair fixed effects and clustered standard errors. Model (2) adds the municipal characteristics used for pair matching. Models (3) and (4) add census and political covariates. Models (1)–(3) cluster by municipality, while Model (4) clusters by matched pair. Models (1) and (2) contain 4,819 observations for reelection, running, and candidate count and 3,197 for vote share. Models (3) and (4) contain 4,781 and 3,170 observations, respectively. The full sample loses 38 villages and the conditional vote-share sample loses 27 observations because the 2010 vote-based controls are missing. Robust standard errors are shown in parentheses. Significantly different from zero at 99 (***) , 95 (**), 90 (*) % confidence.

Table A15 reports the complete IV specification sequence underlying the preferred CATE estimates in Table 2. It uses the same four specifications as the ITT analysis. The first column reports the estimated outcome mean under no recorded participation for compliers in the preferred Model (3) sample. We recover this benchmark with the transformed-outcome IV approach of Abadie (2003), adapting the implementation in the replication code for Goodman et al. (2020). The auxiliary regression uses the same controls, pair fixed effects, and municipality-clustered standard errors as the preferred outcome specification. All models include matched-pair fixed effects and clustered standard errors. Model (2) adds municipal matching covariates, Model (3) adds pre-treatment census and political covariates, and Model (4) replaces municipality clustering with matched-pair clustering in the fully adjusted specification.

The reelection estimate ranges from -0.032 to -0.034 . It is statistically distinguishable from zero at the 5 percent level in all three municipality-clustered specifications and at the 10 percent level with pair-level clustering. Implementation participation also lowers the probability that the incumbent runs for reelection by 2.9 to 3.4 percentage points. The estimates in Models (1) and (2) exclude zero at the 5 percent level, while the fully adjusted municipality-clustered estimate excludes zero at the 10 percent level. The pair-clustered interval includes zero. Conditional vote shares remain virtually unchanged. The estimated effect on the number of candidates is also small and imprecise. In the fully adjusted municipality-clustered specification it is -0.017 with a standard error of 0.028. Thus, the observed change in candidate supply concerns the incumbent's own decision to run rather than a detectable change in the overall number of candidates.

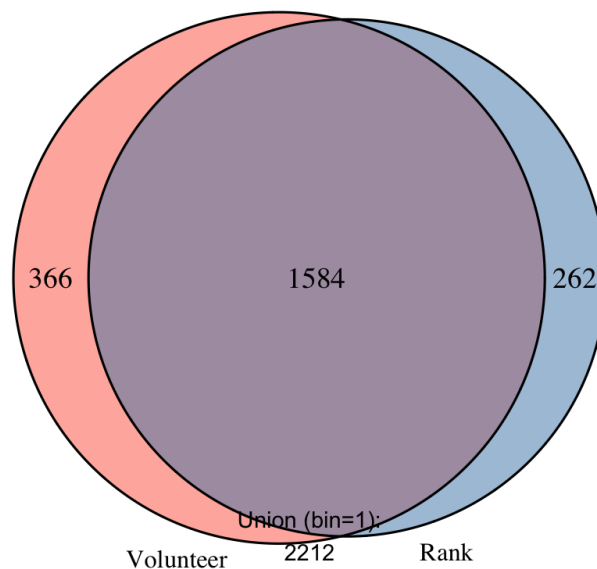
I.2 Alternative Measures of Recorded Implementation Participation

I.2.1 Data Comparison: Ranking and Volunteer Data

The rank-based and volunteer-based indicators do not fully coincide. At the proposal-record level in the experimental analysis sample, 2,212 records appear in at least one source. Of these, 1,584 appear in both sources, 366 only in the volunteer data, and 262 only in the ranking data. Some divergence reflects the sequence of implementation because volunteer teams are formed before proposals are ranked. Missing links across the two administrative files may also account for part of the difference.

The implementation-participation analysis uses villages rather than proposal records as its unit of observation. We therefore collapse multiple records within a village before constructing the treatment indicator. At this level, 2,170 villages appear in at least one source. Of these, 1,543 appear in both, 366 only in the volunteer records, and 261 only in the ranking records. Figure A10 shows this village-level overlap. The main measure is the union of the two sources and therefore includes both sets of source-specific records.

Figure A10: Overlap between Records of Implementation Participation



Notes: The figure uses one observation per village after excluding the matched pair containing Mauban. “Rank” denotes a proposal-ranking record and “Volunteer” denotes a volunteer-team record. The main implementation-participation indicator equals one for the union of the two sources.

I.2.2 Results

The main implementation-participation indicator combines the ranking and volunteer records because neither administrative source is complete on its own. To assess whether the results depend on this choice, Table A16 holds the preferred specification fixed and reconstructs the endogenous treat-

ment in three ways. The first column uses the main union measure. The second codes participation only when a ranking record is observed. The third codes participation only when a volunteer-team record is observed.

The first stage remains strong under all three definitions, although it is largest for the union measure. In the preferred sample, assignment raises the probability of appearing in either source by 0.733, appearing in the ranking records by 0.598, and appearing in the volunteer records by 0.636. The estimated reelection effects range from -0.034 to -0.042 and are statistically distinguishable from zero at the 5 percent level under every definition. The running estimates range from -0.029 to -0.036 and exclude zero at the 90 percent confidence level. Vote-share and candidate-count estimates remain small and statistically indistinguishable from zero. Because the instrument and reduced form are unchanged, alternative treatment definitions primarily rescale the IV coefficient through their different first stages. The stability in sign and inference shows that the findings are not driven by either administrative source alone.

This comparison addresses disagreement between the ranking and volunteer records, but it does not eliminate the broader measurement concern. Villages appearing in neither source may still have experienced initial information, social preparation, or assembly activity. The IV estimates should therefore be interpreted as effects of *recorded implementation participation* under the maintained exclusion restriction, not as effects of any contact with KALAH.

Table A16: Implementation Participation: Robustness to Alternative Measures

	Union (main)	Rank only	Volunteer only
<i>Second-stage outcomes</i>			
Pr(reelection)	-0.034** (0.014)	-0.042** (0.017)	-0.040** (0.016)
Pr(run)	-0.029* (0.015)	-0.036* (0.018)	-0.034* (0.017)
Vote Share	0.006 (0.011)	0.007 (0.014)	0.007 (0.013)
No. candidates	-0.017 (0.028)	-0.021 (0.035)	-0.020 (0.033)
<i>First stage</i>			
Assignment (main sample)	0.733*** (0.042)	0.598*** (0.039)	0.636*** (0.047)
Assignment (vote-share sample)	0.732*** (0.044)	0.606*** (0.041)	0.631*** (0.049)
Pair fixed effects	✓	✓	✓
Clustered SE	✓	✓	✓
Full covariates	✓	✓	✓

^a Notes. Each column reports the preferred Model (3) while changing only the endogenous measure of implementation participation. The main measure equals one when a village appears in either the proposal-ranking records or the volunteer-team records. The remaining columns use each source separately. All models include matched-pair fixed effects, municipality-clustered standard errors, and the full set of municipal, census, and political covariates. The binary outcomes and candidate count use 4,781 villages, and conditional vote share uses 3,170 villages. The first-stage rows use the corresponding estimation samples. The administrative records do not measure initial social-preparation exposure among villages appearing in neither source. Robust standard errors are shown in parentheses. Significantly different from zero at 99 (***), 95 (**), and 90 (*) percent confidence.

I.3 Robustness: Implementation Participation

To check the robustness, we use the omnibus function ‘ivDiag’ in R from [Lal et al. \(2023\)](#) which conducts both OLS and 2SLS estimation and quantify uncertainties using multiple inferential methods. In the figures below, points mark the point estimates and horizontal bars the 95% confidence intervals for both ordinary least squares (OLS) and two-stage least squares (2SLS) estimates of the effect of D on Y , instrumented by Z . Each OLS estimate is accompanied by three inference methods: analytic cluster-robust standard errors and coefficient- and t -ratio block bootstraps—whereas the 2SLS estimate appears with five: those three plus Anderson-Rubin inversion intervals [Chernozhukov and Hansen \(2008\)](#) and the tF procedure of [Lee et al. \(2022\)](#), whose critical value is adjusted by the effective first-stage F . The inset reports the effective Kleibergen–Paap first-stage F statistic, the numbers of observations and clusters. All specifications include the pair fixed effects, standard errors are clustered at the municipality level. Sociodemographic and political controls are included. The fully adjusted specifications contain 4,781 villages for the binary outcomes and 3,170 villages for conditional vote share because 38 villages lack the 2010 vote-based controls and vote share is observed only when an incumbent runs. Computed using ivDiag ([Lal et al., 2024](#)).

We find that the assignment to the KALAH policy has a strong positive effect on policy uptake, as indicated by the high effective first-stage F -statistics (all above $F = 250$), which far exceed the conventional threshold of 10 ([Staiger and Stock, 1997](#)) and the tF threshold of 104.7 required for standard t -ratio inference ([Lee et al., 2022](#)).

The alternative inference procedures support the same qualitative conclusion as the main IV table. The reelection estimate remains negative and is distinguishable from zero under the analytic cluster-robust and most weak-instrument-robust procedures. The running estimate is also negative, although whether its 95 percent interval excludes zero depends on the procedure. The conditional vote-share estimate remains close to zero. The coefficient-bootstrap intervals are wider and sometimes include zero. Taken together, the results support a negative effect of assignment-induced recorded participation on reelection and running, with stronger evidence for reelection and no detectable effect on vote share conditional on running.

Figure A11

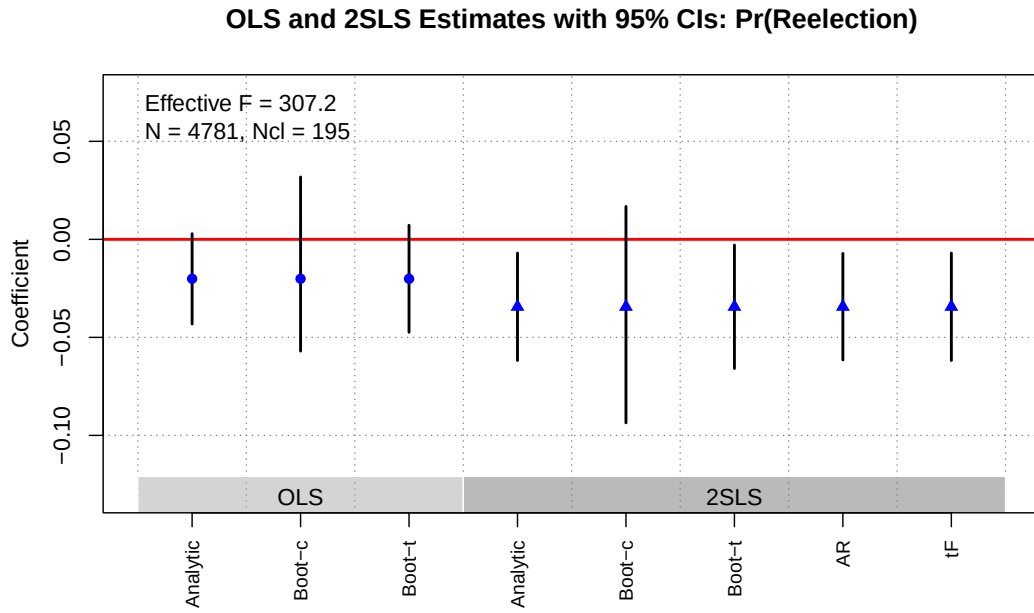


Figure A12

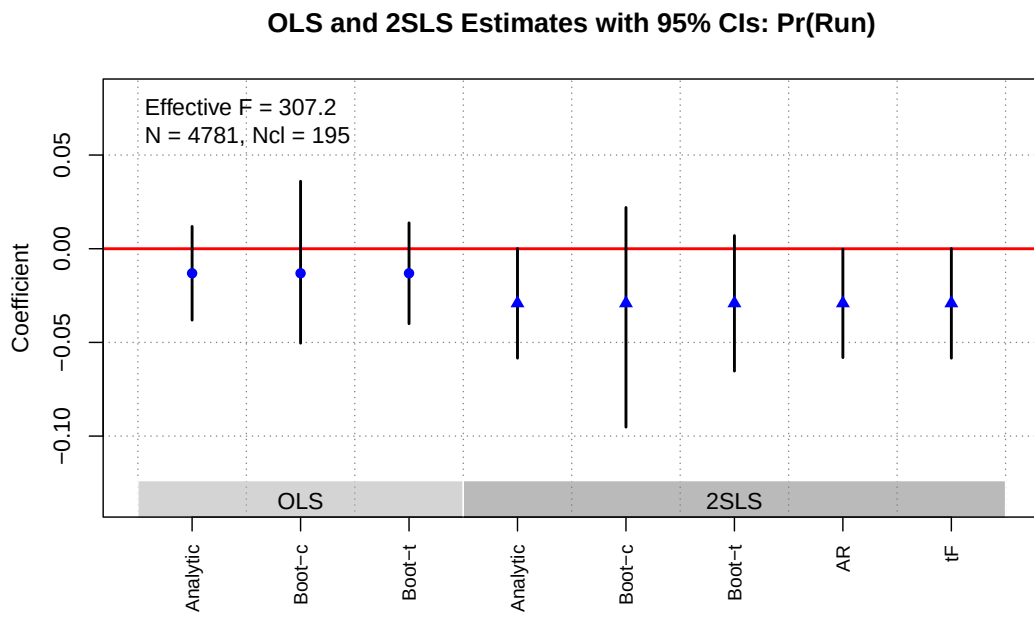
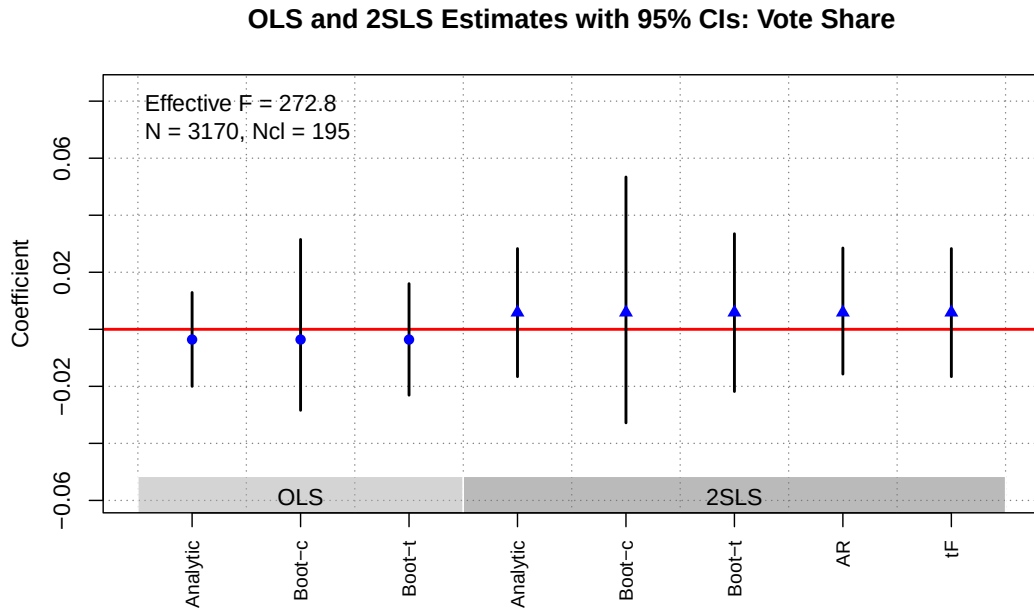


Figure A13



I.4 Evaluating the Exclusion Restriction

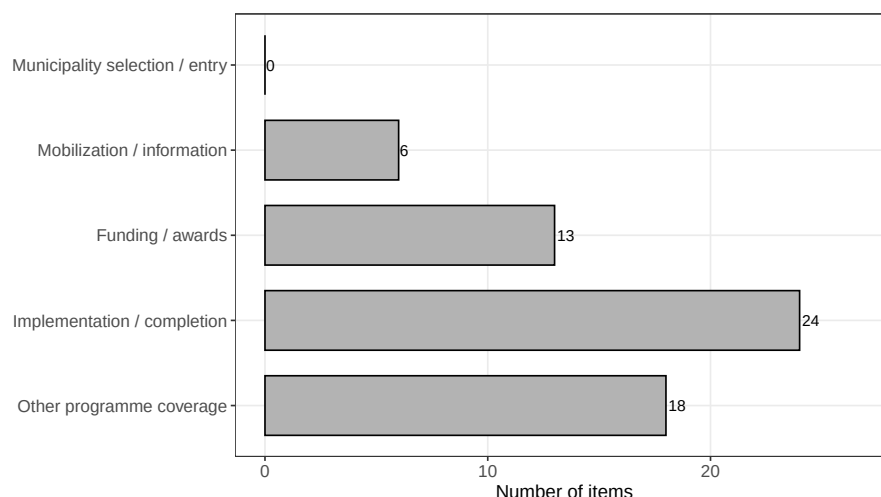
The IV analysis uses municipal assignment, Z_m , as an instrument for recorded implementation participation, M_{im} . The exclusion restriction requires every effect of assignment on the electoral outcome, Y_{im} , to operate through M_{im} . Three possible violations must be distinguished. A *lottery effect* would arise if assignment itself changed voter expectations or electoral behavior, creating a direct $Z_m \rightarrow Y_{im}$ path. An *early-activity effect* would arise if assignment induced meetings or social preparation, denoted by S_{im} , that are omitted from M_{im} but independently affect the outcome, creating a $Z_m \rightarrow S_{im} \rightarrow Y_{im}$ path. This possibility reflects an incomplete measure of implementation rather than an effect of the lottery announcement itself. A *municipality-wide political effect* would arise if assignment changed the municipal mayor’s electoral standing and the political value of ties to the municipal mayor, denoted by P_m , thereby creating a $Z_m \rightarrow P_m \rightarrow Y_{im}$ path even in villages without recorded participation. We examine each channel separately below. We also consider whether assignment affected village mayors through municipality-wide fiscal reallocation or cross-village spillovers.

I.4.1 A Direct Effect of the Lottery

A direct lottery effect would arise if voters learned that their municipality had been selected and changed their expectations of future benefits before any village-level implementation occurred. Assignment would then affect electoral outcomes independently of the participation measure used in the IV analysis. A necessary condition for this channel is that ordinary voters knew about the municipal lottery result.

To probe this condition, we assembled a small corpus of online KALAHİ coverage using Google News RSS searches for “KALAHİ,” “KALAHİ-CIDSS,” and “KC-NCDDP” and classified the resulting headlines into municipality selection/entry, mobilisation/information, funding/awards, implementation/completion, and other programme coverage. The corpus contains articles published between November 2018 and April 2026, with most coverage concentrated in 2023–2026. Figure A14 shows that the corpus is dominated by funding and implementation stories, while none of the coded items concerns municipality entry. The remaining “other programme coverage” items are mostly generic programme or institutional stories rather than reports about municipal entry into the programme. Because the corpus reflects currently indexed online coverage rather than a complete historical newspaper archive, it cannot establish how salient the 2011 lottery was. It provides only suggestive evidence that public reporting about KALAHİ tends to emphasize later, visible stages of the programme rather than municipal selection itself.

Figure A14: Quick Media Audit of Online KALAHl Coverage



Notes: Figure reports counts of unique KALAHl-related items retrieved through Google News RSS searches for “KALAHl,” “KALAHl-CIDSS,” and “KC-NCDDP.” The retrieved items were published between November 2018 and April 2026, with most observations concentrated in 2023–2026. Headlines were classified as municipality selection/entry if they contained terms such as “lottery,” “draw,” “selected to join,” “chosen to join,” “eligible,” or “admitted”; as mobilisation/information if they referred to assemblies, orientations, forums, volunteers, meetings, trainings, or the MIBF; as funding/awards if they referred to approved funds, grants, allocations, awards, or success rates; and as implementation/completion if they referred to projects, construction, turnover, launch, or completion. Remaining items were coded as other programme coverage; these were mostly generic programme-promotion or institutional stories, including items on staff training, DSWD-LGU collaboration, resilience, access to services, or broad community empowerment. The corpus is a quick descriptive probe and not a full historical newspaper archive.

A household survey conducted by [Beatty et al. \(2013\)](#) in 2012 as the baseline for an earlier impact evaluation of KALAHl in the same 198-municipality experimental sample provides contemporaneous but indirect evidence about awareness of the lottery result. Roughly one year after assignment, 30% of households in sampled treatment barangays had never heard of KALAHl. Another 38% had heard of the program but reported no household participation ([Beatty et al., 2013](#)). This limited and uneven awareness weighs against a large and uniform electoral response to the municipal lottery. The H25 module does not, however, ask how respondents first learned about KALAHl and contains no response category for the lottery. It therefore cannot rule out a direct response among households that knew their municipality had been selected.

The qualitative interviews in Appendix [C.3](#) provide a similar, though also indirect, indication. Respondents recalled village meetings and implementation roles more clearly than the municipal assignment event. Together, these sources weigh against, but do not eliminate, a direct effect of the lottery announcement.

I.4.2 Early Village Activity Outside the Participation Measure

The same evidence therefore motivates a separate concern. If early village activities affected electoral outcomes in villages that never formed a recorded volunteer team or submitted a ranked proposal, assignment would affect the outcome through an omitted form of implementation rather than through the lottery announcement itself.

The baseline survey documents such early activity. Fieldwork took place from April to July 2012, roughly one year after the municipal lotteries, and covered one sampled barangay in each of the 198 municipalities and 30 households in each barangay. At that point, none of the sampled barangays had begun receiving KALAHÍ funding. Fifty-four of the 99 sampled treatment barangays had held their first KALAHÍ barangay assembly, which was principally a general-information meeting, and only one had proceeded to a later social-preparation activity (Beatty et al., 2013, 23–26).

Among 2,970 households in the sampled treatment barangays, 929, or 31%, reported that a household member had attended a KALAHÍ activity, meeting, gathering, or information session. Only 10 households in the sampled control barangays reported doing so. Among the 1,069 household members who attended, 81% attended once, 72% took part in a general-information meeting or barangay assembly, and the average activity lasted 169 minutes. A small minority of 22 attendees reported a financial contribution, averaging PHP 299 (Beatty et al., 2013, 26–27). Assignment therefore generated observable, collective activity before funding decisions, with the early burden consisting mainly of residents' time rather than widespread financial contributions.

These activities could also have been associated with the incumbent. The baseline report describes the village mayor as the most trusted government official and notes that about 87% of respondents knew the village mayor or the village mayor's spouse closely, whereas municipal officials were more distant (Beatty et al., 2013). Citizens attributed influence over subproject selection much more often to village mayors than to municipal mayors, 79% compared with 24% (ADB, 2012, 25). The survey does not link household attendance to villages' later administrative participation status or estimate the electoral effect of these early activities. It therefore establishes that exposure preceded the stages in our measure, but not that villages coded as nonparticipants were affected. Recorded participation nevertheless does not encompass every form of contact with the program, so an electoral effect of early village activity remains a distinct concern.

I.4.3 The Municipal Political Channel

Another possible violation of the exclusion restriction is a municipality-wide political channel linking municipal mayors and village mayors. Previous research on an earlier phase of KALAHÍ shows that municipal mayors could benefit electorally from the program despite having little formal control over project selection or funding (Cruz and Schneider, 2017). This municipal effect could reach village mayors through local political networks. Philippine politics is highly clientelistic (Hicken et al., 2018), and village mayors often act as mayoral brokers who distribute goods and

mobilise electoral support (Cruz et al., 2017). If assignment allowed the municipal mayor to claim credit for bringing KALAH! to the municipality, municipal mayors could have used these networks to support allied village mayors, or voters could have rewarded village mayors perceived as politically close to the municipal mayor even absent recorded implementation participation. More generally, assignment may have increased the political value of being associated with the municipal mayor’s local network. At the same time, baseline evidence suggests caution in overstating this channel: fewer than half of respondents in the KALAH! baseline survey said they knew the municipal mayor or anyone in the municipal government closely, and focus groups described municipal officials as relatively distant from many barangays, with little communication between barangay and municipal government (Beatty et al., 2013). We therefore treat this as a plausible but limited threat. To probe this possibility, we use baseline support for the municipal winner as a pre-treatment proxy for whether a barangay belonged to the municipal mayor’s electoral base.

If a municipality-wide mayoral political channel affected village mayor outcomes directly, the treatment effect on village mayor electoral outcomes should be larger in barangays where the municipal winner was already strong before KALAH!. We therefore estimate heterogeneous first-stage and reduced-form regressions by interacting assignment with the municipal winner’s 2010 barangay vote share. The first stage shows whether treatment-induced implementation participation varies with baseline mayoral support. The reduced form then asks whether assignment also has a stronger electoral effect on village mayors in those same barangays, which would be more consistent with a direct municipality-wide political channel.

Constructing the municipal mayor vote share by barangay in 2010 We received the 2010 Statement of Votes by Precinct (SOVP) municipal mayor returns in PDF format from the Commission on Elections. These files do not report barangay identifiers directly, but only province, municipality, and clustered precincts (CPs). We therefore first scraped each SOVP PDF at the CP level and extracted both the CP label and the underlying precinct identifiers listed beneath it (for example, 0017A, 0017B, 0018A). We then parsed the 2016 precinct master file, which contains region, province, municipality, barangay, polling-place, and cluster identifiers, and normalised province and municipality names across the two sources. Next, we matched the SOVP data to the precinct master by linking the underlying precinct identifiers. On the precinct-master side, we decomposed each cluster into its constituent precinct tokens and mapped these tokens to barangays within municipality. On the SOVP side, we likewise decomposed each CP bundle into its constituent precinct tokens. We then matched SOVP tokens to the precinct master within province and municipality. A CP was assigned to a barangay when its matched precinct identifiers pointed uniquely to a single barangay. This token-level approach is useful because cluster definitions may differ between the 2010 SOVP and the 2016 precinct master even when the underlying precinct identifiers are the same. Under this procedure, 6,085 of 6,133 CPs (99.2 percent) map uniquely to one barangay. The remaining cases either split across multiple barangays or have no token

match and are therefore excluded from the final barangay-level aggregation rather than assigned mechanically. The scraping and matching scripts were developed with AI-assisted coding support (Codex, ChatGPT 5.4, Extra High), and we manually checked the resulting matches and ambiguous cases to verify the logic of the procedure.

Table A17: Municipal Political Channel: First-Stage Heterogeneity in Implementation Participation

	Implementation participation
Treatment	0.698*** (0.044)
Baseline mayoral support	-0.044** (0.018)
Treatment x mayoral support	0.078*** (0.027)
Observations	4,537
Matched-pair fixed effects	Yes
Municipality-clustered SE	Yes

^a Notes. The dependent variable is recorded implementation participation. Baseline mayoral support is the 2010 barangay vote share of the municipal winner, standardized to mean zero and unit variance. The analysis uses one observation per village, includes matched-pair fixed effects, and clusters standard errors by municipality. Mayoral support is observed for 4,537 villages in 187 municipalities.

Results We examine both the first stage and the reduced form because they answer different questions. The first stage shows whether treatment-induced implementation participation varies systematically with baseline mayoral support. The reduced form then asks whether assignment also has a stronger electoral effect on village mayors in these same barangays, which would be more consistent with a municipality-wide political channel that bypasses recorded participation. Table A17 shows that treatment strongly increases implementation participation, and that this first-stage effect is somewhat larger in barangays with higher baseline support for the municipal winner in 2010. Thus, villages in municipal-mayor strongholds appear more likely to participate when their municipality is assigned to KALAH. However, Table A18 shows no corresponding heterogeneity in village mayor vote share: the interaction between treatment and baseline mayoral support is small and statistically indistinguishable from zero for the exact-incumbency vote-share measure. This pattern is important for the exclusion restriction. If assignment affected village mayor outcomes through a municipality-wide political channel, such as mayoral credit claiming or the greater political value of being seen as close to the municipal mayor’s local network, one would

expect larger reduced-form effects precisely in barangays where the municipal winner was already electorally strong. We do not observe such a pattern. The evidence therefore suggests that baseline mayoral support shapes the first stage, but does not generate a direct electoral effect of assignment on village mayors independent of recorded implementation participation.

Table A18: Municipal Political Channel: Reduced-Form Heterogeneity in Village Mayor Outcomes

	Re-election	Running	Vote share
Treatment	-0.028*** (0.011)	-0.028*** (0.010)	0.010 (0.009)
Baseline mayoral support	-0.035*** (0.011)	-0.045*** (0.011)	0.011 (0.008)
Treatment x mayoral support	0.026 (0.016)	0.018 (0.016)	0.003 (0.012)
Observations	4,537	4,537	2,999
Matched-pair fixed effects	Yes	Yes	Yes
Municipality-clustered SE	Yes	Yes	Yes

^a Notes. Each column reports reduced-form treatment heterogeneity by baseline mayoral support, measured as the 2010 barangay vote share of the municipal winner and standardized to mean zero and unit variance. The analysis uses one observation per village. Re-election and running are observed for all villages in the estimation sample. Vote share is observed only among incumbents who ran and have vote totals. All specifications include matched-pair fixed effects and cluster standard errors by municipality.

I.4.4 Changes in Spending

Assignment to KALAHÍ could also affect village mayors' electoral outcomes through changes in municipal spending. Municipalities might substitute KALAHÍ resources for other expenditures or redirect spending across villages. We examine this possibility using municipality-level regressions of per-capita expenditure categories on assignment for 2011, 2012, and 2013, including matched-pair fixed effects. Table A19 reports the estimates.

The estimates indicate lower public-services, education, and health spending in 2011 and lower health spending in 2012. None of the differences is statistically distinguishable from zero in 2013. These expenditures cover calendar years, making the timing important. The municipal lotteries took place in May–June 2011, whereas the baseline report dates the first implementation cycle to spring/summer 2012. None of the sampled barangays had begun receiving project funding when interviewed between April and July 2012 (Beatty et al., 2013, 17, 26, 65). The 2011 differences therefore should not be interpreted as spending displaced by projects already financed through

Table A19: Municipal Budget Allocation by Treatment Status, 2011–2013

Outcome	(1) Public Services	(2) Welfare	(3) Housing	(4) Education	(5) Health	(6) Labor	(7) Economic
2011	-337.7* (197.1)	-45.01 (29.21)	9.028 (8.643)	-21.23** (9.985)	-47.44* (24.41)	0.0364 (0.226)	132.7 (94.11)
2012	-306.4 (201.4)	-32.76 (30.76)	8.338 (9.310)	-24.83 (15.35)	-48.06* (26.91)	1.009 (0.641)	82.61 (84.28)
2013	-385.2 (263.8)	-1.094 (30.11)	4.186 (11.05)	-14.24 (11.01)	-44.68 (29.65)	-1.012 (2.074)	122.1 (84.94)
Observations	197	197	197	197	197	197	197

^a Notes. Entries report treatment coefficients from municipality-level OLS regressions of per-capita expenditure categories on an indicator for assignment to KALAHÍ, estimated separately by year. All regressions include matched-pair fixed effects. Robust standard errors are reported in parentheses. Categories follow the municipal budget classifications used in the earlier budget-composition checks. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

this rollout of KALAHÍ. Nevertheless, 2011 includes several months after assignment, during which municipalities could have adjusted spending in anticipation of future grants or counterpart requirements. Nor do the municipal totals establish whether service provision declined, which villages were affected, or whether these changes influenced village mayors' electoral outcomes. Changes in municipal spending remain relevant to exclusion if they affected electoral outcomes independently of recorded participation, but the table alone does not establish such a pathway.

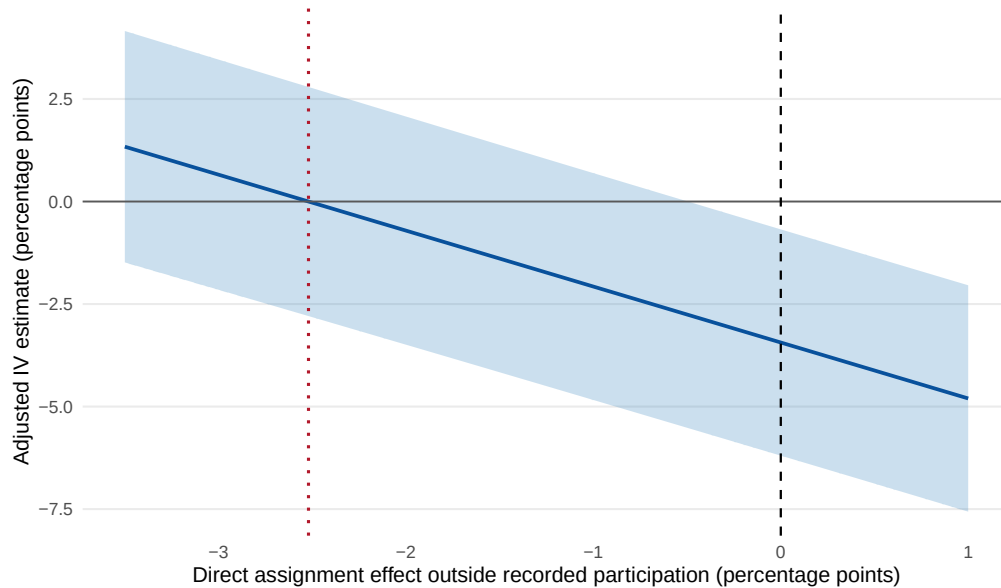
I.4.5 Sensitivity to a Direct Effect of Assignment

The analyses above provide evidence about possible violations of exclusion, but do not establish their absence. We therefore examine how large an effect of assignment outside recorded participation would have to be to change our conclusions. Following [Conley et al. \(2012\)](#) and the implementation in [Felton and Stewart \(2024\)](#), we let θ represent a hypothetical additive effect of assignment on reelection that does not operate through recorded participation. This effect could arise from the lottery announcement, early village activities omitted from our measure, or municipality-wide political and fiscal changes. For each value of θ , we construct the adjusted outcome $Y_{im} - \theta Z_m$ and re-estimate the preferred IV specification. All models retain the covariates, matched-pair fixed effects, and municipality-clustered standard errors with the Stata-style correction used in the main analysis.

Figure [A15](#) shows the adjusted estimates and their 95 percent confidence intervals. With $\theta = 0$, the estimated participation effect is -3.44 percentage points, with a confidence interval of $[-6.20, -0.68]$. An adverse direct effect of assignment of approximately 0.50 percentage points

would make this interval include zero. The estimate would remain statistically significant at the 10 percent level until the adverse direct effect reached approximately 0.82 percentage points. Explaining the point estimate entirely would require a direct effect of -2.52 percentage points, equal to the reduced-form estimate. The analysis thus distinguishes violations sufficient to remove statistical significance from those sufficient to account for the full estimated effect.

Figure A15: Exclusion-Restriction Sensitivity for Incumbent Reelection



Notes: The horizontal axis gives the hypothesised direct effect of assignment on reelection outside recorded implementation participation. The solid line reports the IV estimate after subtracting that direct effect from the outcome. The shaded band is the 95 percent municipality-clustered 2SLS confidence interval using the Stata-style correction. The black dashed line imposes the exclusion restriction. The red dotted line marks the direct effect that accounts for the full reduced-form point estimate. All models include matched-pair fixed effects and the full set of covariates.

Whether effects of these magnitudes are plausible depends on the pathway. For the lottery announcement, uneven awareness of KALAHİ roughly one year after assignment provides some reason to doubt a widespread electoral response to municipal selection alone. This evidence is indirect, however. It neither measures awareness of the lottery specifically nor rules out responses among citizens or incumbents who knew the result.

Early village activity presents a different concern. Most baseline attendees had participated only once, usually in an information meeting, and financial contributions were uncommon (Beatty et al., 2013). Compared with villages that prepared proposals but failed to secure funding, villages that stopped after these initial activities may have incurred fewer costs and developed less concrete expectations of receiving a project. This provides a substantive reason to expect smaller adverse responses to limited early participation. The relevant comparison is with unsuccessful applicants, since funded participants also received benefits that could offset adverse responses to implementation. Nevertheless, missing records do not establish that every village without recorded participation experienced only limited activity.

These considerations make the lottery announcement or limited early activity alone less convincing explanations for the full negative estimate. They do not establish that either effect falls below the smaller thresholds required to weaken statistical significance, nor do they rule out several omitted pathways operating together. The sensitivity analysis shows how the interpretation changes under specified violations of exclusion. It does not estimate those violations or determine their source.

I.5 Profiling Compliers and Noncompliers

The IV estimate for implementation participation identifies a Complier Average Treatment Effect (CATE) for villages whose recorded participation was changed by municipal assignment. We profile these villages to clarify the population represented by the CATE and how assignment changed the composition of participating villages. Compliers participate when assigned to KALAHÍ but would not participate otherwise. Always-takers would participate under either assignment, while never-takers would participate under neither. Under random assignment and monotonicity, participating control villages reveal the characteristics of always-takers and nonparticipating treated villages reveal those of never-takers. Because the full experimental sample is a mixture of these groups and compliers, the complier mean can then be recovered after accounting for the estimated always-taker and never-taker contributions ([Abadie, 2003](#); [Marbach and Hangartner, 2020](#)).

This exercise serves two purposes. The comparison between always-takers and never-takers describes selection into recorded implementation participation, while the characteristics of compliers show which villages assignment added to the participating group ([Bowles and Larreguy, 2025](#)). The profiles also put the local IV estimate in context by showing how compliers compare with the experimental sample. Profiling the strata relies on random assignment and monotonicity, but it does not require the outcome exclusion restriction. It therefore does not validate that restriction, explain the estimated electoral effect, or establish that the CATE generalises to noncompliers ([Angrist and Fernandez-Val, 2013](#); [Marbach and Hangartner, 2020](#)).

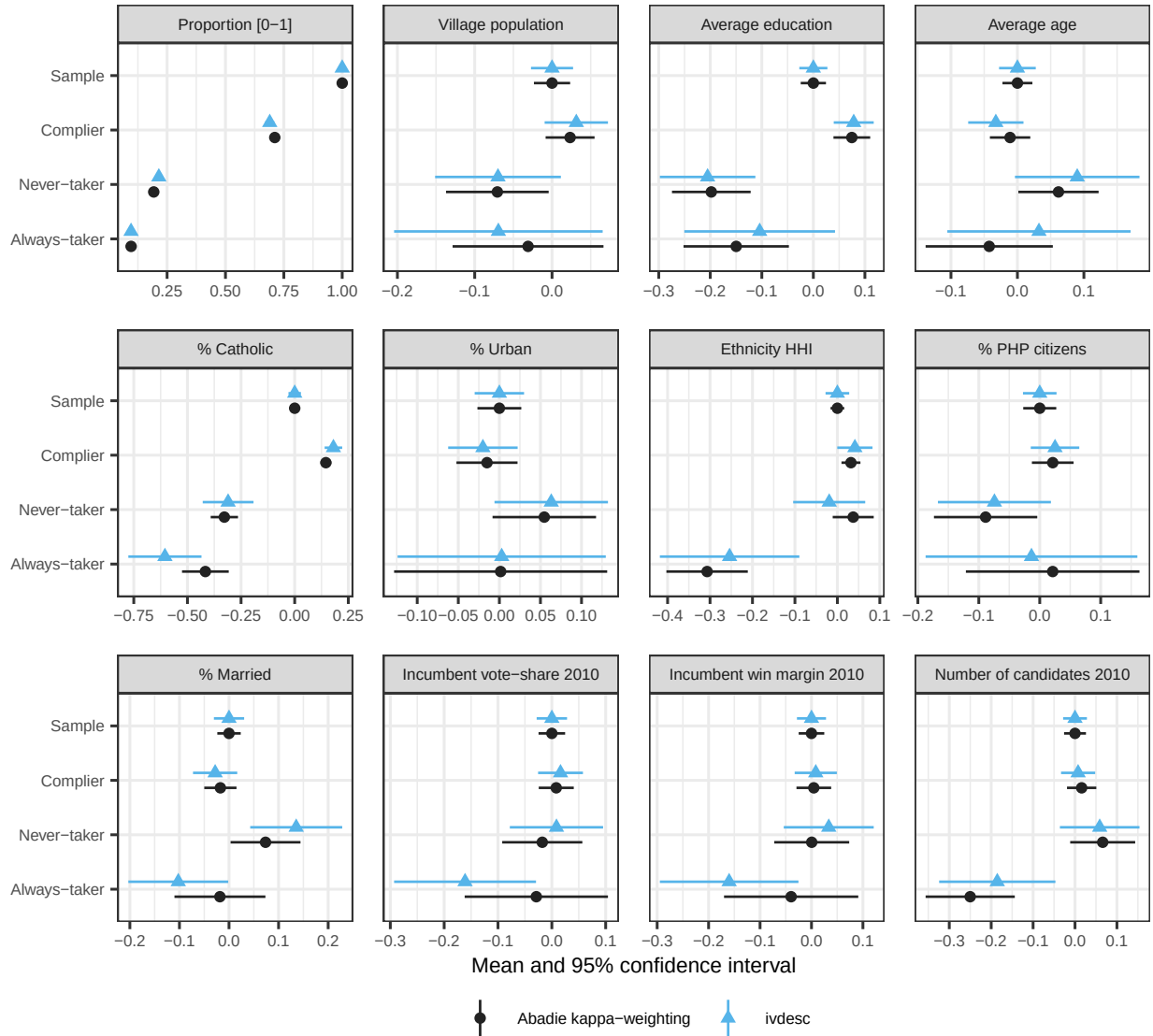
Our preferred implementation follows the κ -weighting procedure of [Abadie \(2003\)](#) used by [Bowles and Larreguy \(2025\)](#). We first standardise each predetermined covariate to have mean zero and unit standard deviation in the full experimental sample. For each covariate, we estimate the probability of assignment using a probit model containing the covariate's linear, quadratic, and cubic terms and matched-pair fixed effects. The estimated assignment probabilities produce κ weights for the three compliance strata, which we use to calculate their mean characteristics. We estimate uncertainty with 500 bootstrap draws that resample villages within matched-pair, assignment, and observed-participation cells. The assignment model and all weighted means are re-estimated in every draw.

As a benchmark, we also report the direct decomposition implemented by `ivdesc` ([Marbach and Hangartner, 2020](#)). This procedure estimates the stratum shares from the observed assignment and participation rates and recovers the complier mean from the corresponding mixture identity. Its confidence intervals use 500 ordinary village-bootstrap draws. The two approaches implement the same principal-strata logic but differ in their adjustment for the matched-pair design and in their resampling procedures.

Figure [A16](#) and the accompanying table show that the two procedures yield similar estimates. The preferred κ -weighted estimates classify 71% of villages as compliers, 19% as never-takers, and 10% as always-takers. Complier villages are larger and more educated than never-taker villages, and they are more educated than always-taker villages. They also have a larger Catholic share

than either group. By contrast, prior incumbent vote share and winning margin are not statistically distinguishable across the compliance strata. The remaining differences vary across comparison groups and should be interpreted descriptively given the number of characteristics examined.

Figure A16: Compliance Profiles for Implementation Participation



Notes: Black circles report subgroup means obtained with the κ -weighting approach of [Abadie \(2003\)](#) and [Angrist and Fernandez-Val \(2013\)](#), with 95% confidence intervals based on 500 conditional village-bootstrap draws within matched-pair, assignment, and observed-participation cells. Blue triangles report the *ivdesc* estimates and 95% confidence intervals from 500 ordinary village-bootstrap draws. The first panel reports estimated stratum shares. All remaining characteristics are standardised to mean zero and unit standard deviation in the full experimental sample.

Table A20: Standardized covariate means by implementation-participation compliance stratum. Parentheses contain standard errors from 500 bootstrap draws. Abadie kappa-weighting uses a conditional village bootstrap within matched-pair, assignment, and observed-participation cells. ivdesc uses an unconditional village bootstrap.

Variable	Method	Always-taker	Never-taker	Complier	Sample
Proportion [0-1]	Abadie kappa-weighting	0.10 (0.00)	0.19 (0.00)	0.71 (0.00)	1.00 (0.00)
	ivdesc	0.10 (0.01)	0.21 (0.01)	0.69 (0.01)	1.00 (0.00)
Village population	Abadie kappa-weighting	-0.03 (0.05)	-0.07 (0.03)	0.02 (0.02)	0.00 (0.01)
	ivdesc	-0.07 (0.07)	-0.07 (0.04)	0.03 (0.02)	0.00 (0.01)
Average education	Abadie kappa-weighting	-0.15 (0.05)	-0.20 (0.04)	0.07 (0.02)	-0.00 (0.01)
	ivdesc	-0.10 (0.07)	-0.21 (0.05)	0.08 (0.02)	-0.00 (0.01)
Average age	Abadie kappa-weighting	-0.04 (0.05)	0.06 (0.03)	-0.01 (0.02)	-0.00 (0.01)
	ivdesc	0.03 (0.07)	0.09 (0.05)	-0.03 (0.02)	-0.00 (0.01)
% Catholic	Abadie kappa-weighting	-0.42 (0.06)	-0.33 (0.03)	0.15 (0.01)	0.00 (0.01)
	ivdesc	-0.60 (0.09)	-0.31 (0.06)	0.18 (0.02)	0.00 (0.01)
% Urban	Abadie kappa-weighting	0.00 (0.07)	0.05 (0.03)	-0.02 (0.02)	0.00 (0.01)
	ivdesc	0.00 (0.06)	0.06 (0.04)	-0.02 (0.02)	0.00 (0.02)
Ethnicity HHI	Abadie kappa-weighting	-0.31 (0.05)	0.04 (0.02)	0.03 (0.01)	-0.00 (0.01)
	ivdesc	-0.25 (0.08)	-0.02 (0.04)	0.04 (0.02)	-0.00 (0.01)
% PHP citizens	Abadie kappa-weighting	0.02 (0.07)	-0.09 (0.04)	0.02 (0.02)	0.00 (0.01)
	ivdesc	-0.01 (0.09)	-0.07 (0.05)	0.03 (0.02)	0.00 (0.01)
% Married	Abadie kappa-weighting	-0.02 (0.05)	0.07 (0.04)	-0.02 (0.02)	0.00 (0.01)
	ivdesc	-0.10 (0.05)	0.14 (0.05)	-0.03 (0.02)	0.00 (0.02)
Incumbent vote-share 2010	Abadie kappa-weighting	-0.03 (0.07)	-0.02 (0.04)	0.01 (0.02)	-0.00 (0.01)
	ivdesc	-0.16 (0.07)	0.01 (0.04)	0.02 (0.02)	-0.00 (0.01)
Incumbent win margin 2010	Abadie kappa-weighting	-0.04 (0.07)	0.00 (0.04)	0.00 (0.02)	0.00 (0.01)
	ivdesc	-0.16 (0.07)	0.03 (0.04)	0.01 (0.02)	0.00 (0.01)
Number of candidates 2010	Abadie kappa-weighting	-0.25 (0.05)	0.07 (0.04)	0.02 (0.02)	-0.00 (0.01)
	ivdesc	-0.19 (0.07)	0.06 (0.05)	0.01 (0.02)	-0.00 (0.01)

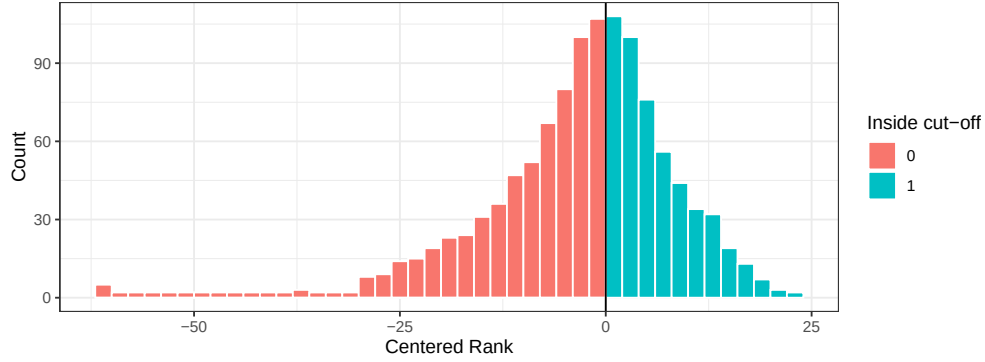
J Further Results: Effect of Funding (RDD)

J.1 RDD: Checking Assumption

J.1.1 Density test

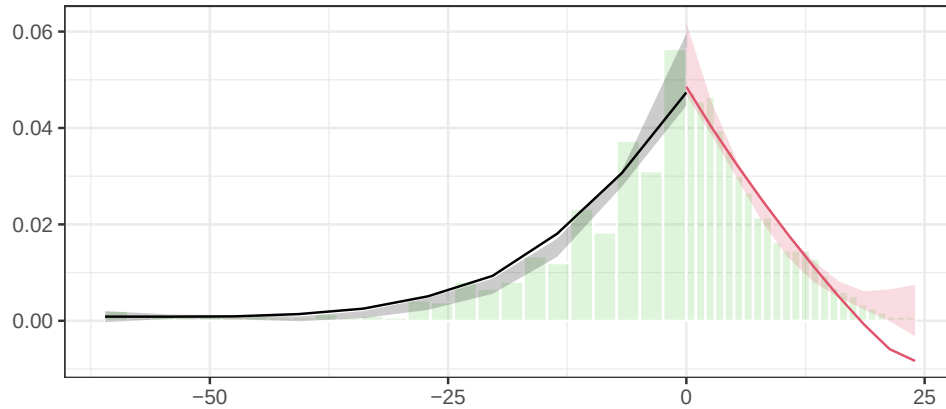
Figure A17 displays the histogram of running variable. From visual inspection, we can see that the distribution looks overall symmetric.

Figure A17: Histogram Running Variable, Full Sample.



To test this notion more formally, Figure A18 shows the estimated density of the centred project rank, $\tilde{r}_{im}$, using a local-polynomial density estimator (jackknife VCE). The solid lines on either side of the cut-off at $\tilde{r} = 0$ represent separate quadratic fits to the histogram bars. The near-perfect overlap at the cut-off indicates no evidence of systematic bunching or manipulation of the rank around the funding threshold.

Figure A18: Discrete-adjusted density test for rank at the funding cut-off.



To test for a discontinuity across different bandwidths, Table A21 reports the discrete-adjusted density-continuity test of Cattaneo et al. (2020) for the centred rank Z_{im} . For each bandwidth—either estimated by the MSE rule or fixed at ± 5 , ± 10 , and ± 20 rank points—the table displays the total effective sample size (sum of the “effective” observations on either side of the cut-off),

the jackknife-robust test statistic, and the associated p -value. All four tests fail to reject density continuity at the 10 percent level, with a global p -value of 0.943. We therefore find no statistically detectable discontinuity at the funding cutoff. Because the running variable is an ordinal rank centered separately within each municipality-cycle, this result does not by itself rule out strategic coordination.

Table A21: Density-discontinuity tests for Z_{im} at the funding cut-off

Window	Bandwidth	Effective N	T-stat	p-value
Global (h est.)	23	1094	0.072	0.943
Fixed ± 5	5	536	0.714	0.475
Fixed ± 10	10	816	-0.236	0.813
Fixed ± 20	20	1066	0.051	0.960

J.1.2 Covariate Balance P-Values for Discontinuity

Table A22 separates the balance checks into two panels. Panel A reports fifteen baseline village, municipal, and political characteristics and the funding-cycle indicator. Panel B reports the KC grant amount, local counterpart contribution, and total project cost, treating the ranking-file amounts as pre-funding proposal characteristics. The source does not separately verify when these amounts were fixed. Figures A19 and A20 present the same checks, ordered by p-value within each panel. Each entry is the clustered p-value on the funded-side indicator from a proposal-level local-linear regression with separate slopes on either side of the midpoint-centered cutoff. We hold the window fixed across characteristics, using the 24.42-rank window selected by the preferred RDHonest vote-share specification and the 11.45-rank window selected by the preferred collapsed support-point specification.

Among the characteristics in Panel A, the number of candidates differs in the RDHonest window ($p = 0.045$), while the married share differs in the collapsed-RD window ($p = 0.049$). Neither difference is present in the other estimator's window. Catholic share and prior incumbent vote share also differ at the 10 percent level in the RDHonest window, with p-values of 0.066 and 0.079. The discontinuities are coded as funded minus unfunded. At the fitted cutoff, the RDHonest-window regressions imply that funded-side proposals come from villages with 2.14 rather than 2.30 prior candidates and a prior incumbent vote share of 62.3 rather than 59.2 percent. The fitted Catholic share is 88.4 percent on the funded side and 85.0 percent on the unfunded side. In the collapsed-RD window, the fitted married share is 34.8 percent on the funded side and 34.0 percent on the unfunded side. Each fitted difference is approximately 0.18–0.20 standard deviations.

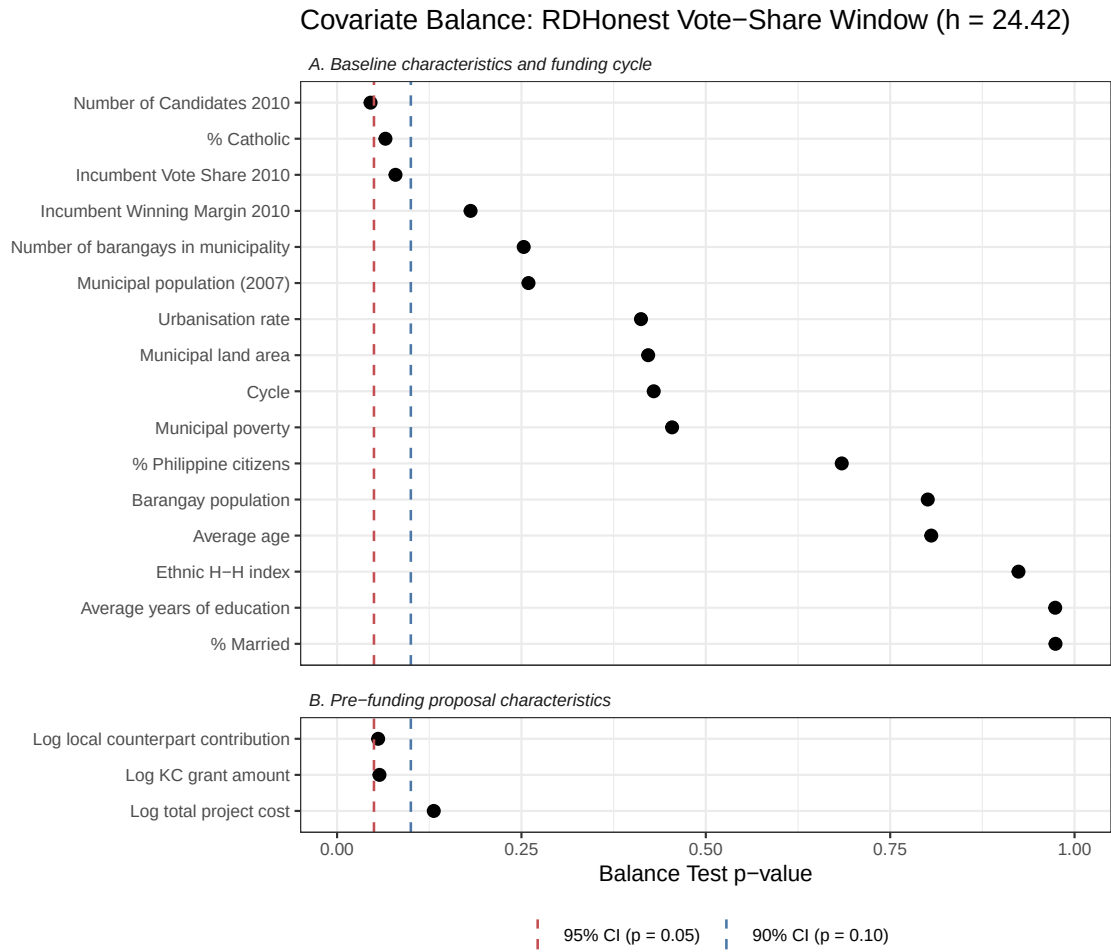
These quantities are fitted discontinuities, not simple differences between the two nearest ranks. At the first unfunded and last funded support points, $X = -0.5$ and $X = 0.5$, prior incumbent vote shares are 59.1 and 58.1 percent, and the numbers of prior candidates are 2.22 and 2.31. These immediate comparisons therefore run in the opposite direction from the fitted differences in the RDHonest window. The corresponding Catholic shares are 87.2 and 85.7 percent, while the married shares are nearly identical at 33.8 and 33.9 percent. The estimated discontinuities consequently reflect the fitted linear trends across the estimator-specific windows rather than a uniformly visible gap between the closest support points. Moreover, all four flagged characteristics enter the preferred covariate-adjusted specification, and the vote-share estimate remains close to five percentage points after adjustment. Overall, two of the 32 predetermined-covariate tests reject at the 5 percent level, the rejections do not recur between the RDHonest and collapsed-RD windows, and the nearest-point comparisons do not display a consistent pattern. We therefore interpret the balance evidence as providing no systematic indication of sorting rather than as demonstrating exact balance.

Table A22: Balance at the Funding Boundary

Characteristic	RDHonest window $h = 24.42$		Collapsed-RD window $h = 11.45$	
	p -value	N	p -value	N
<i>A. Baseline characteristics and funding cycle</i>				
Barangay population	0.801	1,095	0.390	833
Average years of education	0.974	1,095	0.158	833
Average age	0.806	1,095	0.176	833
Urbanisation rate	0.412	1,095	0.219	833
Ethnic H-H index	0.924	1,095	0.346	833
% Philippine citizens	0.684	1,095	0.662	833
% Married	0.974	1,095	0.049	833
% Catholic	0.066	1,095	0.248	833
Incumbent Vote Share 2010	0.079	1,095	0.575	833
Incumbent Winning Margin 2010	0.181	1,095	0.744	833
Number of Candidates 2010	0.045	1,095	0.738	833
Municipal poverty	0.454	1,095	0.228	833
Number of barangays in municipality	0.253	1,095	0.355	833
Municipal land area	0.422	1,095	0.542	833
Municipal population (2007)	0.260	1,095	0.447	833
Cycle	0.429	1,095	0.163	833
<i>B. Pre-funding proposal characteristics</i>				
Log KC grant amount	0.057	1,092	0.009	830
Log local counterpart contribution	0.056	1,087	0.084	829
Log total project cost	0.131	1,092	0.028	830

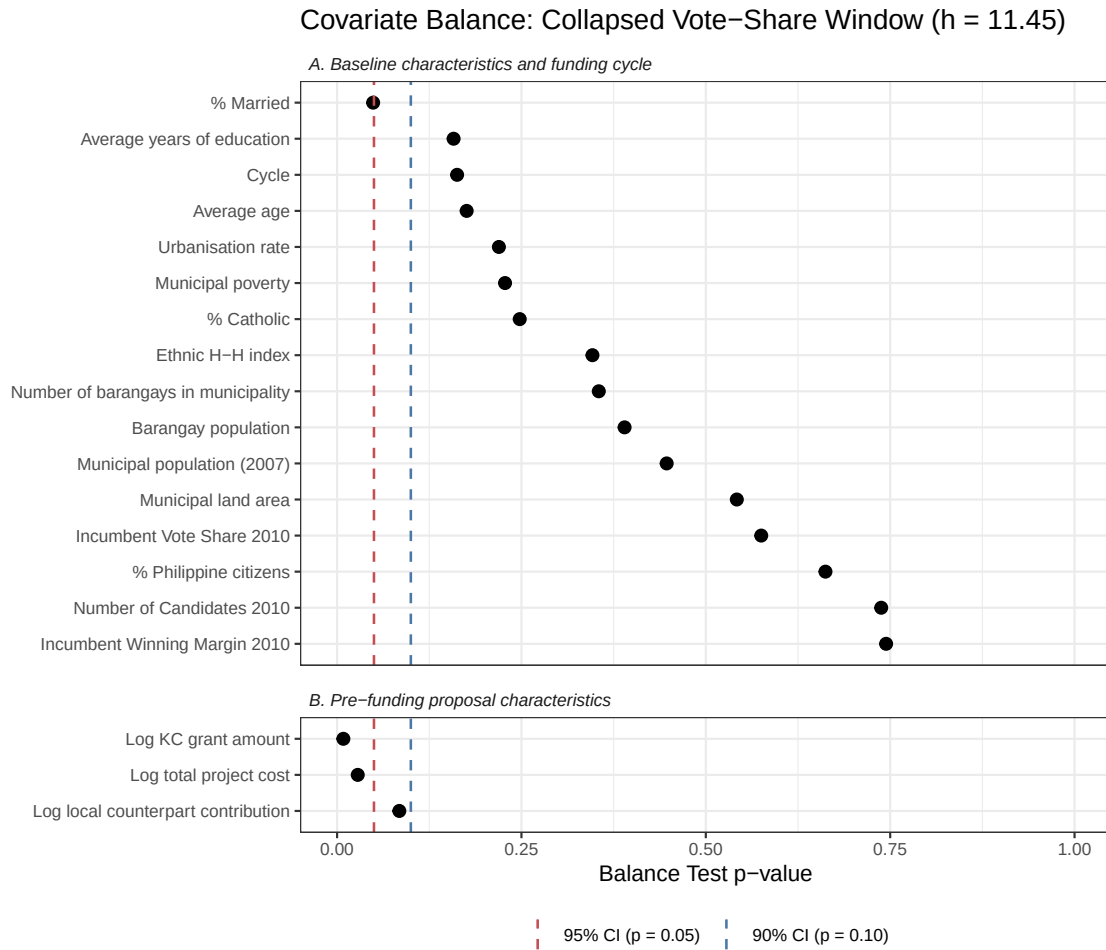
Notes: Entries are p -values for the funded-side indicator in local-linear regressions with separate slopes on either side of the midpoint-centered cutoff. Each window is selected by the corresponding preferred vote-share specification and held fixed across characteristics. Regressions use equal weights and municipality–cycle clustered standard errors. N is the number of proposals with nonmissing data. Values below 0.05 are bold. Panel A retains the funding-cycle indicator alongside baseline village and municipal characteristics. Panel B uses $\log(1 + x)$ of the three ranking-file amounts, treating them as pre-funding proposal characteristics. The source does not separately verify when these amounts were fixed. Actual post-funding costs are not included.

Figure A19: Covariate Balance Using the RDHonest Vote-Share Window.



Notes: Panel A reports baseline characteristics and the funding-cycle indicator. Panel B reports the three ranking-file amounts under the pre-funding interpretation described in Table A22. Points report p-values from local-linear regressions with equal weights and separate slopes on either side of the cutoff. Standard errors are clustered at the municipality-cycle level. Dashed lines mark the 5 and 10 percent significance thresholds.

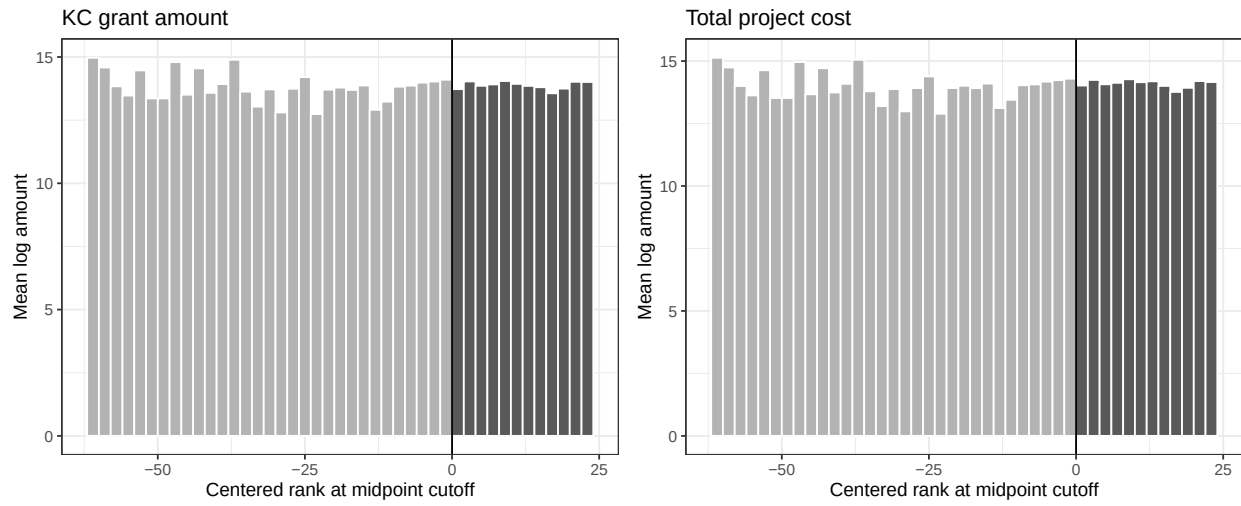
Figure A20: Covariate Balance Using the Collapsed Vote-Share Window.



Notes: Panel A reports baseline characteristics and the funding-cycle indicator. Panel B reports the three ranking-file amounts under the pre-funding interpretation described in Table A22. Points report p-values from local-linear regressions with equal weights and separate slopes on either side of the cutoff. Standard errors are clustered at the municipality-cycle level. Dashed lines mark the 5 and 10 percent significance thresholds.

To unpack the project-cost diagnostics in Figures A19 and A20, Figure A21 plots the KC grant amount and total project cost along the midpoint-centered running variable. Unfunded proposals have higher mean recorded costs immediately outside the cutoff. In the 11.45-rank collapsed-RD window, the estimated discontinuities, defined as funded minus unfunded, are -0.395 log points for the KC grant amount and -0.318 log points for total project cost, with p-values of 0.009 and 0.028. In the 24.42-rank RDHonest window, the corresponding estimates are -0.227 and -0.172 log points, with p-values of 0.057 and 0.131. The local counterpart contribution is positive and differs at the 10 percent level in both estimator-specific windows, but not at the 5 percent level. These comparisons indicate lower recorded KC grant amounts near the funding boundary, with the strongest differences in the collapsed-RD window.

Figure A21: Recorded Project Amounts Along the Funding-Rank Running Variable.



Notes: Bars report binned means of logged amounts from the ranking records within two-rank bins, using the midpoint-centered running variable. The left panel plots the KC grant amount, and the right panel plots total project cost. Light bars are outside the funding cutoff and dark bars are inside the cutoff. The vertical line marks the policy threshold at $X = 0$.

J.2 RDD Main Results, Details

Table A23: Reelection

Pr(reelection)	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: No donut</i>					
Treatment effect	0.086 (0.072)	0.075 (0.066)	0.075 (0.065)	0.084 (0.069)	0.084 (0.070)
Observations	1162	1162	1162	1162	1162
Effective observations	654	699	693	666	654
Optimal bandwidth	9.429	10.767	10.676	9.836	9.648
Smoothness bound M	0.0049	0.0037	0.0037	0.0044	0.0046
<i>Panel B: Donut</i>					
Treatment effect	0.240** (0.089)	0.194* (0.081)	0.197* (0.083)	0.230** (0.086)	0.228** (0.087)
Observations	1053	1053	1053	1053	1053
Effective observations	565	598	608	557	575
Optimal bandwidth	10.186	11.579	11.260	10.533	10.277
Smoothness bound M	0.0049	0.0037	0.0037	0.0044	0.0046
Fixed Effects		✓	✓	✓	✓
Clustered SEs			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. This table reports the preferred/default RDHonest estimates for the outcome 'Reelection'. The smoothness bound uses the published left-side quadratic rule with the 4x multiplier. All estimates use local-linear RDHonest with triangular kernel, Holder smoothness class, and FLCI-optimal bandwidths. Panel B drops the two mass points adjacent to the cutoff. Panel B holds M fixed at the corresponding full-sample calibration in Panel A, so only the estimation sample changes. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors, Model (4) adds census controls, and Model (5) adds political controls. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A24: Reelection: Smoothness-bound sensitivity in Model (5)

Smoothness rule	No donut		Donut	
	M	Estimate (SE)	M	Estimate (SE)
1x	0.0012	0.092 (0.050)	0.0012	0.162** (0.057)
2x	0.0023	0.088 (0.059)	0.0023	0.185** (0.070)
4x (preferred)	0.0046	0.084 (0.070)	0.0046	0.228** (0.087)
8x	0.0093	0.065 (0.081)	0.0093	0.277* (0.111)
16x	0.0185	0.031 (0.087)	0.0185	0.320 (0.138)

^a Notes. This table varies the smoothness bound for the preferred Model (5) RDHonest specification. The 4x row is the preferred/default left-side quadratic calibration. Within each row, the donut estimate uses the same full-sample value of M as the no-donut estimate. Cells report estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A25: Run for reelection

Pr(run)	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: No donut</i>					
Treatment effect	0.083 (0.079)	0.083 (0.069)	0.083 (0.076)	0.102 (0.079)	0.106 (0.080)
Observations	1162	1162	1162	1162	1162
Effective observations	550	581	577	556	559
Optimal bandwidth	7.163	7.848	7.793	7.246	7.315
Smoothness bound M	0.0084	0.0067	0.0067	0.0080	0.0079
<i>Panel B: Donut</i>					
Treatment effect	0.207 (0.100)	0.179 (0.085)	0.183 (0.087)	0.225* (0.087)	0.232** (0.084)
Observations	1053	1053	1053	1053	1053
Effective observations	474	495	485	474	479
Optimal bandwidth	8.030	8.748	8.651	8.042	8.118
Smoothness bound M	0.0084	0.0067	0.0067	0.0080	0.0079
Fixed Effects		✓	✓	✓	✓
Clustered SEs			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. This table reports the preferred/default RDHonest estimates for the outcome 'Run for reelection'. The smoothness bound uses the published left-side quadratic rule with the 4x multiplier. All estimates use local-linear RDHonest with triangular kernel, Holder smoothness class, and FLCI-optimal bandwidths. Panel B drops the two mass points adjacent to the cutoff. Panel B holds M fixed at the corresponding full-sample calibration in Panel A, so only the estimation sample changes. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors, Model (4) adds census controls, and Model (5) adds political controls. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A26: Run for reelection: Smoothness-bound sensitivity in Model (5)

Smoothness rule	No donut		Donut	
	M	Estimate (SE)	M	Estimate (SE)
1x	0.0020	0.097 (0.054)	0.0020	0.138* (0.058)
2x	0.0039	0.104 (0.063)	0.0039	0.178** (0.065)
4x (preferred)	0.0079	0.106 (0.080)	0.0079	0.232** (0.084)
8x	0.0157	0.101 (0.095)	0.0157	0.255 (0.102)
16x	0.0314	0.086 (0.106)	0.0314	0.364* (0.118)

^a Notes. This table varies the smoothness bound for the preferred Model (5) RDHonest specification. The 4x row is the preferred/default left-side quadratic calibration. Within each row, the donut estimate uses the same full-sample value of M as the no-donut estimate. Cells report estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A27: Vote share

Vote Share	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: No donut</i>					
Treatment effect	0.064*** (0.024)	0.059** (0.025)	0.059* (0.031)	0.060* (0.029)	0.049* (0.026)
Observations	780	780	780	780	780
Effective observations	732	653	646	640	665
Optimal bandwidth	61.500	22.247	21.768	21.615	24.415
Smoothness bound M	0.0000	0.0004	0.0004	0.0004	0.0003
<i>Panel B: Donut</i>					
Treatment effect	0.072*** (0.027)	0.065** (0.028)	0.065* (0.030)	0.070** (0.029)	0.056* (0.027)
Observations	713	713	713	713	713
Effective observations	666	587	581	574	600
Optimal bandwidth	61.500	23.106	22.237	21.916	25.199
Smoothness bound M	0.0000	0.0004	0.0004	0.0004	0.0003
Fixed Effects		✓	✓	✓	✓
Clustered SEs			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. This table reports the preferred/default RDHonest estimates for the outcome 'Vote share'. The smoothness bound uses the published left-side quadratic rule with the 4x multiplier. All estimates use local-linear RDHonest with triangular kernel, Holder smoothness class, and FLCI-optimal bandwidths. Panel B drops the two mass points adjacent to the cutoff. Panel B holds M fixed at the corresponding full-sample calibration in Panel A, so only the estimation sample changes. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors, Model (4) adds census controls, and Model (5) adds political controls. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A28: Vote share: Smoothness-bound sensitivity in Model (5)

Smoothness rule	No donut		Donut	
	M	Estimate (SE)	M	Estimate (SE)
1x	0.0001	0.045* (0.024)	0.0001	0.049* (0.024)
2x	0.0002	0.048* (0.024)	0.0002	0.053* (0.025)
4x (preferred)	0.0003	0.049* (0.026)	0.0003	0.056* (0.027)
8x	0.0006	0.051 (0.029)	0.0006	0.060 (0.031)
16x	0.0012	0.051 (0.032)	0.0012	0.063 (0.038)

^a Notes. This table varies the smoothness bound for the preferred Model (5) RDHonest specification. The 4x row is the preferred/default left-side quadratic calibration. Within each row, the donut estimate uses the same full-sample value of M as the no-donut estimate. Cells report estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A29: Number of candidates

No. candidates	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: No donut</i>					
Treatment effect	-0.172 (0.100)	-0.145* (0.072)	-0.144 (0.086)	-0.125* (0.073)	-0.120 (0.088)
Observations	1162	1162	1162	1162	1162
Effective observations	768	1034	1024	1111	985
Optimal bandwidth	12.919	28.407	27.556	61.500	24.009
Smoothness bound M	0.0036	0.0008	0.0008	0.0002	0.0009
<i>Panel B: Donut</i>					
Treatment effect	-0.254* (0.121)	-0.159* (0.082)	-0.159* (0.086)	-0.124* (0.074)	-0.114 (0.089)
Observations	1053	1053	1053	1053	1053
Effective observations	661	923	920	1004	887
Optimal bandwidth	13.631	28.751	28.209	61.500	25.100
Smoothness bound M	0.0036	0.0008	0.0008	0.0002	0.0009
Fixed Effects		✓	✓	✓	✓
Clustered SEs			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. This table reports the preferred/default RDHonest estimates for the outcome 'Number of candidates'. The smoothness bound uses the published left-side quadratic rule with the 4x multiplier. All estimates use local-linear RDHonest with triangular kernel, Holder smoothness class, and FLCI-optimal bandwidths. Panel B drops the two mass points adjacent to the cutoff. Panel B holds M fixed at the corresponding full-sample calibration in Panel A, so only the estimation sample changes. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors, Model (4) adds census controls, and Model (5) adds political controls. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A30: Number of candidates: Smoothness-bound sensitivity in Model (5)

Smoothness rule	No donut		Donut	
	M	Estimate (SE)	M	Estimate (SE)
1x	0.0002	-0.109 (0.073)	0.0002	-0.103 (0.074)
2x	0.0005	-0.113 (0.078)	0.0005	-0.104 (0.079)
4x (preferred)	0.0009	-0.120 (0.088)	0.0009	-0.114 (0.089)
8x	0.0018	-0.133 (0.097)	0.0018	-0.124 (0.101)
16x	0.0037	-0.144 (0.105)	0.0037	-0.147 (0.118)

^a Notes. This table varies the smoothness bound for the preferred Model (5) RDHonest specification. The 4x row is the preferred/default left-side quadratic calibration. Within each row, the donut estimate uses the same full-sample value of M as the no-donut estimate. Cells report estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

J.3 Robustness checks

This section examines whether the funding RD results depend on the two support points at the boundary, the estimation window, or the location of the cutoff.

J.3.1 Donut-hole RD

If actors strategically moved proposals across the funding line, such intervention should be most consequential for the last funded and first unfunded proposals because the smallest rank change can alter their funding status. Table [A31](#) probes this concern by replicating both panels of the main RD table after dropping these two support points. The RDHonest curvature bounds and the collapsed-RD bandwidths are held at their corresponding full-sample values, so the comparison changes the estimation sample rather than the tuning parameters. This exercise only addresses manipulation concentrated at the immediate boundary and requires extrapolation across a wider gap. We therefore interpret it as a diagnostic rather than as an alternative preferred design. The table note provides the full implementation details.

Table A31: Donut-Hole RDD Estimates of KALAH! Funding on Electoral Outcomes

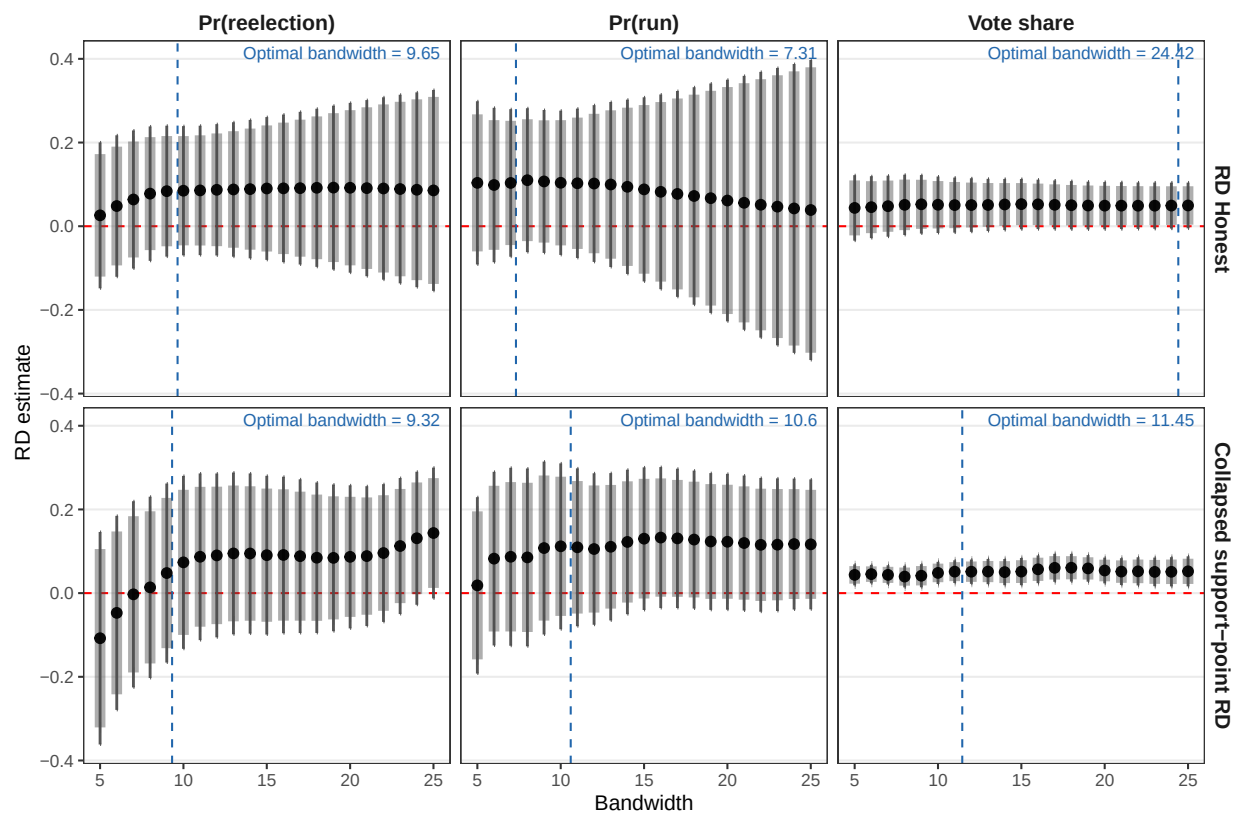
	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: RD Honest</i>					
Pr(reelection)	0.240** (0.089)	0.194* (0.081)	0.197* (0.083)	0.230** (0.086)	0.228** (0.087)
Pr(run)	0.207 (0.100)	0.179 (0.085)	0.183 (0.087)	0.225* (0.087)	0.232** (0.084)
Vote Share	0.072*** (0.027)	0.065** (0.028)	0.065* (0.030)	0.070** (0.029)	0.056* (0.027)
No. candidates	-0.254* (0.121)	-0.159* (0.082)	-0.159* (0.086)	-0.124* (0.074)	-0.114 (0.089)
<i>Panel B: Collapsed support-point RD</i>					
Pr(reelection)	0.306* (0.161)	0.312* (0.169)	0.312* (0.169)	0.330** (0.167)	0.324* (0.166)
Pr(run)	0.252** (0.124)	0.218* (0.131)	0.218* (0.131)	0.230* (0.135)	0.231* (0.137)
Vote Share	0.054 (0.046)	0.068 (0.042)	0.068 (0.042)	0.068* (0.039)	0.062 (0.041)
No. candidates	-0.212 (0.148)	-0.215 (0.136)	-0.215 (0.136)	-0.185 (0.152)	-0.200 (0.147)
Fixed Effects		✓	✓	✓	✓
Clustered SEs (Panel A)			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. This table replicates Table 3 after dropping the two mass points adjacent to the cutoff: the first unfunded proposal and the last funded proposal. The running variable is midpoint centered, so the omitted boundary pair has $|X| = 0.5$. Panel A reports RDHonest local-linear inference with a triangular kernel, Holder smoothness class, and FLCI-optimal bandwidth. For each outcome and model, Panel A holds M fixed at the corresponding full-sample calibration used in Table 3; only the estimation sample changes. That preferred/default bound fits a quadratic on the full-sample unfunded side within 25 ranks and multiplies the implied absolute second derivative by four. Panel B collapses the donut sample to one unweighted mean outcome for each remaining score value, then applies local-linear rdrobust inference to the collapsed support-point data, holding the main and bias-correction bandwidths fixed at their full-sample MSE-optimal values so that, as in Panel A, the donut comparison changes only the sample. For covariate-adjusted Panel B columns, the outcome is first adjusted on the raw donut sample using the listed predetermined controls and fixed effects, then collapsed to score-level means. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors in Panel A, Model (4) adds census controls, and Model (5) adds political controls. Panel B does not use clustered standard errors, so its Models (2) and (3) are identical. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

J.3.2 Bandwidth Sensitivity

Figure A22 assesses sensitivity to fixed bandwidths from 5 to 25 using the two discrete-score estimators reported in the main RD table. The upper row reports RDHonest estimates using the preferred covariates, municipality–cycle clustered EHW standard errors, and the maintained left-side quadratic curvature bound with the four-times multiplier. Thin and thick intervals denote 95 and 90 percent honest confidence intervals. The lower row first adjusts each outcome using the preferred controls, collapses the data to one unweighted mean for each score value, and then estimates a local-linear RD on the support-point data; its intervals are robust bias-corrected intervals from `rdrobust`. The dashed vertical lines report the estimator- and outcome-specific optimal bandwidths. Re-election and running estimates are positive across all bandwidths under RDHonest and across nearly all bandwidths in the collapsed analysis, but their 95 percent confidence intervals include zero throughout. Vote-share estimates are positive throughout. All 21 collapsed support-point estimates are statistically distinguishable from zero at the 5 percent level, 12 of the 21 RDHonest estimates are distinguishable from zero at the 10 percent level. Thus, the bandwidth exercise reinforces the main pattern: the evidence is consistently positive but substantially more precise for conditional vote share than for re-election or running.

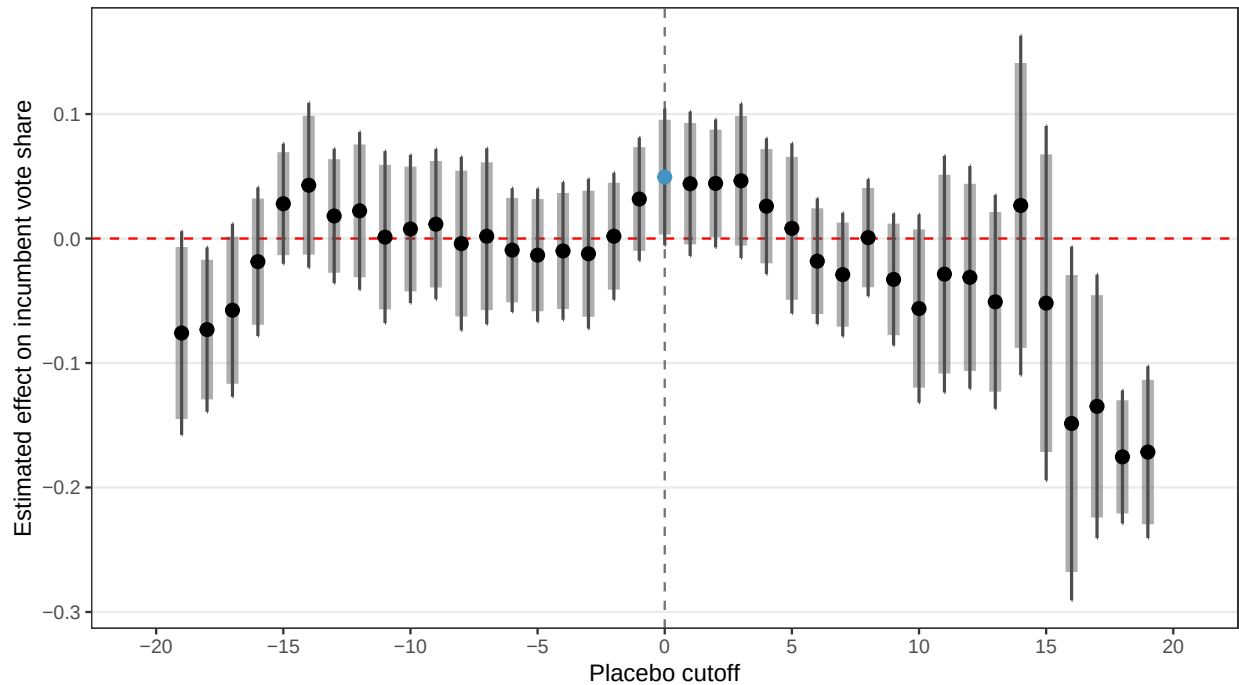
Figure A22: Bandwidth Robustness of RD Estimates



J.3.3 Placebo Cutoffs

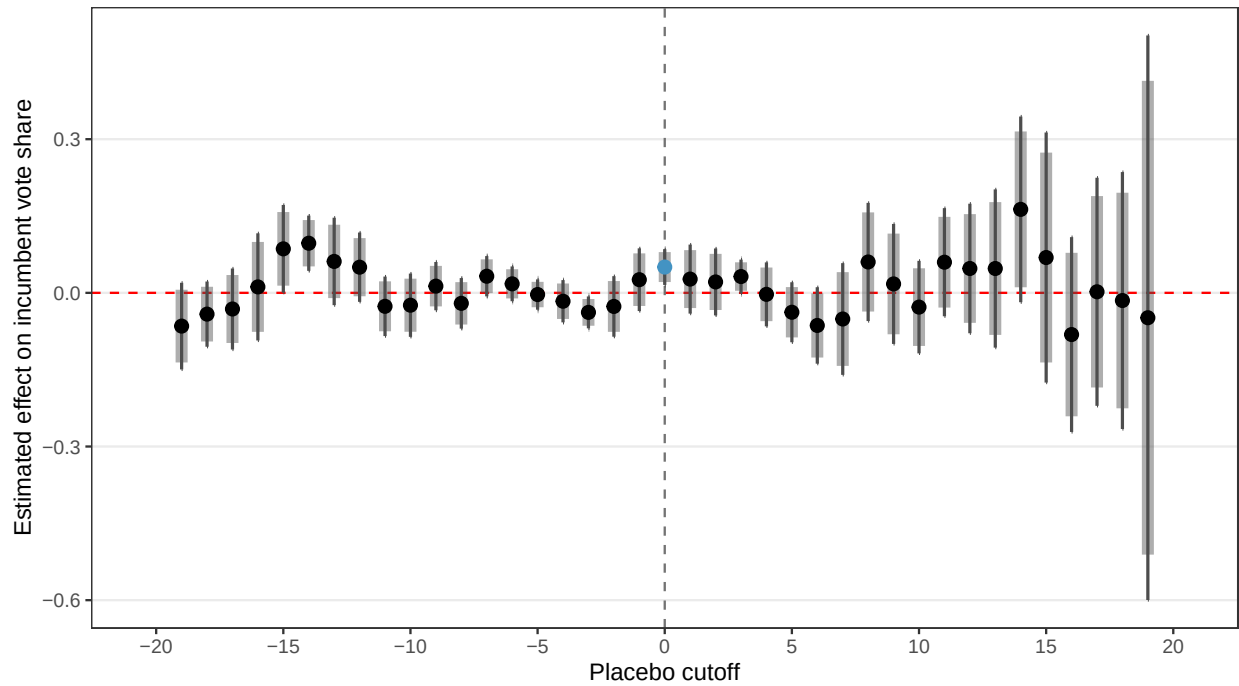
Figures A23 and A24 re-estimate the preferred conditional vote-share specification at every integer cutoff from -19 to 19 along the discrete rank score. Each estimate uses the complete clean sharp-RD sample, and the estimate at zero corresponds to the true funding cutoff.

Figure A23: Placebo Cutoffs: RD Honest



Notes: Points report Model (5) local-linear RDHonest estimates for incumbent vote share conditional on running. Thin intervals are 95 percent honest confidence intervals and thick intervals are 90 percent honest confidence intervals. The curvature bound is held at the preferred four-times left-side calibration, while the FLCI-optimal bandwidth is selected separately at each cutoff and held fixed across confidence levels. EHW standard errors are clustered by municipality-cycle. The blue point marks the true funding cutoff.

Figure A24: Placebo Cutoffs: Collapsed Support-Point RD



Notes: The outcome is first adjusted using the fixed effects and predetermined controls in Model (5), then collapsed to one unweighted mean at each distinct score value. Points report local-linear, robust bias-corrected `rdrobust` estimates using a cutoff-specific MSE-optimal bandwidth. Thin intervals are 95 percent confidence intervals and thick intervals are 90 percent confidence intervals. Inference is unclustered because the unit is the collapsed support point. The blue point marks the true funding cutoff.

J.4 Honest inference for discrete running variables

The RD running variable is a project rank and therefore takes only a finite number of values. This raises a concern that conventional local-polynomial inference may understate uncertainty when few support points lie near the cutoff. [Kolesár and Rothe \(2018\)](#) show that clustering by the running variable does not generally solve this problem and recommend inference procedures that account directly for possible specification bias. We therefore report honest confidence intervals that account separately for sampling uncertainty and for the largest approximation bias allowed by a bound on how sharply the conditional expectation function may bend.

For this implementation, we recode the running variable as $X = \text{midpoint cutoff rank} - \text{proposal rank}$, where the midpoint lies between the last funded proposal and the first unfunded proposal. Thus, funded proposals have positive values of X , unfunded proposals have negative values of X , and no proposal lies exactly at $X = 0$. The replication code verifies this coding directly before estimation. Table [A32](#) reports the local-linear RDHonest estimates using a triangular kernel, the Holder smoothness class, the FLCI-optimal bandwidth criterion, and Eicker-Huber-White standard errors clustered at the municipality-cycle level. The FLCI criterion chooses the bandwidth that minimizes the length of the bias-aware confidence interval, thereby balancing sampling noise against the worst-case approximation bias. The specification includes the same preferred predetermined covariates as the main RD table, including pair and cycle fixed effects where feasible.

The additional tuning parameter is the curvature bound, called K in some applied papers and M in RDHonest. [Imbens and Wager \(2019\)](#) propose relating local and global smoothness by fitting a global quadratic and scaling its fitted curvature by a factor such as two or four. Following the applied left-side calibration in [Goldsmith-Pinkham et al. \(2023\)](#), our preferred specification adapts this approach by fitting the quadratic among unfunded proposals within 25 ranks of the cutoff. We take the absolute implied second derivative and multiply it by four for the baseline bound. We use the unfunded side because it provides the counterfactual trend for proposals that narrowly missed funding and avoids using many funded proposals farther from the margin to discipline a local smoothness assumption. This bound is a maintained smoothness assumption rather than a directly validated feature of the data, so Table [A33](#) reports sensitivity to multipliers from one to sixteen. Tables [A34](#) and [A35](#) report the same multiplier grid for reelection, running, and vote share. Table [A36](#) also reports an unclustered nearest-neighbor version as a secondary check. The clustered EHW version is the preferred specification because it matches the main RD design.

The resulting honest RD estimate for personal reelection remains positive. Under the baseline curvature bound, the estimated effect is 8.4 percentage points, although its confidence interval includes zero. The candidate-count estimate is -0.120 , with a 95 percent honest confidence interval of $[-0.302, 0.062]$, so it does not provide precise evidence of a change in candidate entry. The broader personal-outcome sensitivity tables show a similar pattern. Without a donut hole, personal reelection and running remain positive but imprecise across the full left-side multiplier grid, while

personal vote share is positive and marginal for smaller curvature bounds. Appendix Table A31 reports the corresponding five-column donut-hole estimates and repeats both the RDHonest and collapsed support-point panels. The RDHonest donut estimates hold M fixed at the corresponding full-sample calibration, isolating the effect of removing the boundary observations. Because this exercise removes the closest treated and control mass points, we interpret it as a diagnostic rather than the preferred specification. Overall, the comparison reinforces the value of reporting the transparent curvature grid alongside the complementary support-point estimates.

Table A32: Main honest RD inference for the discrete funding rank.

Outcome	RD estimate	Std. error	Maximum bias	95% honest CI	p-value	Bandwidth	Effective observations	Mass points in bandwidth	Maximal leverage	Smoothness bound M
Personal incumbency	0.084	(0.070)	0.038	[-0.071, 0.239]	0.295	9.648	653,504	20	0.008	0.0046
Number of candidates	-0.120	(0.088)	0.031	[-0.302, 0.062]	0.198	24.009	984,589	48	0.006	0.0009

Notes: Estimates use RDHonest local linear RD with triangular kernel, Hölder smoothness class, FLCI-optimal bandwidth, and municipality-cycle clustered EHW standard errors. The RDHonest running variable is X = midpoint cutoff rank minus proposal rank, where the midpoint cutoff lies between the last funded proposal and first unfunded proposal; funded proposals have positive values of X and unfunded proposals have negative values. Confidence intervals account for worst-case bias under the maintained second-derivative bound M . Baseline M equals four times the absolute implied second derivative from a quadratic fit estimated on the unfunded side of the cutoff within 25 ranks, using the covariate-adjusted outcome. Results should be interpreted as honest conditional on this curvature bound; sensitivity to M is reported separately.

Table A33: Honest RD sensitivity to the curvature bound.

Multiplier on curvature	Personal M	Personal estimate (SE)
1	0.0012	0.092 (0.050)
2	0.0023	0.088 (0.059)
4	0.0046	0.084 (0.070)
8	0.0093	0.065 (0.081)
16	0.0185	0.031 (0.087)

Notes:

Rows vary the multiplier on the absolute implied second derivative from a left-side quadratic fit among unfunded proposals within 25 ranks of the cutoff. All rows use the preferred covariate-adjusted RDHonest specification with FLCI-optimal bandwidths and municipality-cycle clustered EHW standard errors. Cells report estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style.

Table A34: Personal-outcome honest RD sensitivity to the left-side smoothness bound.

Outcome	1x	2x	4x	8x	16x
Personal incumbency	0.092 (0.050)	0.088 (0.059)	0.084 (0.070)	0.065 (0.081)	0.031 (0.087)
Personal running	0.097 (0.054)	0.104 (0.063)	0.106 (0.080)	0.101 (0.095)	0.086 (0.106)
Personal vote share	0.045* (0.024)	0.048* (0.024)	0.049* (0.026)	0.051 (0.029)	0.051 (0.032)

Notes: Cells report RD estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. All estimates use RDHonest local linear RD with triangular kernel, Holder smoothness class, FLCI-optimal bandwidths, municipality-cycle clustered EHW standard errors, and the preferred predetermined covariates. The sample keeps the boundary mass points: the first unfunded proposal and the last funded proposal. Columns 1x through 16x use the left-side calibration: a quadratic among unfunded proposals within 25 ranks of the cutoff, with the implied absolute second derivative multiplied by the column multiplier.

Table A35: Personal-outcome honest RD sensitivity after dropping the boundary pair.

Outcome	1x	2x	4x	8x	16x
Personal incumbency	0.162** (0.057)	0.185** (0.070)	0.228** (0.087)	0.277* (0.111)	0.320 (0.138)
Personal running	0.138* (0.058)	0.178** (0.065)	0.232** (0.084)	0.255 (0.102)	0.364* (0.118)
Personal vote share	0.049* (0.024)	0.053* (0.025)	0.056* (0.027)	0.060 (0.031)	0.063 (0.038)

Notes: Cells report RD estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. All estimates use RDHonest local linear RD with triangular kernel, Holder smoothness class, FLCI-optimal bandwidths, municipality-cycle clustered EHW standard errors, and the preferred predetermined covariates. The donut sample drops the two mass points adjacent to the cutoff: the first unfunded proposal and the last funded proposal. At each multiplier, M is held fixed at its corresponding full-sample calibration. Columns 1x through 16x use the left-side calibration: a quadratic among unfunded proposals within 25 ranks of the cutoff, with the implied absolute second derivative multiplied by the column multiplier.

Table A36: Nearest-neighbor RDHonest robustness check.

Outcome	RD estimate	Std. error	Maximum bias	95% honest CI	p-value	Bandwidth	Effective observations	Mass points in bandwidth	Maximal leverage	Smoothness bound M
Personal incumbency	0.084	(0.068)	0.039	[-0.069, 0.238]	0.290	9.754	660.759	20	0.008	0.0046
Number of candidates	-0.120	(0.071)	0.033	[-0.272, 0.032]	0.124	25.074	998.727	49	0.006	0.0009

Notes: This secondary check uses RDHonest with nearest-neighbor standard errors and no clustering. It is included only as a robustness check and should not be interpreted as matching the main clustered RD design.

J.4.1 RD Honest Robustness: Using the full left-hand side data to calibrate M

Table A37 repeats the five RDHonest specifications from the main table after fitting the calibration quadratic over the entire unfunded-side support, rather than within 25 ranks. The vote-share result is robust to this change, while the broader calibration implies much wider bandwidths and attenuates the running estimates toward zero.

Table A37: RDD Estimates of KALAHİ Funding on Electoral Outcomes: Full Left-Side Calibration

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>RD Honest: Full left-side calibration</i>					
Pr(reelection)	0.088*	0.061	0.061	0.065	0.061
	(0.049)	(0.046)	(0.040)	(0.040)	(0.042)
Pr(run)	0.017	-0.004	-0.004	0.001	0.006
	(0.047)	(0.043)	(0.046)	(0.047)	(0.047)
Vote Share	0.071**	0.060**	0.061**	0.061**	0.047*
	(0.026)	(0.024)	(0.028)	(0.026)	(0.024)
No. candidates	-0.154**	-0.135*	-0.136*	-0.126	-0.109
	(0.074)	(0.068)	(0.078)	(0.075)	(0.072)
Fixed Effects		✓	✓	✓	✓
Clustered SEs			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. This table repeats Panel A of Table 3, changing only the range used to calibrate the smoothness bound M. The calibration quadratic uses the entire observed unfunded-side support, from $X = -61.5$ to $X = -0.5$, rather than the preferred 25-rank window; M equals four times the implied absolute second derivative. All other choices match Panel A of Table 3: local-linear RDHonest inference, triangular kernel, Holder smoothness class, FLCI-optimal bandwidth, and the same specification ladder. Model (2) adds municipality and cycle fixed effects together, Model (3) adds municipality-cycle clustered EHW standard errors, Model (4) adds census controls, and Model (5) adds political controls. Standard errors are shown in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

Table A38 further varies the multiplier on this full-left-side curvature estimate from one to sixteen. We can see that vote-share effects are robust up until a multiplier of eight.

Table A38: Honest RD sensitivity to the curvature bound using the full unfunded-side support.

Multiplier	Pr(reelection)		Pr(run)		Vote share	
	M	Estimate (SE)	M	Estimate (SE)	M	Estimate (SE)
1	0.000054	0.061 (0.040)	0.000053	0.004 (0.045)	0.000032	0.045* (0.024)
2	0.000108	0.061 (0.040)	0.000107	0.004 (0.045)	0.000063	0.045* (0.024)
4	0.000217	0.061 (0.042)	0.000213	0.006 (0.047)	0.000127	0.047* (0.024)
8	0.000433	0.077 (0.044)	0.000427	0.027 (0.048)	0.000254	0.050* (0.025)
16	0.000866	0.091* (0.047)	0.000853	0.060 (0.048)	0.000507	0.050 (0.028)

Notes: Rows vary the multiplier on the absolute implied second derivative from a quadratic fit over the entire observed unfunded-side support, from $X = -61.5$ to $X = -0.5$. All estimates use the preferred covariate-adjusted RDHonest specification with FLCI-optimal bandwidths and municipality-cycle clustered EHW standard errors. Cells report estimates with standard errors in parentheses; confidence intervals are omitted to match the regression-table style. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

J.5 External Validity of the Funding RD

Table A39 describes the populations that lead from the experimental sample to the local funding-RD comparison. The first four columns report mean predetermined characteristics in the population, among villages appearing in any ranked sequence, among villages in clean sequences, and among villages with a proposal inside the preferred RDHonest vote-share bandwidth. The final column estimates the corresponding covariate profile at the funding cutoff using the local-linear procedure of Mountjoy (2026). Because the running variable is discrete and the cutoff lies between adjacent ranks, this final column is an estimated local profile rather than an observed group.

Table A39: Describing the Funding-RD Sample and Marginal Proposals

	Observed samples				Estimated local population
	Population	All sequences	Clean sequences	RD bandwidth	RD cutoff
Population	1,326.46	1,310.24	1,322.19	1,319.55	1,282.00
Urban	1.97	1.97	1.97	1.97	1.99
Education	2.76	2.77	2.76	2.77	2.75
Age	26.22	26.21	26.10	26.13	26.10
Catholic	84.60	87.00	87.99	87.60	89.68
Married	33.75	33.68	34.05	34.06	33.79
Ethnicity HHI	0.83	0.83	0.83	0.83	0.83
Vote share 2010	0.62	0.61	0.60	0.60	0.60
Winning margin 2010	0.29	0.28	0.26	0.26	0.25
N candidates 2010	2.24	2.20	2.25	2.24	2.25
Villages	4,819	1,804	1,129	1,063	–
Ranked proposals	–	1,846	1,162	1,095	–

Note:

The population column contains the 4,819 villages in the experimental sample after excluding the pair containing Mauban. All-sequence villages have at least one ranked proposal, clean-sequence villages belong to the 53 sequences used by the sharp funding RD, and the RD-bandwidth column uses the 24.42-rank FLCI-optimal bandwidth from the preferred RDHonest vote-share specification. Observed-sample means give each village equal weight; if a village has multiple proposals, it enters a column once. The ranked-proposal row reports the underlying proposal observations. Following Mountjoy (2026), the RD-cutoff column transforms each predetermined covariate X to $-X(1 - Z)$ and reports the coefficient on the funded-side indicator Z from a local linear regression with triangular weights, municipality-cycle fixed effects, and side-specific municipality-cycle slopes. Because rank is discrete and the funding cutoff lies at the midpoint between adjacent ranks, the cutoff column is an estimated local covariate profile rather than an observed subsample.

Table A40 compares proposals inside each preferred vote-share bandwidth to the remaining proposals in the clean sharp-RD sample on pre-treatment covariates from the 2010 census and the 2010 election. Panel A uses the 24.42-rank FLCI-optimal bandwidth from Model (5) of the RDHonest analysis. Panel B uses the 11.45-rank MSE-optimal bandwidth from the collapsed support-point analysis. Both comparisons include all clean-sequence proposals regardless of whether the incumbent runs and vote share is observed. P-values come from regressions of each

covariate on a close-to-cutoff indicator, with standard errors clustered at the municipality–cycle level. No difference is statistically significant at the 5 percent level in either panel. At the 10 percent level, the Catholic share differs in Panel A, while education and the ethnicity HHI differ in Panel B.

Table A40: Selection of Proposals: Close to the Cutoff vs. Rest of the RD Sample

	Non-close sample	Close sample	P-value
Panel A: RD Honest ($h = 24.42$)			
Population	1366.72	1315.58	0.738
Urban	1.93	1.97	0.408
Education	2.65	2.78	0.335
Age	25.56	26.17	0.279
Catholic	94.27	87.26	0.068*
Married	33.75	34.03	0.819
Ethnicity HHI	0.86	0.83	0.278
Vote share 2010	0.60	0.60	0.756
Winning margin 2010	0.27	0.26	0.863
N candidates 2010	2.39	2.24	0.164
Panel B: Collapsed support-point RD ($h = 11.45$)			
Population	1275.93	1335.35	0.653
Urban	1.97	1.97	0.929
Education	2.71	2.80	0.095*
Age	25.87	26.24	0.174
Catholic	89.57	86.91	0.228
Married	33.62	34.16	0.374
Ethnicity HHI	0.86	0.81	0.067*
Vote share 2010	0.59	0.61	0.205
Winning margin 2010	0.24	0.27	0.273
N candidates 2010	2.30	2.22	0.146

Note:

The clean sharp-RD sample contains $N = 1,162$ proposals. Panel A classifies 1095 proposals as close and 67 as non-close. Panel B classifies 833 proposals as close and 329 as non-close. The comparison does not condition on incumbent candidacy or observed vote share. P-values are from regressions of each covariate on a close-to-cutoff indicator, with standard errors clustered at the municipality-cycle level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

J.6 Counting the Number of Mass Points in the RD Score

As Cattaneo et al. (2024) point out, a primary challenge with discrete scores (like project ranks) is the lack of denseness near the cutoff, which makes it impossible to invoke standard large-sample limit arguments without imposing parametric assumptions. Table A41 reports the number of distinct score values in each complete-case outcome sample and within the baseline MSE-optimal bandwidth.

Table A41: Mass-point diagnostics for the RD running variable.

Outcome	N	K	N/K	h	Effective N	K_h	$K_{h,-}$	$K_{h,+}$
<i>Panel A: Personal Incumbency</i>								
Pr(reelection)	1162	86	13.512	6.046	571	12	6	6
Pr(run)	1162	86	13.512	5.650	571	12	6	6
Vote Share	780	75	10.400	8.217	454	16	8	8
<i>Panel B: Candidate Entry</i>								
No. candidates	1162	86	13.512	8.492	694	16	8	8

^a Notes. The table reports the number of observations and unique score values for each complete-case outcome sample using the midpoint-centered funding rank. K denotes the number of mass points and h the MSE-optimal bandwidth from the baseline `rdrobust` specification. Effective N and K_h are the numbers of observations and mass points within h . $K_{h,-}$ and $K_{h,+}$ give the corresponding counts to the left and right of the cutoff.

J.7 Complementary Collapsed Support-Point Analysis

Panel B of Table 3 reports the complete five-specification collapsed support-point analysis. Table A42 isolates the unadjusted version and reports its robust confidence interval, number of support points, and selected bandwidth. Following Cattaneo et al. (2024), the procedure collapses the data to one unweighted mean outcome at each distinct midpoint-centered score before estimating the local discontinuity. The unadjusted vote-share estimate is 4.6 percentage points but is imprecise. After adding fixed effects and predetermined controls in Table 3, the estimates range from 5.0 to 5.4 percentage points. All four adjusted estimates exclude zero at the 5 percent level, and the fully adjusted estimate has a p -value of 0.005. Because each score receives equal weight, this evidence shows that the main result is not driven by ranks containing more proposals. The collapsed analysis uses robust bias-corrected inference on the score-level data rather than municipality–cycle clustered inference. We therefore interpret its greater precision together with, rather than as a replacement for, the clustered honest-RD results.

Table A42: RD estimates using collapsed mass-point data.

Outcome	Estimate	SE	p -value	95% CI	K	h
<i>Panel A: Personal Incumbency</i>						
Pr(reelection)	0.080	(0.110)	0.465	[-0.136, 0.297]	86	9.319
Pr(run)	0.119	(0.081)	0.139	[-0.039, 0.278]	86	10.244
Vote Share	0.046	(0.031)	0.140	[-0.015, 0.108]	75	10.066
<i>Panel B: Candidate Entry</i>						
No. candidates	-0.121	(0.093)	0.193	[-0.304, 0.061]	86	10.160

^a Notes. Following Cattaneo, Idrobo, and Titiunik’s guidance for discrete running variables, each row collapses the data to unique values of the midpoint-centered rank score by taking the unweighted mean outcome at each mass point, then estimates a sharp RD using `rdrobust` on the collapsed support-point data. The table reports robust bias-corrected estimates, robust standard errors, p -values, and 95 percent confidence intervals.

J.8 Alternative Identification Strategy: Local Randomization RDD

Table A43 reports local-randomization RD estimates for the same outcomes using the covariate-selected window around the cutoff. Personal re-election and vote share are positive and statistically distinguishable from zero in this window, while the running estimate is positive but imprecise. The candidate-entry estimate is negative and statistically distinguishable from zero under local-randomization inference. Because this procedure does not cluster inference by municipality–cycle, we do not treat that result as evidence of a candidate-entry effect.

Table A43: Local-randomization RD estimates.

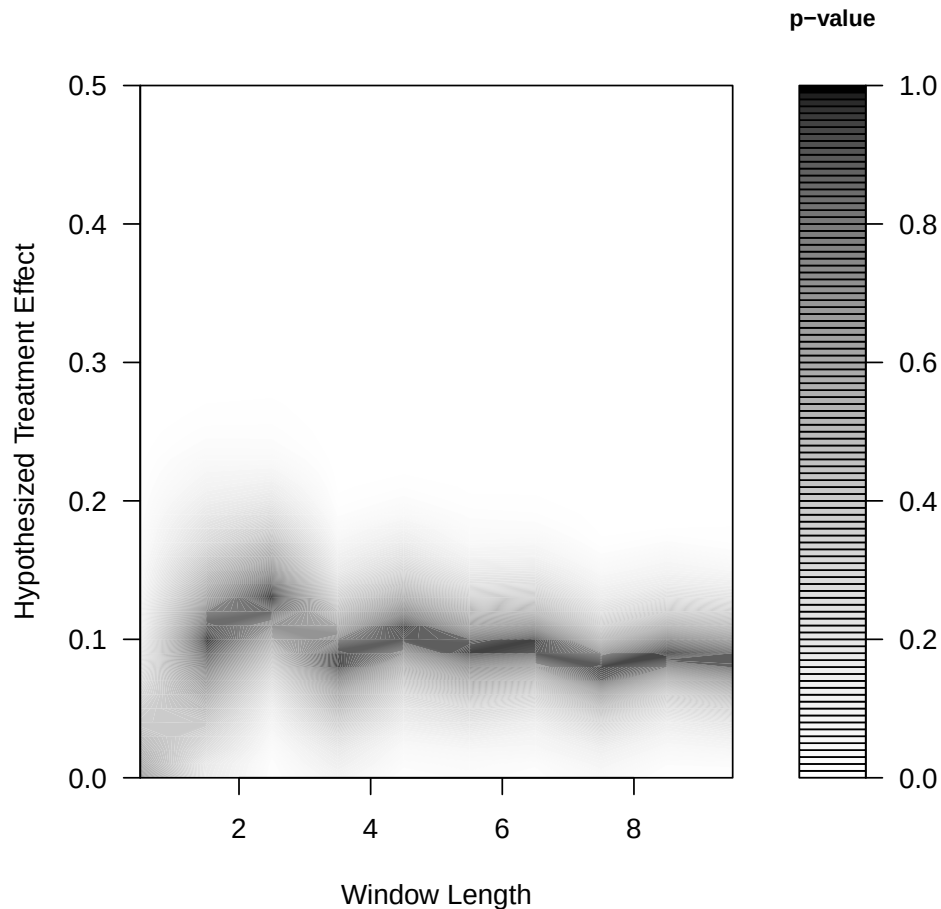
Outcome	Window	Eff. N	Control mean	Treated mean	Difference	p_F	p_N
<i>Panel A: Personal Incumbency</i>							
Pr(reelection)	[-4.5, 4.5]	497 (248; 249)	0.423	0.530	0.107**	0.020	0.017
Pr(run)	[-4.5, 4.5]	497 (248; 249)	0.617	0.675	0.058	0.188	0.178
Vote Share	[-4.5, 4.5]	321 (153; 168)	0.570	0.632	0.062***	0.006	0.008
<i>Panel B: Candidate Entry</i>							
No. candidates	[-4.5, 4.5]	497 (248; 249)	2.306	2.104	-0.202***	0.004	0.004

^a Notes. The window is selected using `rdwinselect` with pre-treatment covariates, half-integer windows aligned to the discrete support points, 1,000 randomization-test replications, and cutoff zero. Each row then reports randomization inference from `rdrandinf` for the corresponding outcome using the selected window. Effective N reports total observations in the window, with left and right counts in parentheses. The columns p_F and p_N report Fisher and Neyman p-values. Significance stars are based on Neyman p-values: *** 1 percent, ** 5 percent, * 10 percent.

J.8.1 Sensitivity Analysis

To ensure our results are not driven by the specific window chosen, we conduct a sensitivity analysis using `rdsensitivity`. This procedure tests various treatment effects across a range of window sizes. The resulting figure (Figure A25) shows the window length on the horizontal axis and the hypothesized treatment effect on the vertical axis. The shading represents the p-value: dark regions indicate effect sizes that the data supports ($p > 0.05$), while white regions indicate effects that are statistically rejected.

Figure A25: Local Randomization RD Sensitivity Plot



In the recommended window ($W = [-4.5, 4.5]$), the estimated increase in personal reelection probability is 10.7 percentage points. Figure A25 shows how the randomization-inference conclusions vary across alternative windows and hypothesized effects; we treat this as a sensitivity diagnostic rather than evidence that significance is invariant to every local neighborhood.

J.9 Alternative Identification Strategy: Fuzzy RD

The sharp funding RD uses municipality-cycles in which actual funding follows a clean ranking boundary. Here we relax that restriction and treat crossing a reconstructed boundary as an instrument for actual funding, measured by `pme_Prioritized`. The analysis focuses on incumbent vote share conditional on running, the outcome for which the main discrete-score RD produces the most precise evidence. For outcome Y , actual funding D , and centered rank X , `RDHonest` estimates the reduced-form discontinuity τ_Y , the first-stage discontinuity τ_D , and their fuzzy ratio τ_Y/τ_D .

Covariate adjustment mirrors the fuzzy estimator rather than adjusting only the outcome. In every specification with controls, we separately residualize Y and D using the same predetermined covariates and then estimate the discontinuity ratio using the two adjusted variables. We likewise calibrate separate smoothness bounds for the adjusted outcome and funding equations. Following the quadratic-plus-multiplier calibration of [Imbens and Wager \(2019\)](#) and the left-side implementation of [Goldsmith-Pinkham et al. \(2023\)](#), each bound equals four times the absolute coefficient on the quadratic term from a fit over the available unfunded-side range up to 33 ranks from the cutoff. This is a maintained smoothness assumption, not a data-free guarantee. All estimates use local-linear regression, a triangular kernel, the Hölder smoothness class, and an FLCI-optimal bandwidth. Models (3)–(5) use EHW inference clustered by municipality-cycle. The 90 percent intervals in [Figure A29](#) are evaluated at the bandwidth selected for the corresponding 95 percent interval, so both intervals surround the same estimate.

Our main fuzzy sample contains 64 municipality-cycles whose sequence begins funded and whose observed rank support is contiguous. We locate the cutoff halfway between the funded and unfunded support points at the first funded-to-unfunded transition. This is the same assignment boundary as centering the score between the two adjacent ranks; the closest observations therefore have scores 0.5 and -0.5 . [Figure A26](#) displays the sequences included in this preferred analysis, while [Figure A27](#) displays the broader descriptive sample of 72 municipality-cycles containing at least one funded and one unfunded proposal.

Figure A26: Funding Sequences in the Preferred Fuzzy RD Sample

Each row is one of the 64 municipality-cycles included in the preferred fuzzy RD analysis. The sample retains sequences that begin funded and have contiguous observed rank support. Projects are ordered by their observed rank. Blue tiles denote funded proposals and gray tiles denote unfunded proposals. The dashed red segment marks the first observed funded-to-unfunded transition used to reconstruct the assignment cutoff. Later departures from the funded-to-unfunded pattern are treated as noncompliance.

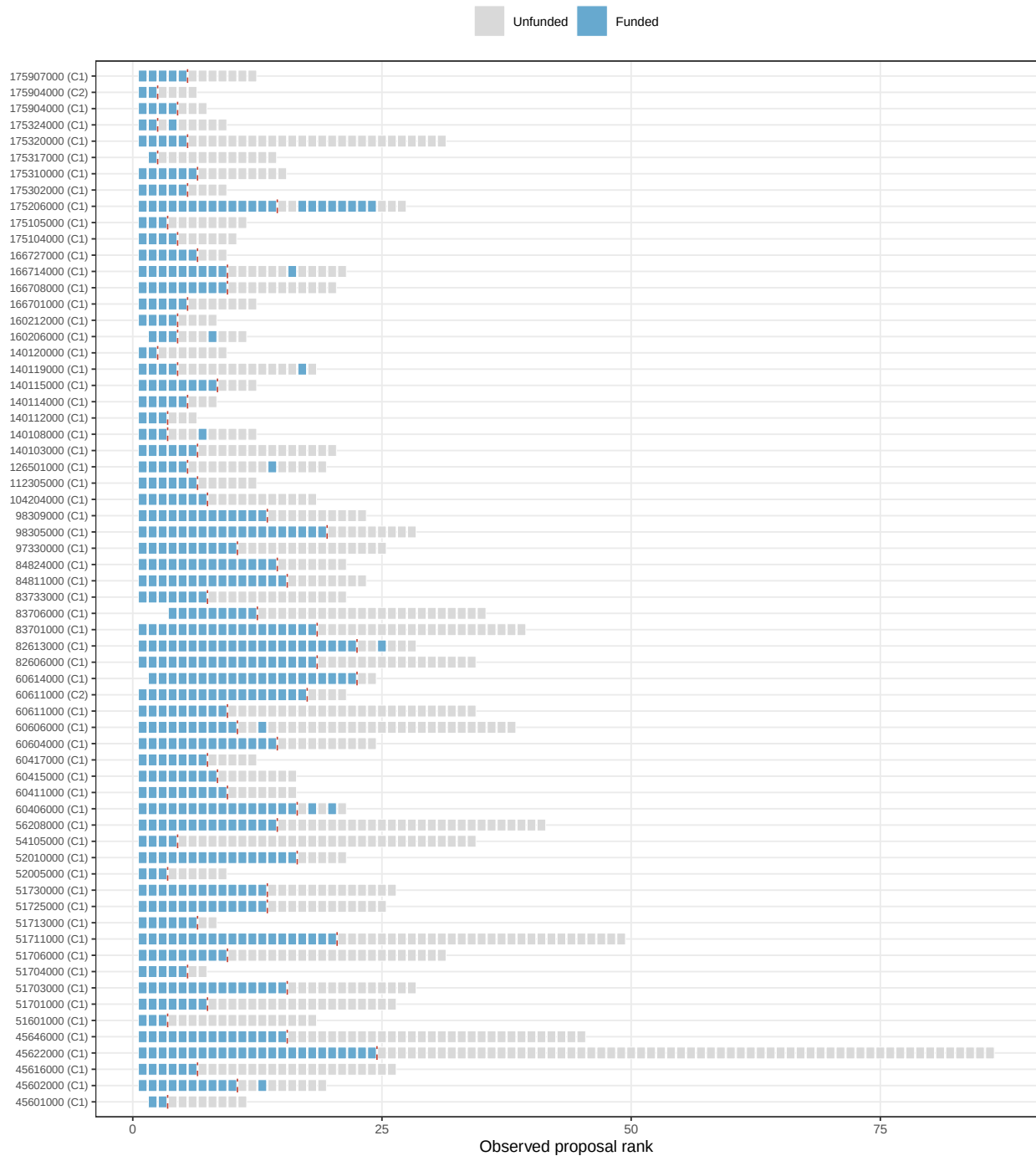


Figure A27: Funding Sequences Eligible for the Fuzzy RD

Each row is a municipality-cycle containing at least one actually funded and one actually unfunded ranked proposal. Projects are ordered by their observed rank. Blue tiles denote funded proposals and gray tiles denote unfunded proposals. The dashed red segment marks the first observed funded-to-unfunded transition. The main fuzzy specification retains the subset of 64 sequences that begin funded and have contiguous rank support; the sensitivity analysis progressively admits the remaining sequences.

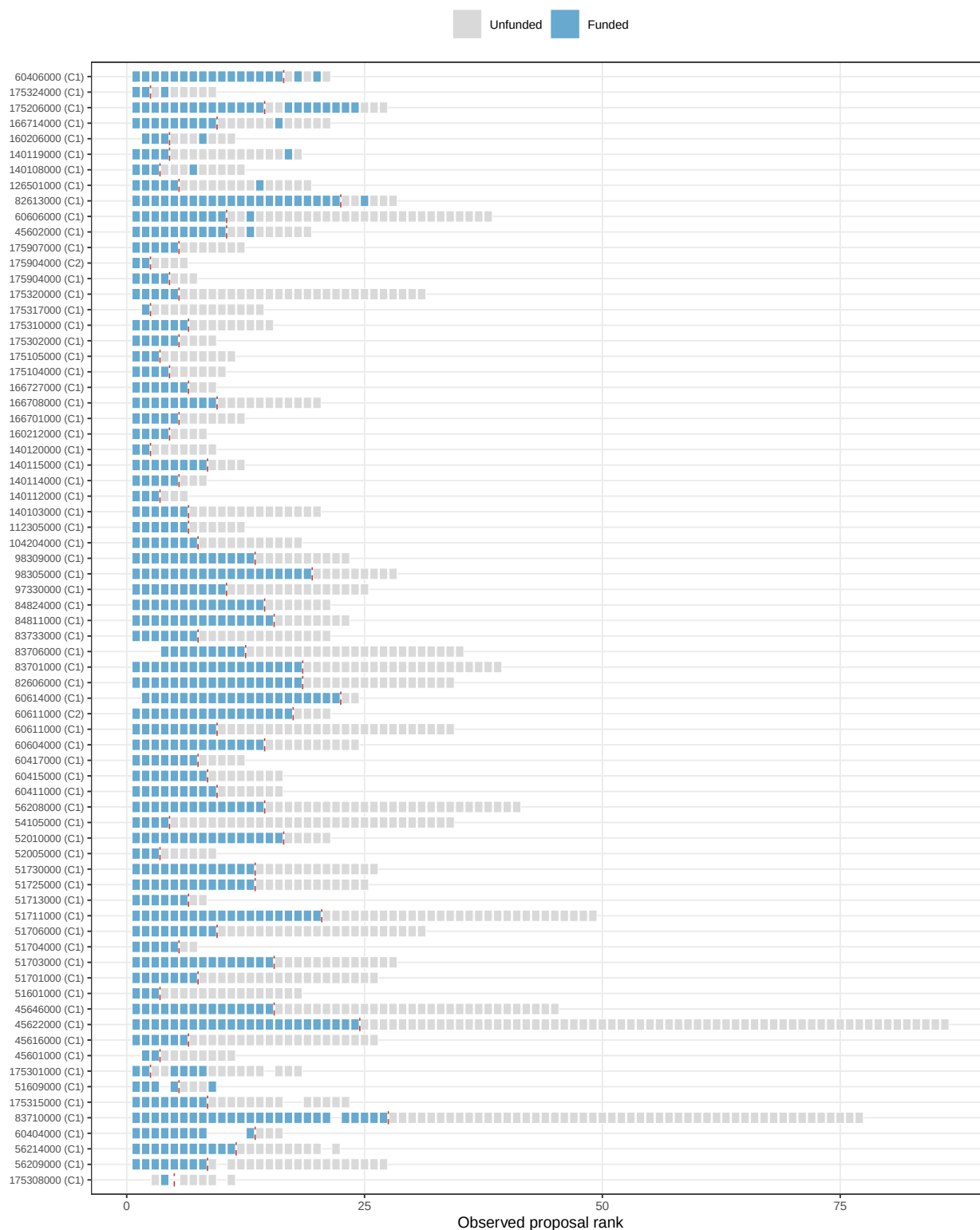
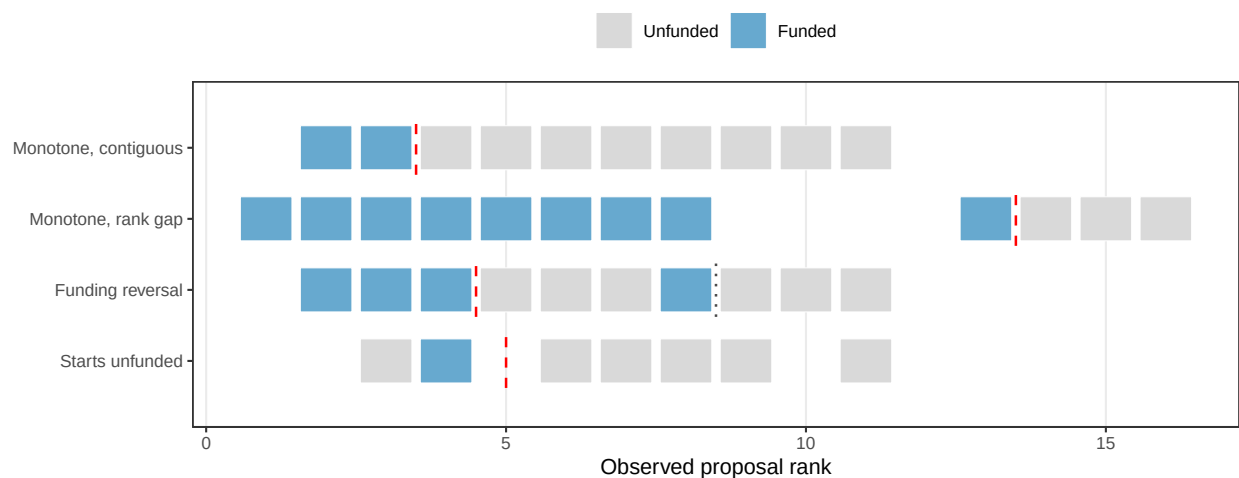


Figure A28 isolates the four observed sequence features that motivate the sample and cutoff checks. A monotone sequence with contiguous ranks identifies a unique boundary between adjacent funded and unfunded proposals and most closely resembles a sharp assignment rule. With a rank gap, the funding change remains visible but the exact location of the latent boundary between the two observed support points is not. A funding reversal instead creates treatment noncompliance: crossing the reconstructed boundary changes the probability of funding without determining receipt perfectly, and the first and last funded-to-unfunded transitions need not coincide. Finally, a sequence that begins unfunded does not satisfy the preferred requirement that the highest-ranked observed proposal is funded. The preferred 64-cycle sample admits contiguous sequences that begin funded, including later reversals, and treats those reversals as noncompliance. The broader checks then admit rank gaps and sequences that begin unfunded rather than selecting a single favorable reconstruction.

Figure A28: Illustrative Funding-Sequence Types

The figure shows one observed municipality-cycle for each sequence type used to motivate the fuzzy-RD sensitivity analysis. Blue tiles denote funded proposals and gray tiles denote unfunded proposals. Dashed red lines mark the first funded-to-unfunded transition; dotted gray lines mark the last transition when it differs from the first. The selected rows are illustrations rather than estimates.



For each sequence, the first transition is the first observed funded rank followed by an unfunded rank, and the last transition is defined analogously. The two coincide for monotone sequences but can differ after a funding reversal. When adjacent observed ranks straddle the boundary, the cutoff is their midpoint; if ranks are missing, the sensitivity analysis places the cutoff immediately after the funded support point, at the midpoint of the two observed support points, or immediately before the unfunded support point. Under every reconstruction, the centered score is positive for better-ranked proposals and negative for worse-ranked proposals, with the package cutoff normalized to zero.

Table A44 reports the five-specification ladder. The first-stage discontinuity is between 0.963 and 0.967. The vote-share estimates range from 0.043 to 0.062 and remain positive as fixed effects, clustering, census controls, and political controls are added. The preferred fixed-effects

Table A44: Fuzzy RDHonest Estimates of KALAHl Funding on Incumbent Vote Share

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
Actual KALAHl funding	0.062** (0.027)	0.054* (0.025)	0.054* (0.030)	0.054* (0.027)	0.043 (0.025)
95% honest CI	[0.002, 0.122]	[-0.002, 0.110]	[-0.010, 0.118]	[-0.004, 0.111]	[-0.012, 0.097]
N	938	938	938	938	938
N effective	727.7	710.3	713.0	759.5	727.4
First stage	0.963	0.967	0.967	0.966	0.966
Bandwidth	17.28	16.56	16.13	19.79	17.61
Maximum bias	0.014	0.014	0.013	0.012	0.012
<i>M</i> outcome	0.0005768	0.0006216	0.0006208	0.0004331	0.0005067
<i>M</i> funding	0.0001303	0.0000206	0.00002034	0.00005436	0.00002189
Pair and cycle fixed effects		✓	✓	✓	✓
Clustered SEs			✓	✓	✓
Census controls				✓	✓
Political controls					✓

^a Notes. The outcome is incumbent vote share conditional on running. Fuzzy RDHonest instruments actual KALAHl funding with crossing the first funded-to-unfunded boundary in the current 64-cycle sample. The cutoff is the midpoint between the adjacent funded and unfunded support points, so the closest scores are 0.5 and -0.5. In models with covariates, RDHonest separately adjusts vote share and actual funding using the same covariates before forming their discontinuity ratio. Separate outcome and funding curvature bounds equal four times the absolute quadratic coefficient from left-side fits over ranks -33 to 0. All models use local linear regression, a triangular kernel, the Hölder smoothness class, and an FLCI-optimal bandwidth. Models (3)–(5) use municipality-cycle clustered EHW standard errors. Stars use the bias-aware honest p-value: *** 1 percent, ** 5 percent, * 10 percent.

and clustered estimates in Models (3)–(5) range from 0.043 to 0.054. Two of these three estimates are statistically distinguishable from zero at the 10 percent level, while the fully adjusted estimate remains positive but is less precise.

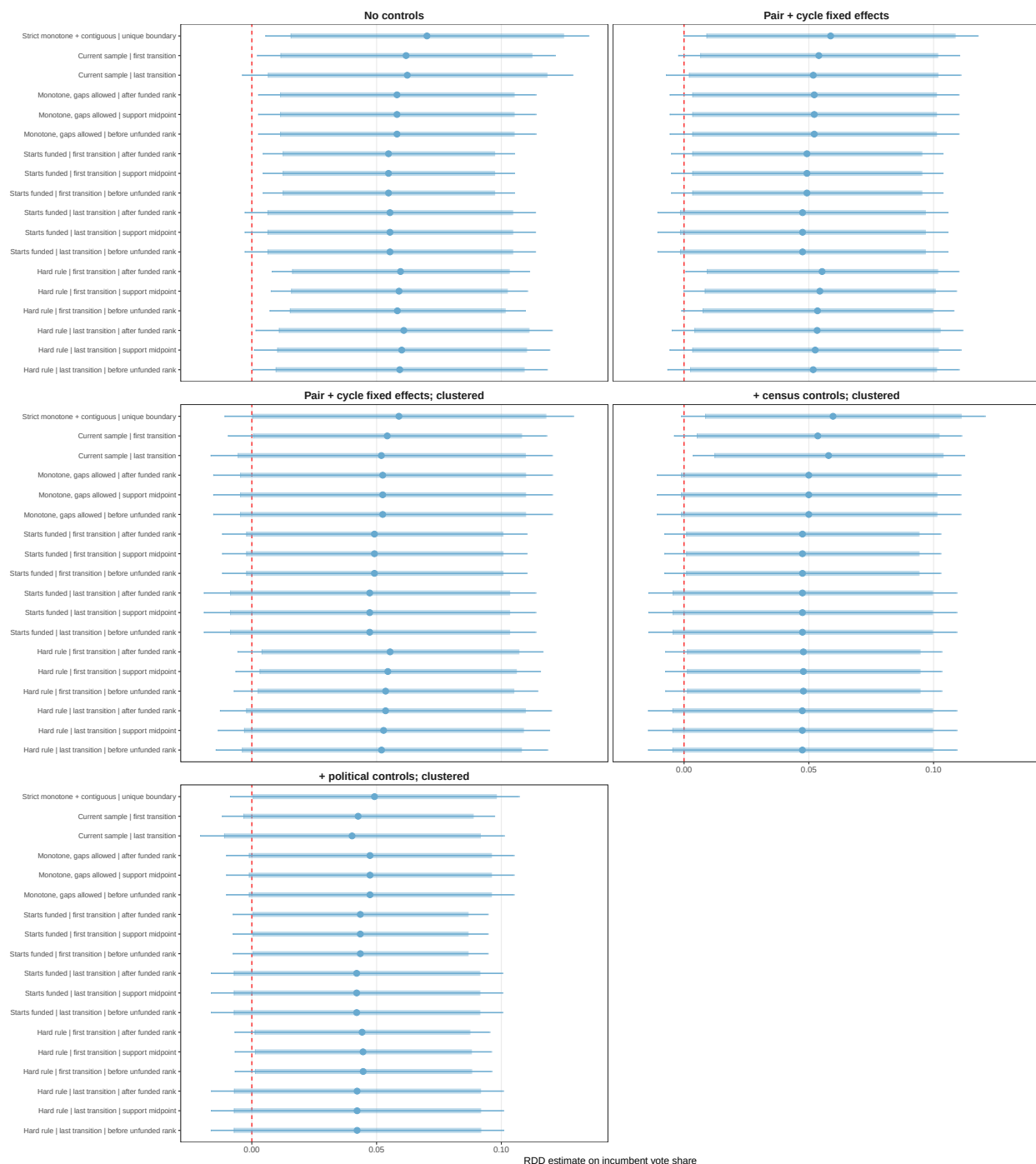
J.9.1 Fuzzy RD: sensitivity

We next vary which observed funding sequences are admissible and how their boundary is reconstructed. The 18 rules below comprise: (i) the unique boundary in the strict monotone-contiguous sample; (ii) the first and last transition in the current sample; (iii) three placements around the first transition when monotone sequences may have gaps; and (iv) the same three placements around both the first and last transitions when reversals are allowed, first among sequences that begin funded and then among all sequences with at least one funded and one unfunded proposal. The three placements are immediately after the funded support point, halfway between observed support points, and immediately before the unfunded support point. They coincide when the boundary ranks are adjacent. We exclude a cutoff selected to maximize the realized first stage from the reported sensitivity set because that choice would be data-adaptive.

Figure A29 presents all 18 reconstructions for each of the five specifications. Every one of the 90 point estimates is positive. Their ranges are 0.055–0.070 without controls, 0.047–0.059 with pair and cycle fixed effects, 0.047–0.059 after municipality-cycle clustering, 0.047–0.060 with census controls, and 0.040–0.049 with political controls. The 90 percent honest interval excludes zero in 18, 15, 5, 9, and 7 of the 18 reconstructions, respectively. The corresponding counts at 95 percent are 14, 1, 0, 1, and 0. Thus, among the 54 estimates that include fixed effects and municipality–cycle clustering, 21 exclude zero at the 10 percent level, and all lie between 4.0 and 6.0 percentage points. Across the sensitivity set, the first stage ranges from 0.890 to 1.000. The sign and magnitude are therefore stable across defensible sequence and cutoff definitions, while statistical precision depends on clustering and covariate adjustment. This is why we emphasize the robustness of the positive pattern rather than the stars attached to any single reconstruction.

Figure A29: Fuzzy RD Sensitivity across Funding Sequences and Specifications

Points are fuzzy RDHonest estimates of actual KALAH! funding on incumbent vote share conditional on running. Rows vary the municipality-cycle inclusion rule, the first versus last funded-to-unfunded transition, and the placement of the cutoff when observed ranks are not adjacent. Panels apply the five specifications in Table A44. Thin intervals are 95 percent honest confidence intervals; thick translucent intervals are 90 percent honest confidence intervals evaluated at the same FLCI-optimal bandwidth. All models use separate left-side curvature bounds for the outcome and funding equations; Models (3)–(5) cluster EHW standard errors by municipality-cycle.



K Further Results: Mechanisms

This section provides the fuller evidence behind the mechanism discussion in Section 6.4. We proceed in two steps. First, we ask whether fewer incumbents sought another term because they anticipated voters' responses to implementation or because implementation directly reduced the value of campaigning and serving. The winning-margin interactions show whether the effects were larger among electorally vulnerable incumbents. Surveys and interviews then help assess whether village mayors' own costs account for that pattern (Appendix K.1).

Second, we consider the three accountability mechanisms (Appendix K.2). We use differences in the funding effect across BRT roles to assess whether outcomes may have changed citizens' beliefs or the standards they applied to village mayors. Survey and interview evidence then describes citizens' assessments, expectations, and participation costs. Throughout, we distinguish evidence from the impact evaluation in the experimental sample, other KALAHİ implementations, and our later interviews. Visibility and attribution are discussed in Appendix C.3, and early exposure outside the participation measure in Appendix I.4.2.

K.1 Anticipated Electoral Responses or Direct Costs

Why did fewer incumbents seek another term? Under the accountability explanation, village mayors reconsider running because they anticipate an unfavorable voter response to implementation. An alternative explanation is that implementation makes holding office less attractive to village mayors independently of voters' reactions. We assess these explanations using evidence on which incumbents were less likely to run and how officials evaluated their experience of implementation.

K.1.1 Prior Electoral Strength and Candidacy

We first examine whether the effects of program access and implementation participation on candidacy vary with the village mayor's previous winning margin. If unsuccessful experiences contribute to these effects, the three accountability mechanisms yield a common prediction for this exercise. A failure may change citizens' assessment of the village mayor, fall short of a higher performance standard, or leave participation costs uncompensated. Village mayors who anticipate these responses may reconsider seeking another term, especially when their previous winning margin was narrow. An incumbent who previously won narrowly may expect even a modest loss of support to reduce the probability of reelection, whereas an incumbent with a larger majority may expect to retain sufficient support.

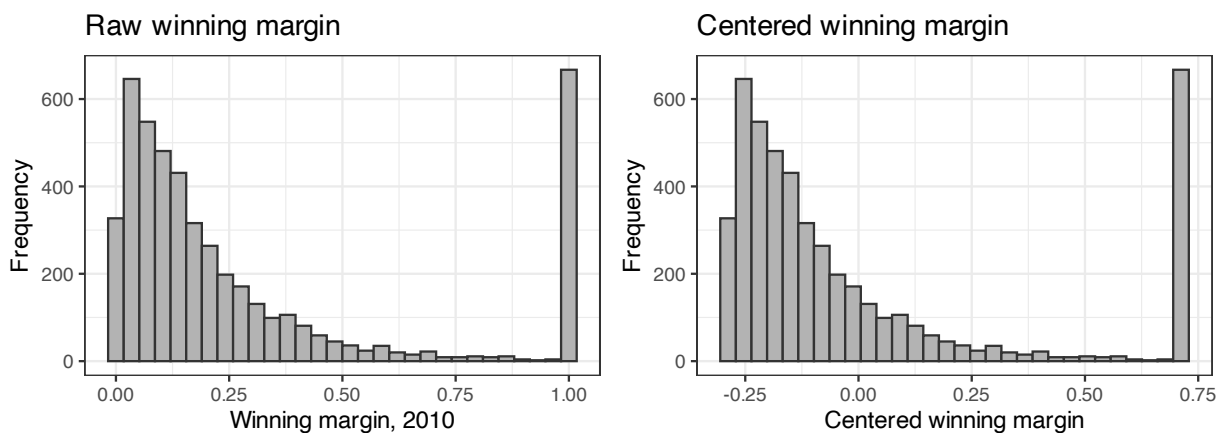
Personal costs could produce the same pattern without an anticipated change in voter support. For this explanation to affect the decision to run, implementation must change the anticipated costs or benefits of another term, or the resources available to campaign. Costs already incurred without continuing consequences would not suffice. Experience with KALAHİ might, for example, lead a village mayor to expect greater administrative demands during another term or leave fewer resources available for the next campaign.

These consequences need not be greater for narrow winners. If a smaller prior margin indicates

a lower expected probability of winning, even identical additional burdens could make campaigning less worthwhile for less secure incumbents. The interaction therefore cannot distinguish anticipated voter responses from changes in village mayors' own expected returns. The relevant question for the personal-cost explanation is whether implementation changed what village mayors expected from continued service, not simply whether it required work.

We center the 2010 winning margin at its sample mean, $m_i - \bar{m}$. Figure A30 shows the raw and centered distributions. In both analyses below, the coefficient on the treatment gives its effect at the mean prior margin, while the interaction shows how the effect varies with prior electoral strength. A 10-percentage-point increase in the margin changes the estimated treatment effect by one tenth of the interaction coefficient.

Figure A30: Incumbent Winning Margin in 2010, Raw and Centered



K.1.1.1 Program Access We first ask whether program access reduced the probability of running more among incumbents who previously won narrowly. The ITT uses the original randomized assignment without conditioning on subsequent participation or funding. Table A45 reports six specifications using the same 4,781 villages in 98 matched-pair strata. Models 1 and 2 omit pair fixed effects and use Eicker–Huber–White and municipality-clustered standard errors, respectively. Models 3–5 include pair fixed effects, with Eicker–Huber–White, municipality-clustered, and pair-clustered standard errors. Model 6 retains pair fixed effects and pair clustering and adds census controls for population, education, age, urbanization, religion, ethnic concentration, citizenship, and marital status, together with the 2010 number of candidates. We estimate

$$\text{Run}_i = \alpha + \beta_1 \text{ASSIGNED}_m + \beta_2 \text{MARGIN}_{ci} + \beta_3 (\text{ASSIGNED} \times \text{MARGIN}_c)_{im} + \mathbf{X}'_i \boldsymbol{\gamma} + \alpha_{p(i)} + u_i.$$

Here, $\alpha_{p(i)}$ denotes the matched-pair fixed effects included in Models 3–6 and $\mathbf{X}_i$ denotes the census controls and prior number of candidates included in Model 6.

Table A45: Intention-to-Treat Effect of Program Assignment on Candidacy by Prior Winning Margin

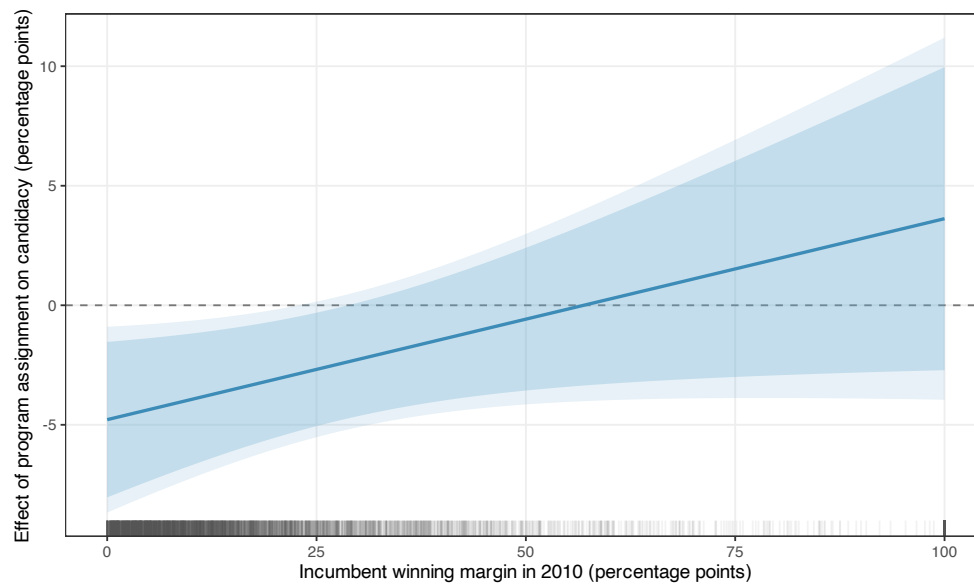
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Program assignment	-0.02*	-0.02*	-0.02	-0.02**	-0.02	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Centered winning margin, 2010	-0.21***	-0.21***	-0.21***	-0.21***	-0.21***	-0.16***
	(0.03)	(0.04)	(0.03)	(0.03)	(0.04)	(0.04)
Assignment $\times$ winning margin	0.07	0.07	0.08*	0.08*	0.08	0.08*
	(0.04)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Interaction p -value	0.100	0.147	0.089	0.099	0.122	0.088
Pair fixed effects	No	No	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	Yes
Clustering	None	Municipality	None	Municipality	Pair	Pair
Observations	4,781	4,781	4,781	4,781	4,781	4,781
Clusters	–	195	–	195	98	98

Notes: The outcome is incumbent candidacy in 2013. Each model interacts original municipal assignment with the centered 2010 winning margin. All models include the main effect of prior winning margin. Models 3–6 include matched-pair fixed effects. Models 2 and 4 cluster standard errors by municipality, while Models 5 and 6 cluster by matched pair. Model 6 adds the census controls and the 2010 number of candidates. All models use the same 4,781 villages in 195 municipalities spanning 98 matched-pair strata. The sample excludes 38 villages without a prior winning margin. No model conditions on observed mobilization, funding, or subsequent candidacy. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Models 3–5 yield the same point estimates because they differ only in the standard-error calculation. At the mean prior winning margin, Model 5 estimates that program assignment reduces candidacy by 2.3 percentage points, with a pair-clustered standard error of 1.4 points and a p -value of 0.103. After adding the census controls and prior candidate count in Model 6, the corresponding effect is –2.4 percentage points, with a p -value of 0.100. The interaction estimates in Models 5 and 6 imply that a 10-percentage-point increase in the prior margin makes the assignment effect approximately 0.8 percentage points less negative. Their p -values are 0.122 and 0.088, respectively.

Figure A31 plots the effects implied by Model 6 across the observed range of prior winning margins. For an incumbent with a prior margin of zero, the estimated effect of assignment is –4.8 percentage points. It becomes less negative as the prior margin increases and reaches zero at a margin of approximately 57 percentage points. The point at which the fitted effect equals zero should not be interpreted as a threshold in incumbents’ decisions to run. The confidence bands widen at larger margins, where the data are sparse. The pattern is consistent with less secure incumbents being more likely to reconsider running, although the interaction is imprecisely estimated and does not reveal why they do so.

Figure A31: Intention-to-Treat Effect on Candidacy by Prior Winning Margin



Notes: The figure displays Model 6 in Table [A45](#), which includes matched-pair fixed effects, census controls, and the 2010 number of candidates. The lighter and darker bands show the 95 and 90 percent confidence intervals, respectively, based on standard errors clustered by matched pair. Marks along the horizontal axis show the observed prior winning margins.

K.1.1.2 Implementation Participation We next examine the corresponding pattern for implementation participation. Following the interacted 2SLS approach discussed by [Zhao et al. \(2025\)](#), we instrument participation and its interaction with the centered prior margin using municipal assignment and the assignment–margin interaction. Interpreting these estimates as participation effects requires the IV assumptions discussed in the main text, including exclusion and monotonicity, as well as the linear specification of effect variation used here. Random assignment alone does not establish this interpretation. We estimate

$$\begin{aligned} \text{Run}_i = & \alpha + \beta_1 \text{PARTICIPATION}_i + \beta_2 \text{MARGIN}_{ci} \\ & + \beta_3 (\text{PARTICIPATION} \times \text{MARGIN}_c)_i + \mathbf{X}_i' \boldsymbol{\gamma} + \alpha_{p(i)} + u_i. \end{aligned}$$

where $\alpha_{p(i)}$ denotes the matched-pair fixed effects included in Models 3–6 and $\mathbf{X}_i$ denotes the census controls and prior number of candidates included in Model 6. As in the ITT analysis, the census controls measure population, education, age, urbanization, religion, ethnic concentration, citizenship, and marital status.

Table A46: Complier-Average Effect of Implementation Participation on Candidacy by Prior Winning Margin

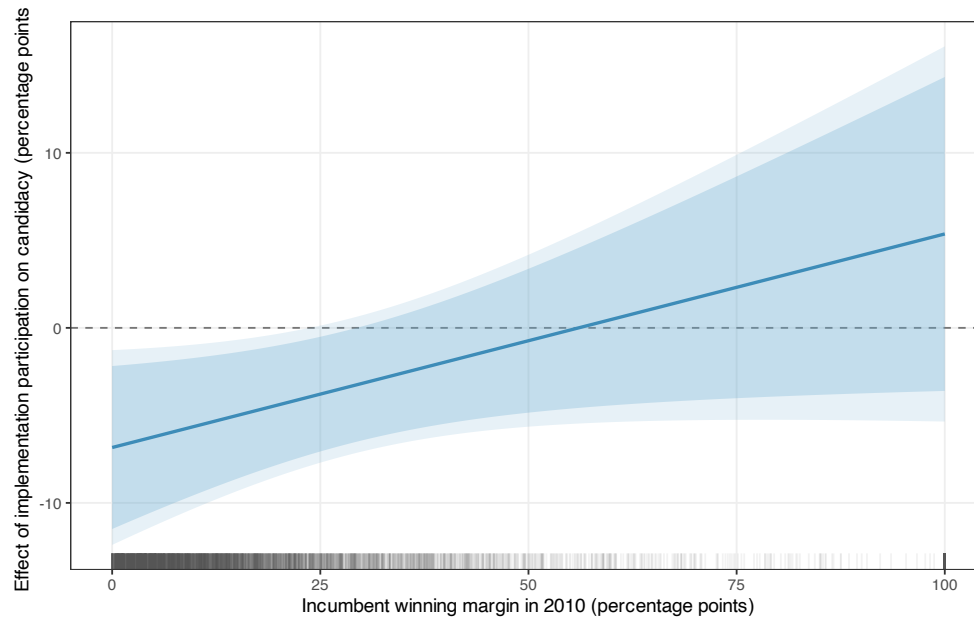
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Implementation participation	-0.03* (0.02)	-0.03* (0.02)	-0.03 (0.02)	-0.03** (0.01)	-0.03* (0.02)	-0.03* (0.02)
Centered winning margin, 2010	-0.22*** (0.03)	-0.22*** (0.04)	-0.22*** (0.04)	-0.22*** (0.04)	-0.22*** (0.04)	-0.18*** (0.04)
Participation $\times$ winning margin	0.10* (0.06)	0.10 (0.07)	0.11* (0.07)	0.11* (0.07)	0.11 (0.07)	0.12* (0.07)
Interaction p -value	0.098	0.147	0.083	0.098	0.116	0.086
Pair fixed effects	No	No	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	Yes
Clustering	None	Municipality	None	Municipality	Pair	Pair
Observations	4,781	4,781	4,781	4,781	4,781	4,781
Clusters	–	195	–	195	98	98

Notes: The outcome is incumbent candidacy in 2013. Each model instruments implementation participation and its interaction with the centered 2010 winning margin using assignment and the assignment–margin interaction. All models include the main effect of prior winning margin. Models 3–6 include matched-pair fixed effects. Models 2 and 4 cluster standard errors by municipality, while Models 5 and 6 cluster by matched pair. Model 6 adds the census controls and the 2010 number of candidates. All models use the same 4,781 villages in 195 municipalities spanning 98 matched-pair strata. The sample excludes 38 villages without a prior winning margin. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Models 3–5 yield the same point estimates because they differ only in the standard-error calculation. At the mean prior winning margin, Model 5 estimates that implementation participation reduces candidacy by 3.2 percentage points, with a pair-clustered standard error of 1.9 points and a p -value of 0.095. After adding the census controls and prior candidate count in Model 6, the corresponding effect is –3.3 percentage points, with a p -value of 0.093. The interaction estimate in Model 5 implies that a 10-percentage-point increase in the prior margin makes the estimated effect 1.1 percentage points less negative, with a p -value of 0.116. With controls, the estimate is 1.2 percentage points and excludes zero at the 10 percent level ($p = 0.086$).

Figure A32 plots the effects implied by Model 6 across the observed range of prior winning margins. For an incumbent with a prior margin of zero, the estimated effect of implementation participation is –6.8 percentage points. The fitted effect becomes less negative as the prior margin increases and reaches zero at approximately 56 percentage points. As in the ITT figure, the point at which the fitted effect equals zero should not be interpreted as a threshold in incumbents’ decisions to run. Among the pair-clustered specifications, the interaction excludes zero at the 10 percent level with controls but not without them. The confidence bands are wide, particularly at large prior margins, so we treat the pattern as suggestive.

Figure A32: Complier-Average Effect of Implementation Participation on Candidacy by Prior Winning Margin



Notes: The figure displays Model 6 in Table A46, which includes matched-pair fixed effects, census controls, and the 2010 number of candidates. The lighter and darker bands show the 95 and 90 percent confidence intervals, respectively, based on standard errors clustered by matched pair. Marks along the horizontal axis show the observed prior winning margins.

The estimated reductions are larger among previously less secure incumbents, as any of the three accountability mechanisms could imply if village mayors anticipate an unfavorable response. The pattern is also consistent with implementation changing the expected personal return to another term. Neither incumbents' expectations nor the effect of unsuccessful participation alone is observed, and the ITT and CATE interactions use the same lottery variation rather than providing independent tests. We therefore turn to officials' assessments of the program and accounts of their work to examine whether the available evidence supports the personal-cost explanation.

K.1.2 Barangay-Official Assessments of Benefits and Negative Impacts

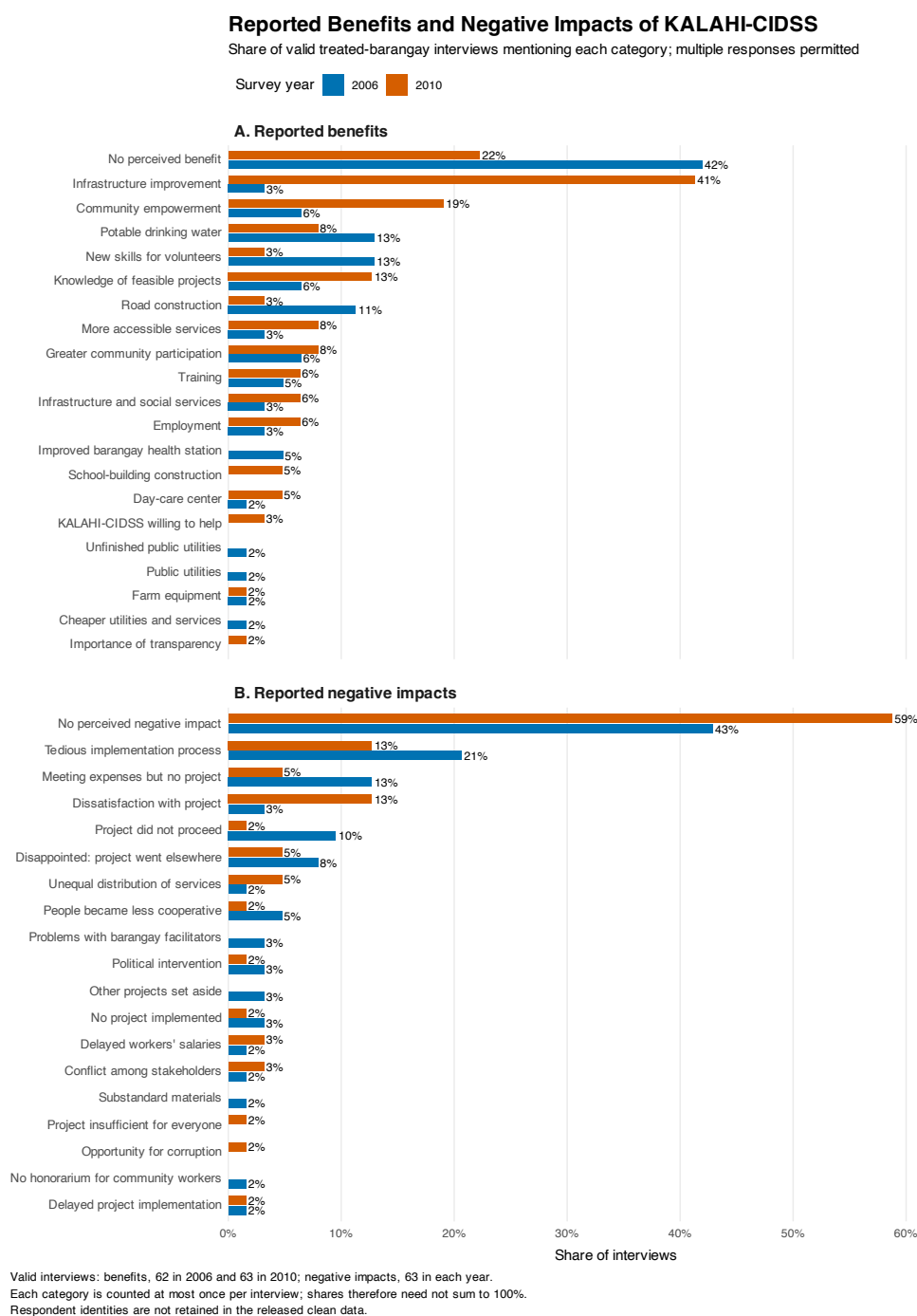
The personal-cost interpretation of the winning-margin interactions requires more than evidence that implementation was demanding. It requires consequences for the expected return to another term or the resources available to campaign. Officials' assessments provide indirect evidence by showing whether reported burdens were accompanied by dissatisfaction with the program. We use barangay-official surveys from the evaluation of KALAH!s original phase, with a baseline in 2003 before implementation in the sampled municipalities and follow-ups in 2006 and 2010 (Labonne, 2013). Our analysis uses treated-barangay observations from both follow-ups. These data concern an earlier phase, not the 198-municipality experiment analyzed in the main text.

The questionnaire asks about satisfaction with KALAH! (K75), benefits gained by the barangay (K76), and losses or negative impacts experienced by the barangay (K77). The latter questions concern the community, not the respondent's personal costs of holding office. They can therefore describe officials' experience of implementation but cannot directly test the personal-cost explanation. Although the questionnaire records respondent designations, the released clean panel retains neither RESP1 nor another designation variable. We describe these observations as barangay-official interviews rather than verified village mayor responses.

For benefits and negative impacts, we use the three response fields retained for each question in the released data. Although the 2010 questionnaire provides five response lines, additional answers cannot be recovered from these fields. We count each category at most once per interview and divide by the number of interviews with at least one valid response to that question. Code 98, indicating no perceived benefit or negative impact, is retained as a substantive response. Nonresponse, not-applicable, and do-not-know codes are excluded. There are 62 valid benefit interviews in 2006 and 63 in 2010, and 63 valid negative-impact interviews in each year. Because multiple responses were possible, category shares need not sum to 100 percent.

Figure A33 shows that a tedious process was reported in 20.6 percent of interviews in 2006 and 12.7 percent in 2010. Meeting expenses without a project were mentioned in 12.7 and 4.8 percent, respectively. No negative impact was reported in 42.9 and 58.7 percent. Reported benefits included infrastructure, water, skills, and community participation, while 41.9 percent in 2006 and 22.2 percent in 2010 reported no benefit. Some officials therefore reported costs without a project, but negative assessments were not universal. Reported expenses could also reflect citizens' costs, as in the third accountability mechanism. The responses identify neither who paid nor whether costs outweighed benefits.

Figure A33: Benefits and Negative Impacts Reported in Barangay-Official Interviews



Notes: Authors' calculations from the 2006 and 2010 treated-barangay observations in the earlier impact evaluation. Shares are calculated separately by question and year using interviews with valid responses. Multiple categories may be reported, but each category is counted once per interview. Respondents cannot be restricted to verified village mayors in the released data.

To compare the two sets of responses directly, Table A47 restricts each year to interviews with valid answers to both questions. In 2006, the same proportion reported no benefits and no negative impacts. In 2010, 58.7 percent reported no negative impacts, compared with 22.2 percent reporting no benefits, and the mean number of recorded benefits exceeded that of negative impacts. These comparisons describe the frequency of reported effects, not whether their value outweighed their costs.

Table A47: Benefits and Negative Impacts Reported by Barangay Officials

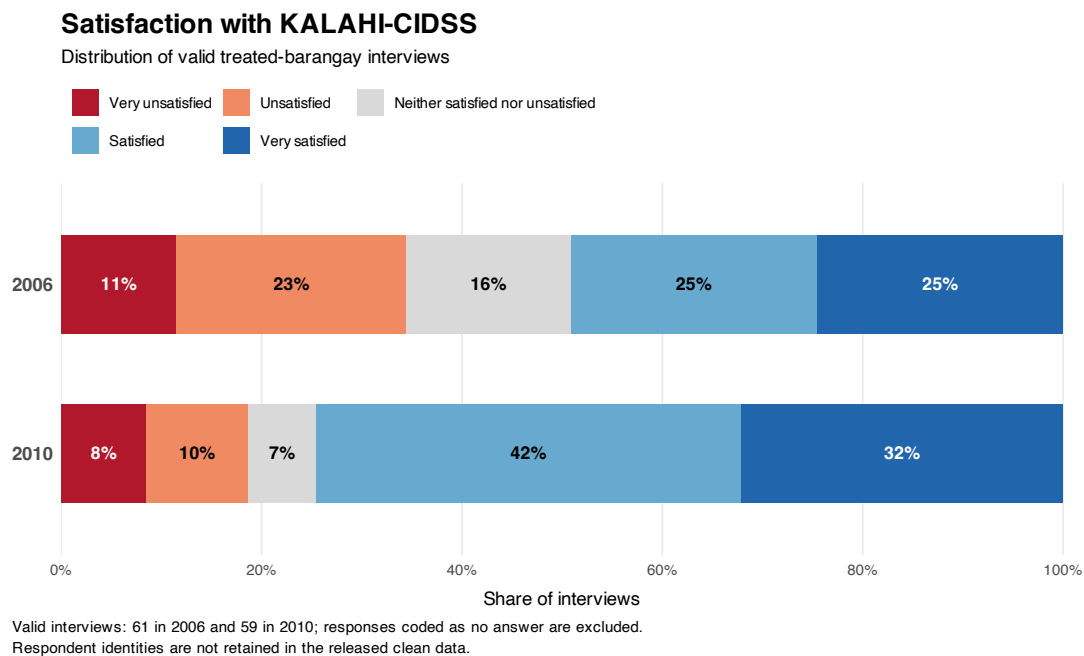
	2006	2010
Interviews with valid answers to both questions	61	63
No benefits reported	26 (42.6%)	14 (22.2%)
No negative impacts reported	26 (42.6%)	37 (58.7%)
Mean number of recorded benefits	0.87	1.38
Mean number of recorded negative impacts	0.82	0.57
Total recorded benefit mentions	53	87
Total recorded negative-impact mentions	50	36

Notes. Authors' calculations from treated-barangay interviews in the earlier KALAH! evaluation (Labonne, 2013). Within each year, every row uses the same interviews with valid responses to both K76 and K77. Percentages are in parentheses. Each distinct category is counted once per interview using the three response fields retained for each question. Explicit reports of no benefit or no negative impact (code 98) count as zero. Interviews without a valid answer to either question are excluded. The questionnaire provides five response lines, but additional responses are unavailable in the released panel. Counts describe reported effects on the barangay, not their relative importance or officials' personal costs. Respondents cannot be restricted to verified village mayors.

Figure A34 examines whether officials nevertheless evaluated the program favorably. Among 61 valid interviews in 2006 and 59 in 2010, satisfied and very satisfied responses account for 49.2 and 74.6 percent, respectively. Unsatisfied and very unsatisfied responses account for 34.4 and 18.6 percent. Favorable assessments were thus common, particularly in 2010, despite the burdens reported in these surveys. These responses provide little support for an account in which implementation left officials generally and persistently dissatisfied with the program.

This evidence helps qualify the personal-cost interpretation but does not rule it out. Satisfaction with KALAHÍ need not imply that serving another term was personally worthwhile. The surveys do not measure anticipated demands on village mayors during a subsequent term, their campaign resources, or the reasons some chose not to run. Nor do these separate yearly distributions establish changes in individual officials' satisfaction. They weigh against a general account of discouragement while leaving open the possibility that continuing costs affected particular incumbents.

Figure A34: Satisfaction with KALAHÍ in Barangay-Official Interviews



Notes: Authors' calculations from K75 in the earlier evaluation's treated-barangay sample. The five substantive satisfaction categories are retained. Missing, nonresponse, and not-administered codes are excluded. Percentages describe valid interviews in each wave rather than verified village mayor responses or individual changes over time.

K.1.3 Village Mayors' Workload and Candidacy

We next examine evidence of costs borne by officials themselves. Our 2019 interviews document substantial work by village mayors. One village mayor described his role succinctly as “The captain is the workhorse.” Another described visiting households to encourage participation and making a point “not to get tired.”³⁸ These accounts establish demands on village mayors’ time, rather than only on community volunteers.

The ADB assessment also notes that village officials were often expected to subsidize assembly costs personally because, until recently, those costs could not be charged to village budgets (ADB, 2012, p. 18). This is evidence that officials could incur financial costs as well as devote time to implementation. It does not quantify the additional costs caused by KALAHÍ or show how much village mayors, as distinct from other officials, paid. These accounts support the possibility of direct costs to incumbents, but do not establish that the costs outweighed the perceived benefits of remaining in office.

The interviews contain no account of a village mayor deciding not to seek reelection because of these demands. Asked about participating in another KALAHÍ cycle, one village mayor replied, “Sure, why not. That will be value-adding to the barangay.”³⁹ Willingness to join another program cycle is not, however, a decision to seek another term. The interviews were conducted with serving officials and residents several years after the election studied here, not with a sample of village mayors who withdrew. The absence of such an account cannot rule out costs as a reason for withdrawal.

A tracer study in Enrique Villanueva similarly describes demanding and stressful volunteer work alongside continued public engagement. Some former volunteers who became barangay officials took on greater responsibilities, yet the report states that “none of them complained about their current role” (Songco, 2016, pp. 10, 24). This selected group shows that demanding participation could coexist with a willingness to serve. It does not represent incumbents deciding whether to run again. Across these sources, the evidence documents officials’ personal costs more directly than their effect on candidacy.

Taken together, the evidence documents demands on village mayors but provides little indication that those demands reduced the value of seeking another term. The winning-margin interactions are also consistent with village mayors anticipating voters’ responses. We therefore treat direct costs as a possible but less supported explanation and turn next to the mechanisms through which citizens may have responded.

³⁸Transcribed and translated Nueva Ecija interviews, file d8861f, p. 7, and file ddb3be, p. 23.

³⁹Transcribed and translated Nueva Ecija interviews, file 2d611b, p. 4.

K.2 Mechanisms of Electoral Accountability

We next ask why citizens might respond to implementation outcomes. We begin with the funding effect across village mayors' BRT roles because the three mechanisms differ in whether formal involvement should condition voters' responses. We then turn to survey and interview accounts of citizens' assessments, expectations, and costs.

Under voter learning, a prominent role may connect success or failure more closely to the village mayor and make the outcome more informative about competence. Raised performance standards yield a similar initial prediction if chairing the BRT leads citizens to expect more from the village mayor. Absent differences in prior information, both mechanisms therefore provide reasons to expect a larger funding effect for chairs than for members or village mayors outside the team.

Citizens' participation costs do not by themselves imply either ordering. Residents may devote time or resources to implementation without obtaining the anticipated benefit, whether the village mayor chairs the BRT or not. Receiving funding may offset those burdens by providing a valued project. If benefits, participation costs, and attribution are comparable across roles, this explanation gives no reason to expect systematically larger or smaller funding effects for chairs. Explaining such differences through citizens' costs would require those conditions to differ across role groups.

These comparisons depend on the stated assumptions and cannot uniquely identify a mechanism. We therefore also examine survey and interview accounts of how citizens understood implementation outcomes, what they expected from the village mayor, and the costs they incurred. This evidence helps assess whether the responses required by each explanation were present. Expressions of disappointment alone cannot distinguish learning from unmet expectations or participation costs.

K.2.1 Funding Effects by Village Mayor's BRT Role

We compare funding effects on vote share across village mayors not recorded on the BRT, members who never chaired, and chairs.⁴⁰ We also compare prior electoral profiles to assess whether chairs were more established incumbents, as the learning interpretation might suggest. Prior electoral strength does not directly measure citizens' information about the village mayor.

The estimates below concern vote share among incumbents who run. Roles were not randomly assigned and may reflect earlier program experiences, so these comparisons do not identify the effect of assigning greater responsibility. Because BRT roles were established before funding, costs associated with preparing and presenting proposals were incurred by both funded and narrowly unfunded village mayors within each role and therefore cancel from the within-role RD contrast.

Table A48 re-estimates the five main specifications separately by role. Each RDHonest estimate uses a subgroup-specific left-side curvature bound and FLCI-optimal bandwidth. The preferred

⁴⁰Roles summarize service on the three-member presentation team across observed cycles. Chairs served as chair in at least one cycle. The categories measure formal BRT service, not whether village mayors had any implementation role (Appendix D.3).

funding effects are 8.5 percentage points for village mayors not on the BRT, 7.9 for members, and –1.5 for chairs. The corresponding collapsed estimates are 7.9, 12.3, and –3.0 points. Only the member estimates exclude zero at the 10 percent level. These separate tests do not establish differences across roles.

Table A48: Funding RD Effects on Incumbent Vote Share by Village Mayor’s BRT Role

Role	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
<i>Panel A: RD Honest</i>					
Not on BRT	0.093 (0.058)	0.088* (0.052)	0.088 (0.060)	0.108* (0.061)	0.085 (0.066)
Member	0.099* (0.049)	0.084* (0.043)	0.084* (0.045)	0.086 (0.050)	0.079* (0.043)
Chair	0.031 (0.043)	0.007 (0.039)	0.007 (0.058)	-0.006 (0.053)	-0.015 (0.058)
<i>Panel B: Collapsed support-point RD</i>					
Not on BRT	0.086 (0.119)	0.088 (0.063)	0.088 (0.063)	0.080 (0.066)	0.079 (0.056)
Member	0.034 (0.051)	0.089*** (0.034)	0.089*** (0.034)	0.111** (0.044)	0.123** (0.050)
Chair	-0.011 (0.052)	-0.007 (0.054)	-0.007 (0.054)	-0.007 (0.057)	-0.030 (0.051)

Notes: The dependent variable is incumbent vote share conditional on running. The three role groups are mutually exclusive. Not on BRT means that the village mayor was not recorded as serving on the three-member team that presented the village proposal. Member means that the village mayor served on the BRT but was never recorded as chair. Chair means that the village mayor chaired the BRT in at least one observed cycle. All village mayors remained associated with village implementation, so the categories measure formal BRT participation rather than whether the village mayor had any implementation role. The corresponding numbers of available vote-share observations are 215, 218, and 263. Each cell is estimated separately within the indicated subgroup. Models follow the specification ladder in the main RD table. Panel A uses RDHonest with the left-side quadratic curvature rule and four-times multiplier. The curvature bound and FLCI-optimal bandwidth are calibrated separately for every role–model cell. Models (3)–(5) cluster EHW standard errors by municipality–cycle. Panel B collapses observations to support-point means and uses unclustered inference. The chair sample contains only one program cycle, so its constant cycle fixed effect is absorbed. Because role is not randomly assigned, vote share is observed only for incumbents who run, and each estimate uses a smaller subgroup, these results are suggestive. Standard errors are in parentheses. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

To assess the learning explanation, Table A49 examines whether chairs were already more electorally established. They had the highest average 2010 vote share (61.4 percent), largest winning margin (27.3 points), and fewest candidates (2.19). Compared with members, their prior vote share was 3.4 points higher ($p = 0.011$) and their winning margin 4.7 points larger ($p = 0.045$). They also faced fewer candidates than either other group. The table reports unadjusted village-level pairwise tests with Eicker–Huber–White standard errors.

The distributions nevertheless overlap substantially (Figures A35 and A36). Median winning margins are similar, at 13.8 points for chairs, 14.1 for village mayors not on the BRT, and 12.5 for members. Single-candidate races account for 13.0, 10.4, and 9.7 percent, respectively. The higher chair averages partly reflect the upper tail rather than a sharp separation between groups.

The smaller estimated effects for chairs do not follow this simple role-based prediction. Their stronger prior electoral profiles offer one possible interpretation. If citizens already knew more

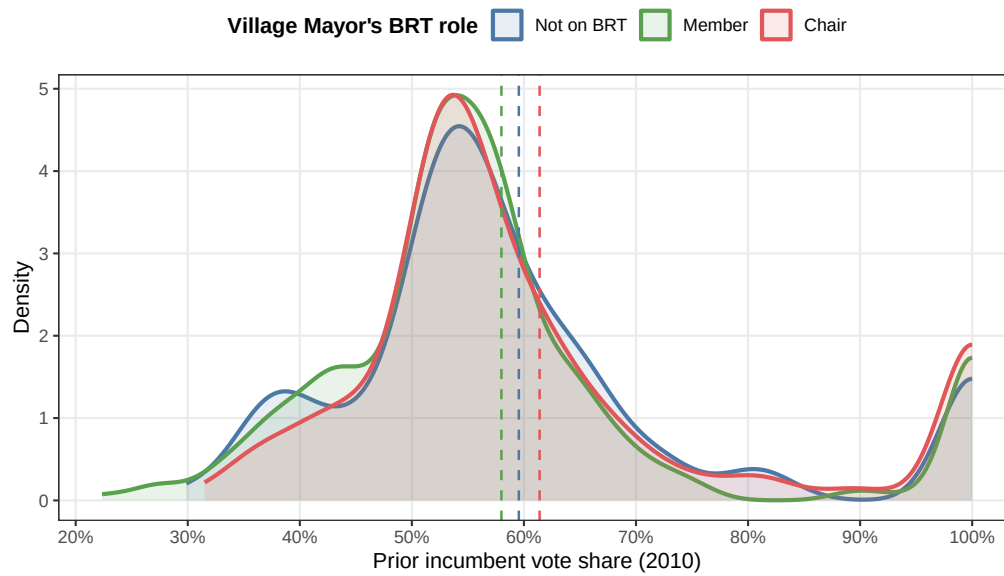
Table A49: Prior Electoral Strength by BRT Role

Statistic	Group means			Pairwise differences		
	Not on BRT	Member	Chair	Member – not on BRT	Chair – not on BRT	Chair – member
Winner vote share, 2010 (percent)	59.5	58.0	61.4	-1.55 (1.33)	1.86 (1.29)	3.41** (1.33)
Winning margin, 2010 (percentage points)	25.3	22.7	27.3	-2.65 (2.27)	2.02 (2.26)	4.67** (2.32)
Number of candidates, 2010	2.32	2.31	2.19	-0.01 (0.07)	-0.13** (0.06)	-0.12* (0.06)
Incumbent vote share if unfunded, 2013 (percent)	56.8	54.3	58.2			
Villages	345	298	378			
Unfunded vote-share villages	124	117	143			

^a Notes. The table reports descriptive means after retaining one observation per village. The unfunded vote-share row likewise retains one unfunded observation per village. Pairwise cells report mean differences with standard errors in parentheses. The tests use estimatr with village-level EHW standard errors and the Stata-style HC1 correction. Not on BRT means that the village mayor was not recorded as a member of the three-member presentation team; member denotes service without chairing; and chair denotes chairing in at least one observed cycle. All village mayors remained associated with village implementation. The BRT categories are mutually exclusive, and role assignment is not random. Significance levels: *** 1 percent, ** 5 percent, * 10 percent.

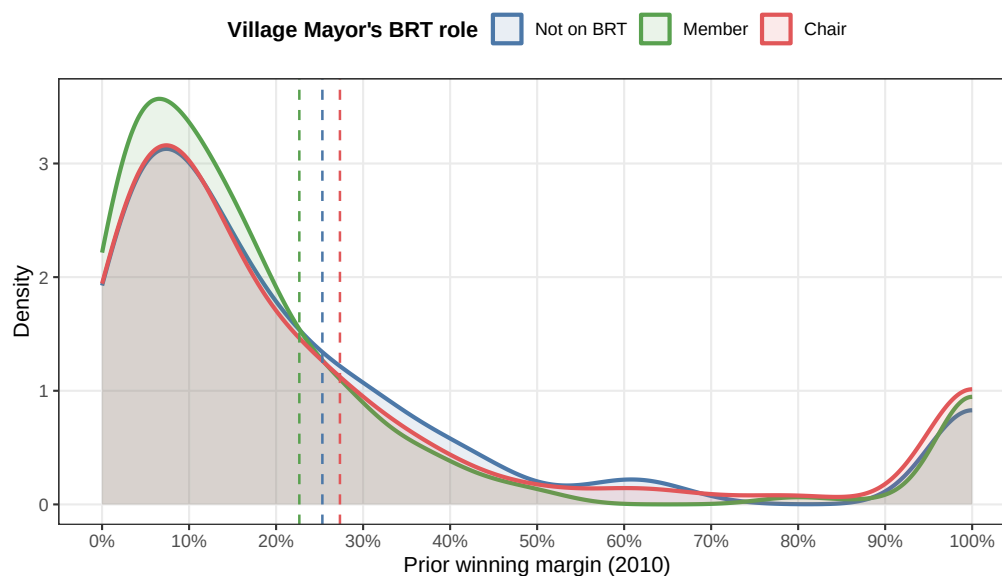
about these more established incumbents, funding may have conveyed less new information about their competence. Prior electoral strength does not directly measure citizens' information or changes in their beliefs, however. The point estimates also do not match the larger chair effect implied by raised standards, while citizens' participation costs alone predict no smaller effect for chairs. Because the subgroup differences are imprecise and roles were not randomly assigned, these patterns cannot identify a mechanism. They help interpret the funding effect but do not explain the negative average effect of participation.

Figure A35: Prior Incumbent Vote Share by Village Mayor's BRT Role



Notes: Curves show kernel-density estimates of the incumbent village mayor's 2010 vote share for three mutually exclusive BRT role groups. The sample retains one observation per village from the clean funding-RD sequences. Dashed vertical lines show role-specific means. BRT roles were not randomly assigned.

Figure A36: Prior Winning Margin by Village Mayor's BRT Role



Notes: Curves show kernel-density estimates of the incumbent village mayor's 2010 winning margin for three mutually exclusive BRT role groups. The winning margin is the village mayor's vote share minus the runner-up's vote share. The mass at 100 percentage points consists almost entirely of single-candidate races, with one recorded two-candidate race. The sample retains one observation per village from the clean funding-RD sequences. Dashed vertical lines show role-specific means. BRT roles were not randomly assigned.

K.2.2 Citizens' Responses to Implementation Outcomes

We next examine survey and interview accounts of how citizens responded to unsuccessful proposals, funded projects, and the demands of participation. We draw on household surveys and qualitative fieldwork from the impact evaluation in our experimental sample (Beatty et al., 2018). We also use an ADB assessment and evaluation synthesis from earlier KALAHI implementations (ADB, 2012; Labonne, 2013), alongside our 2019 interviews with residents and officials in eight barangays outside the experimental sample (Appendix C.3).

For voter learning, we examine accounts linking implementation outcomes to assessments of the village mayor's competence or effort. For raised standards, the relevant evidence is whether participation changed what citizens believed the village mayor should deliver. For citizens' costs, we examine reports of time and resources spent without the anticipated benefit and whether residents connected those burdens to the village mayor. Disappointment after unsuccessful participation may reflect unmet expectations, costs incurred without a benefit, or both. We distinguish residents' statements from officials' accounts of residents' responses.

Why Participation Did Not Always Produce Funding We first use the interviews to examine how respondents understood why proposals received funding or were unsuccessful. This helps assess the learning and expectations explanations. If outcomes were understood as entirely beyond the village mayor's influence, they would provide little information about competence or basis for expecting the village mayor to secure funding. Accounts of actions that could affect the decision would make both interpretations more plausible, without distinguishing them. This exercise does not establish the electoral consequences of participation costs.

Respondents described both opportunities for village mayors to influence outcomes and constraints beyond their control. One village mayor emphasized the need to “show that your project is important” through a good presentation. A second described proposals failing because of missing documents or right-of-way problems. In another interview, a respondent described preparing a proposal that did not proceed because the barangay could not afford the required counterpart contribution.⁴¹

Respondents also differed in their accounts of bargaining among barangays. In a joint village mayor–secretary interview, one respondent explained, “Before the vote, barangays talk to one another.” Others emphasized formal procedures, need, and proposal quality, or described yielding to barangays with more pressing needs.⁴² Success was therefore not understood as a simple reflection of competence. Village mayors' actions could influence funding decisions, but these also depended on documentation, resources, and decisions by other barangays. These accounts support

⁴¹Transcribed and translated Nueva Ecija interviews, file d8861f, p. 6, file 2d611b, p. 3, and file 807cb2, pp. 12–13. Page numbers refer to the combined transcript.

⁴²Transcribed and translated Nueva Ecija interviews, file 807cb2, pp. 15–17, file d8861f, pp. 6–7, and file ddb3be, pp. 25–26. The combined transcript in file 807cb2 does not consistently identify which of the two respondents is speaking.

the possibility that outcomes informed evaluations and expectations, but whether citizens actually used them in that way requires further evidence.

Experiences of Unsuccessful Participation We next examine accounts of unsuccessful proposals in three villages discussed in the ADB assessment. Under the learning explanation, we would expect citizens to interpret failure as evidence about the village mayor's competence or effort. Under raised standards, dissatisfaction should reflect what citizens had come to expect from the process. Under the cost explanation, it could instead reflect time or resources spent without a project. Frustration alone cannot distinguish these responses.

Participants in all three villages described frustration after losing, and the report states that "their first impulse was to discontinue participation in future cycles." Volunteers found returning after the MIBF to "face their neighbors and friends" particularly difficult (ADB, 2012, pp. 25–26).

Reflection sessions and encouragement from facilitators and municipal officials persuaded the villages to continue. Two subsequently received funding. The third participated in three regular cycles and an additional pilot cycle without receiving a grant. Its proposals failed twice because of a missing certified deed of land donation and twice because unresolved follow-up requirements prevented admission to the MIBF. Residents regarded capacity building and project grants as "one integrated package." These are accounts of substantial participation without funding, rather than an absence of implementation activity.

Residents did not necessarily attribute unsuccessful proposals to the incumbent. After receiving funding, residents in the first two villages described their earlier losses as "the luck of the draw." Representatives of the third blamed other villages and preferred an entitlement to funding (ADB, 2012, p. 26). These accounts do not indicate that residents interpreted failure as evidence of the village mayor's competence or effort, as the learning explanation would require in these cases. They are more directly consistent with unmet expectations or participation without the anticipated benefit, but establish neither a change in performance standards nor an electoral response.

The earlier impact-evaluation synthesis similarly warns that time spent on unsuccessful proposals may discourage future participation and recommends better management of expectations (Labonne, 2013). This conclusion draws on qualitative evidence rather than direct measures of changes in performance standards. The accounts are consistent with unmet expectations and participation costs, without distinguishing them or establishing an electoral penalty.

Perceived Electoral Responses We then examine whether respondents in our 2019 interviews connected implementation outcomes to evaluations of village mayors or electoral choices. For learning, the relevant accounts would treat success or failure as evidence about the incumbent. For raised standards, they would describe what the village mayor should have delivered and why. For citizens' costs, they would connect electoral dissatisfaction to burdens borne during participation. Appreciation of a useful project alone is consistent with material benefits and does not distinguish

these mechanisms.

Some officials believed that delivered projects increased their electoral support. Asked whether two road projects aided his reelection, one village mayor replied tentatively that they probably had, noting, “They didn’t even give me an opponent.” Another village mayor described residents as “really really happy” with new street lights and said they thanked him for his support.⁴³ These accounts indicate perceived rewards for delivery, without revealing which mechanism produced them.

Accounts of unsuccessful participation more directly suggest evaluations of the incumbent. In one joint village mayor–secretary interview, a respondent reported that residents thought the “leadership was weak” when proposals did not proceed. Another village mayor considered it possible that voters would be upset when available funding did not produce a project because of “neglect or lack of effort.” A barangay secretary suggested that the previous village mayor’s electoral defeat might reflect residents tiring of waiting for projects.⁴⁴ The first two accounts invoke qualities of the incumbent, as the learning explanation would suggest. The account linking electoral defeat to delays is also consistent with unmet expectations. None directly observes a change in citizens’ beliefs or establishes why an incumbent did not seek reelection.

Residents’ own statements provide limited evidence about expectations. Two respondents said they expected the village mayor to obtain KALAHÍ projects, whereas a third said, “We don’t expect any to come.”⁴⁵ Respondents thus differed in their expectations of receiving a project. These statements do not establish that participation raised the standard residents applied to the village mayor, rather than revealing an expectation they already held.

The accounts therefore offer evidence of evaluations consistent with learning and of expectations directed at village mayors, but not direct observations of changing beliefs or standards. The accounts quoted here do not directly connect citizens’ own participation costs to their votes. Several followed questions suggesting an electoral interpretation, and most descriptions of electoral consequences came from officials rather than citizens reporting their own choices. They illustrate how implementation outcomes could inform citizens’ evaluations of incumbents without establishing their effect on voting or candidacy. We examine the evidence on citizens’ costs more closely next.

Citizens’ Participation Costs Finally, we examine reports of delays, unpaid work, and expenditures to assess whether participation imposed costs on citizens. We use household surveys and qualitative work from the impact evaluation in our experimental sample, together with the earlier

⁴³Transcribed and translated Nueva Ecija interviews, file d8861f, p. 5, and file ddb3be, p. 24. The former question explicitly asked whether KALAHÍ aided reelection. The latter asked whether residents appreciated the project and were grateful to the village mayor.

⁴⁴Transcribed and translated Nueva Ecija interviews, file 807cb2, pp. 12–13, file d8861f, p. 7, and file d82222, p. 19. The first question asked whether voters had punished the previous village mayor. The second introduced the researchers’ interpretation that project outcomes could generate electoral rewards or penalties. The third statement was the secretary’s interpretation of the previous village mayor’s defeat.

⁴⁵Transcribed and translated Nueva Ecija interviews, file 2f2f08f, pp. 28–29, file 4450af2, p. 34, and file 2ef321, p. 28.

ADB assessment. Under the cost mechanism, citizens evaluate the benefits received in relation to the burdens of participation. Dissatisfaction with costs incurred without a benefit would be consistent with this explanation, even without learning about competence or raised performance standards. Learning and raised standards do not require such costs, although they can coexist with them. Evidence of participation costs is therefore necessary for this explanation, but does not establish their electoral effect.

In the 2015 third-round survey, 45 percent of respondents in treatment municipalities considered the process too slow, while 11 percent considered it too burdensome (Beatty et al., 2018, p. 26). The distinction matters. Delays were a common criticism, but relatively few respondents described the overall process as excessively burdensome. The accompanying qualitative work covered 12 treatment barangays that had completed at least two rounds of the community process and received a subproject. It therefore describes funded villages, not those that stopped before applying or participated without receiving funding. Receiving a subproject did not necessarily mean that it was completed or usable. Both sources were collected after the 2013 election and cannot establish how costs affected voting at that election.

An assessment of earlier KALAHİ implementations provides more detailed accounts of the resources involved. A village councilor serving as a community volunteer described working on a road subproject for eight months without pay, during which his wife supported the family. Other volunteers reported lost income and using personal funds for meetings (ADB, 2012, pp. 23–24). These accounts document costs to citizens and volunteers. They do not necessarily describe costs borne by village mayors, which we examine separately below.

The surveys and accounts establish that participation could be costly even before a benefit arrived. Combined with the frustration reported after unsuccessful proposals, they make citizens' participation costs a plausible explanation for dissatisfaction without a change in beliefs or standards. They do not show that these costs outweighed benefits or caused an electoral response. Nor does the relatively small share describing the process as too burdensome rule out a stronger response among unsuccessful participants, whom the qualitative sample does not represent.

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